Analysis Note:

Unbinned measurement of the charged-current inclusive cross section in vN in collisions with public MINERvA data

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Code and file references: github.com/josephbaileyy/MINERvA-OmniFold

Abstract

We perform unbinned, multi-dimensional cross-section measurements of medium-energy charged-current neutrino-nucleus events measured by the MINERvA experiment. The machine learning-based method OmniFold is used to enable these measurements. We first reproduce a published double-differential cross section measurement in muon transverse and longitudinal momentum, recovering both the spectrum and its uncertainty budget with a fully independent method. We then perform the first simultaneous three-, four-, and five-observable measurements of MINERvA data, adding available energy ($E_{\rm avail}$), three-momentum transfer (q_3), and hadronic invariant mass (W); each reproduces its lower-dimensional predecessor under marginalization. These higher-dimension results recover, through an independent method, the established MINERvA deficit of low-momentum-transfer multinucleon interactions, and localize a residual high-$E_{\rm avail}$ excess to the deep-inelastic region, where no current generator reproduces the data. We also extend the fiducial volume by relaxing the muon kinematic cuts and measure the full phase space using a point-cloud transformer to process variable-length inputs. Quantitative results, the full uncertainty budgets, and the validation suite are described in detail.

Executive summary

This note documents the application of *unbinned machine-learning unfolding* (OmniFold [1, 2]) to MINERvA's medium-energy neutrino data — first as a faithful reproduction of a published flagship measurement, then as its extension to simultaneous three-, four- and five-observable cross sections and to the first full-phase-space charged-current (CC)-inclusive measurement, neither of which a binned method has produced at MINERvA. The executive summary outlines the whole story; everything afterwards is the supporting technical record.

Traditional unfolding corrects detector effects with a migration matrix between histogram bins, so each added observable multiplies the number of bins and the method drowns in dimensions.

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OmniFold instead trains a classifier to learn, event by event, a weight that morphs the simulation into the data, and pulls that weight back to the true (pre-detector) level through the simulation's truth–reco pairing. No binning is imposed until the very end, so adding an observable is just adding an input column. This analysis uses gradient-boosted decision trees for the fixed-dimension measurements and validates the choice extensively (closure tests, coverage, calibrated-classifier checks, and neural-network cross-checks).

Uncertainty-remediation status. The validated 2D reproduction, higher-dimensional central values, closure tests, marginalisation anchors, and shape-localisation statements remain current. Numerical N-D uncertainty budgets, covariance-based significances, the former unified-throw adoption, and PET–GBDT precision comparisons are withheld until the corrected production and its validation gates are complete.

Results overview:

- **Reproduction.** Our unbinned measurement of the muon transverse and longitudinal momentum, $d^2\sigma/(dp_T\, dp_{\|})$, matches the published measurement [3]: total cross-section ratio 1.011, 94.1 % of bins within 10 %, per-bin pulls sub-1σ, and a fully independent uncertainty budget whose median per-bin uncertainty (6.87 %) matches the published 6.86 % (§5, §4, Fig. 7).
- **First 3D/4D/5D MINERvA measurements.** Adding available energy (E_{avail}), momentum transfer (q_3), and hadronic mass (W) produces the first simultaneous three-, four- and five-observable unfolds of MINERvA data; every result reproduces its lower-dimensional predecessor when marginalized — a built-in cross-check (§7).
- **Known physics recovered independently.** All four truth-level generators (GENIE [4], MINERvA Tune [5], NuWro [6], GiBUU [7]) under-predict the data at low E_{avail}; pion final-state-interaction dials move the spectrum by $< 1\,\%$ while adding Valencia 2p2h [8, 9] fills the quasielastic–Δ dip — the established MINERvA low-recoil/2p2h deficit [5, 10], re-found by an independent method (§7.7, Fig. 23).
- **A new localization of the remaining excess.** The high-E_{avail} excess that 2p2h cannot explain is localised in the (E_{avail}, W) plane: its positive data-minus-generator difference concentrates in the deep-inelastic corner ($W \geq 1.8\,\text{GeV}$), and all four generators underpredict that region. Exact significances are withheld while the corrected covariance is regenerated (§8, Fig. 31).
- **A methodological finding.** The standard covariance adds matrices built for one systematic source at a time, which omits cross-source response terms. The corrected unified-throw test perturbs all reweightable sources simultaneously using actual asymmetric endpoints, one fixed estimator seed, and mean-centered throws; it reports the joint mean shift separately. The earlier numerical adoption is withdrawn pending corrected production (§4.4). **[BPN: do neutrino people really sum in blocks and not all 160? I know that collider people add in blocks, but I thought neutrino people always do it the way you do it?] [JRB: Here in the MINERvA codebase, they block-sum the systematic bands.][BPN: It looks like they add matrices? Where do I see that they do it in blocks?] [JRB: Adding the matrices is what I meant by block-summing. `GetTotalErrorMatrix` (L1373) loops over the named systematic sources and accumulates `covmx += GetSysErrorMatrix(name)` (L1382–1385); each `GetSysErrorMatrix` is the covariance built from that one source's universes, with every other source held at nominal.**

Summing those per-source matrices retains within-source bin–bin correlations but omits cross-source terms because no universe moves two sources together. The corrected unified throw is the direct test of that approximation; its numerical result is pending.]

- **Toward raw-data unfolding.** The other unfolds compress each event into engineered scalars; here a point-cloud transformer instead takes reconstructed calorimeter clusters together with the muon kinematics and produces an absolutely normalized 4D central cross section. Closure validates the central extraction, and the truth cloud supports post-hoc E_{avail} projection. A fair PET–GBDT precision comparison is withheld until the fully retrained data+MC bootstrap ensemble is complete (§9, §9.3). **[BPN: Awesome! Can you now clean up this bullet point to reflect our discussion? I'd probably call this 'full phase space' since it measures everything that can be measured and then maybe the next one should be 'larger fiducial volume' or similar? - see how I rephrased the abstract.] [JRB: Cleaned up above. On naming: do you mean that "full phase space" is synonymous with the point-cloud/raw data inputs? If so, I probably need to change the way I use this throughout the whole paper, since I used full phase space to mean no cuts.]**
- **A first full-phase-space cross section.** **[BPN: See above bullet!]** Re-running the same 2D $(p_T, p_{\|})$ muon double-differential unfold with the muon kinematic cuts removed from the signal definition (they hide a third of the rate) gives a full-phase-space CC-inclusive cross section, $\sigma_{\text{FPS}} = 4.502 \times 10^{-38}\,\text{cm}^2/\text{nucleon}$, anchored to the restricted measurement at 0.57 % where they overlap, with a three-prior envelope declaring the extrapolation cost cell-by-cell (§10). The same extension is then realized with the point-cloud engine at full statistics on the 5D grid, with two-tier (measured/extrapolated) reporting by reconstruction efficiency and a demonstration that PET retraining noise sits an order of magnitude below the prior envelope (§10.1).

This note is organized as follows. Sections 2–3 define the samples and the method; §4 and App. A are the uncertainty machinery (the appendix documents the analysis-specific procedures — variance sources, coverage and closure toys, the covariance construction — for reproduction); §5 is the 2D reproduction against the published measurement; §6 is the validation suite; the physics is in §7–8; §9 is the point-cloud track; §10 is the full-phase-space measurement and §11 the outlook.

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1 Introduction and motivation

MINERvA (Main Injector Experiment for ν-A) is a dedicated neutrino–nucleus scattering experiment at Fermilab, situated in the NuMI beamline immediately upstream of the MINOS near detector [11]. Over roughly a decade of operation, ending in 2019, MINERvA recorded high-statistics samples in both the low-energy and medium-energy (ME) NuMI beam configurations [12] and in neutrino-dominated (forward horn current, FHC) and antineutrino-dominated beam modes. Its physics goal is the precision measurement of neutrino–nucleus interaction cross sections on a range of nuclear targets in the few-GeV region most relevant to accelerator-based oscillation experiments: these cross sections, and the nuclear effects that shape them, are a leading systematic uncertainty for experiments such as NOvA [13] and DUNE [14] **[BPN: Add in competitors T2K and HyperK?]**, and the few-GeV regime mixes quasistatic, resonant, and deep-inelastic processes that no single model describes cleanly. This note uses the public ME FHC CC-inclusive data product; the detector itself is described in §2.

The MINERvA ME FHC charged-current (CC) inclusive double-differential cross section in muon transverse and longitudinal momentum, $d^2\sigma/(dp_T\,dp_{\|})$, was published by Ruterbories *et al.* [3] using D'Agostini iterative Bayesian unfolding (IBU) [15] with a binned response matrix. This note reproduces that measurement with *unbinned* OmniFold [1, 2], an unfolding technique based on iterated reweighting with machine-learned likelihood ratios that operates on per-event observables rather than on a fixed histogram binning. Recent advances in machine learning have enabled unbinned unfolding techniques; see Refs. [16, 17] for overviews and Ref. [18] for cross-experiment practical guidance. These methods fall into two broad families: those that iteratively reweight a starting simulation, of which OmniFold [1, 2, 19] is the most widely adopted, and those that generate new samples directly with a neural network [20–23]. OmniFold was first demonstrated on collider data by the H1 collaboration [24, 25] and has since been adopted across a number of experiments; the application closest to this work is the OmniFold study on simulated T2K near-detector data [26]. These techniques lift the binned-response limitations of widely-used classical unfolding methods [27] like IBU, the SVD method [28], and TUnfold [29].

Why unbinned:

- **No binning at unfold time.** OmniFold learns event weights in the full observable space; the result is histogrammed only at the end, so the binning can be chosen (or refined) after unfolding without re-running.
- **Natural path to higher dimensions.** Adding observables is adding input features, not multiplying response-matrix bins — this is the enabling property for the available-energy extension (§7) that has no practical IBU analog.
- **Different regularization** **[BPN: Are you or your LLM British??]** **[JRB: Apparently. Fixed throughout.]** OmniFold's regularizer is the machine learning model capacity in addition to the iteration count in common with IBU. Comparing the two approaches on the same data quantifies the method-dependence of the result (§5).

Goal and scope. The goal is a faithful, fully uncertainty-quantified reproduction of the published 2D cross section as the demonstrator for unbinned OmniFold on MINERvA data, followed by its higher-dimensional extensions. This note covers the experiment and samples (§2), the method and pipeline (§3), the 2D result and its comparison to the published data and to GENIE MINERvA Tune v1 (§5), the uncertainty budget (§4), the validation suite (§6), the three- and four/five-observable extensions with available energy and q_3 (§7), the $(E_{\rm avail}, W)$ study that localises the remaining deep-inelastic-tail excess (§8), the point-cloud transformer track (§9), the full-phase-space measurement (§10), and a summary and outlook (§11). Appendix A is the complete statistical methodology and Appendix C the MINERvA measurement-dimensionality survey.

2 The MINERvA experiment, data and simulation samples

2.1 Detector and beam

MINERvA [11] was a fine-grained scintillator-tracker neutrino experiment on the NuMI beamline at Fermilab. The central tracker is built from hexagonal planes of triangular polystyrene scintillator strips in three stereo orientations, surrounded by electromagnetic and hadronic calorimetry; the magnetized MINOS near detector, 2 m downstream, serves as the muon spectrometer, providing the muon momentum and charge for MINOS-matched tracks. This analysis uses the *medium-energy* (ME) neutrino-mode (FHC) exposure with $\langle E_\nu \rangle \sim 6\,\mathrm{GeV}$. The neutrino flux is the *a posteriori* constrained NuMI ME prediction of Ref. [12] (PPFX hadron-production weights plus the neutrino–electron-scattering constraint), identical to the flux used by the published measurement [3].

2.2 Data and simulation samples

The data are the twelve ME FHC playlists (ME1A–ME1G, ME1L–ME1P), corresponding to 1.057×10^{21} protons on target — the same exposure as Ref. [3]. After the selection below the data sample contains 4 119 797 reconstructed events.

The reference simulation is GENIE 2.12.6 [4] with the MINERvA Tune v1 applied as per-event weights through the MAT (MINERvA Analysis Toolkit) reweighters: the flux-and-CV weight, the GENIE dial weights, the low-recoil 2p2h enhancement fit, and the non-resonant-pion reduction. Detector response is the full Geant4-based [30] MINERvA simulation underlying the production AnaTuples. The simulated sample comprises 32 849 103 true signal events (POT-normalized to data exposure event-by-event), processed through the identical reconstruction and selection as the data.

2.3 Signal definition and event selection

The signal definition and phase space follow Ref. [3] exactly. The truth-level signal is a charged-current ν_μ interaction with vertex in the central-tracker fiducial volume, with muon phase space

$$\theta_\mu < 20^\circ, \qquad 1.5 < p_\parallel < 60\,\mathrm{GeV/c}, \qquad p_T < 4.5\,\mathrm{GeV/c}.$$

The fiducial volume is the tracker region $5980 < z < 8422\,\mathrm{mm}$ within an 850 mm apothem, containing 3.2353×10^{30} nucleons (the fixed tracker-geometry constant).

Reconstructed events are selected by requiring: a reconstructed vertex in the same tracker fiducial volume and apothem; a muon track with $\theta_\mu < 20^\circ$ matched to a track in MINOS with a passing MINOS fit; no dead-time overlap; and a neutrino (rather than antineutrino) candidate. The MINOS-match requirement is implemented as `isMinosMatchTrack==1 && minos_trk_is_ok==1` (the current MINERvA-101 tutorial enforces the same `isMinosMatchTrack` match; we additionally require a passing MINOS fit). This yields a predicted genuine-background rate — charged-current events of other flavours or wrong sign, neutral-current interactions, and out-of-fiducial-vertex events; out-of-phase-space signal (*fakes*) is treated separately — of 0.35 % of the selected sample (evaluated on playlist 1A). This is the same sub-percent scale as the 8655 (0.2 %) predicted background events of Ref. [3], whose definition counts only the wrong-flavour, wrong-sign, and neutral-current components;

exact agreement is not expected given the definitional and selection differences, and neither prediction carries an uncertainty. Backgrounds (reconstructed events whose true interaction fails the signal definition) and *fakes* (reconstructed in phase space, true muon outside it) are POT-scaled and subtracted from the reconstructed-data target before unfolding (§3).

2.4 Per-event analysis trees

Unbinned unfolding needs per-event records rather than histograms. A dedicated event loop (`MINERvA101/MINERvA-101-Cross-Section/runEventLoopOmniFold.cpp`), built on the MAT systematic-universe machinery, writes four flat trees per playlist: `mc_truth_denom` (every true signal event — the efficiency denominator), `mc_signal_reco` (reconstructed, truth-matched simulated events), `mc_background`, and `data`. Each entry carries the truth and reconstructed observables (p_T, $p_\parallel$, and in the extended runs E_{avail}, q_3, W), the MnvTune central-value weight, and — in the `_universes_full` variants — the full set of systematic-universe weights and laterally-shifted observables (§4). Two invariants make the result truth-tree authoritative:

- **Truth-tree-authoritative reco gate**: a reconstructed signal event is filled only when its $(\mathrm{run}, \mathrm{subrun}, \mathrm{event})$ key is present in the truth denominator; truth-only events enter as native OmniFold "miss" entries.
- **Completeness** $c =$ (events in the OmniFold input)/(all truth events) is 1.000000 by construction at ME FHC (32 849 103 entries), so the cross-section completeness division is a no-op self-check.

Because the event loop selects one global playlist flux/calibration from the first event, it is always run per playlist and the twelve outputs merged afterwards (with a `TTree::SetMaxTreeSize`-aware merger for the > 100 GB universe dumps). The flux histogram (`pTmu_reweightedflux_integrated`) comes from the corresponding baseline histogram event loop and is POT-weighted across playlists; playlist-to-playlist flux differences are a $\sim 0.2\,\%$ effect.

2.5 Reco-level control distributions

Figure 1 compares the selected data to the POT-scaled simulation (signal + background) at reconstruction level in all five observables. The selected event count exceeds the MnvTune-v1 prediction by $\sim 12\,\%$ in overall normalization, with a clear trend in p_T (the data/MC ratio rises from 0.88 in the lowest bin to ~ 1.2 above 1 GeV/c) — the reco-level image of the normalization deficit quantified at truth level in §5.4 and §7.5. The simulation describes the *shapes* well, which is what the unfolding relies on; the rate offset is what the data pull corrects.

The shapes themselves are set by the beam and the acceptance. The $p_\parallel$ spectrum peaks at 4–6 GeV/c — the image of the $\langle E_\nu \rangle \approx 6$ GeV NuMI medium-energy peak — and the spectrometer-match requirement removes the lowest-$p_\parallel$ muons; p_T peaks near 0.7 GeV/c and falls by over an order of magnitude across the measured range. On the hadronic side the sample is dominated by low-recoil (quasi-elastic and $2p2h$) events: E_{avail} is concentrated below $\sim$0.5 GeV with a long inelastic tail collected by the wide upper bins, q_3 tracks the momentum transfer of the same population, and reconstructed W rises through the smeared quasi-elastic and $\Delta(1232)$ region before the deep-inelastic continuum takes over above $\sim$2 GeV. The absolute background is at the few-percent level under the muon projections and is concentrated at high recoil (highest E_{avail}, q_3 and W bins). Beyond the p_T trend noted above, the ratio panels also show a mild enhancement in the lowest E_{avail} bins — the reco-level trace of the low-recoil excess examined at truth level in §5.4.

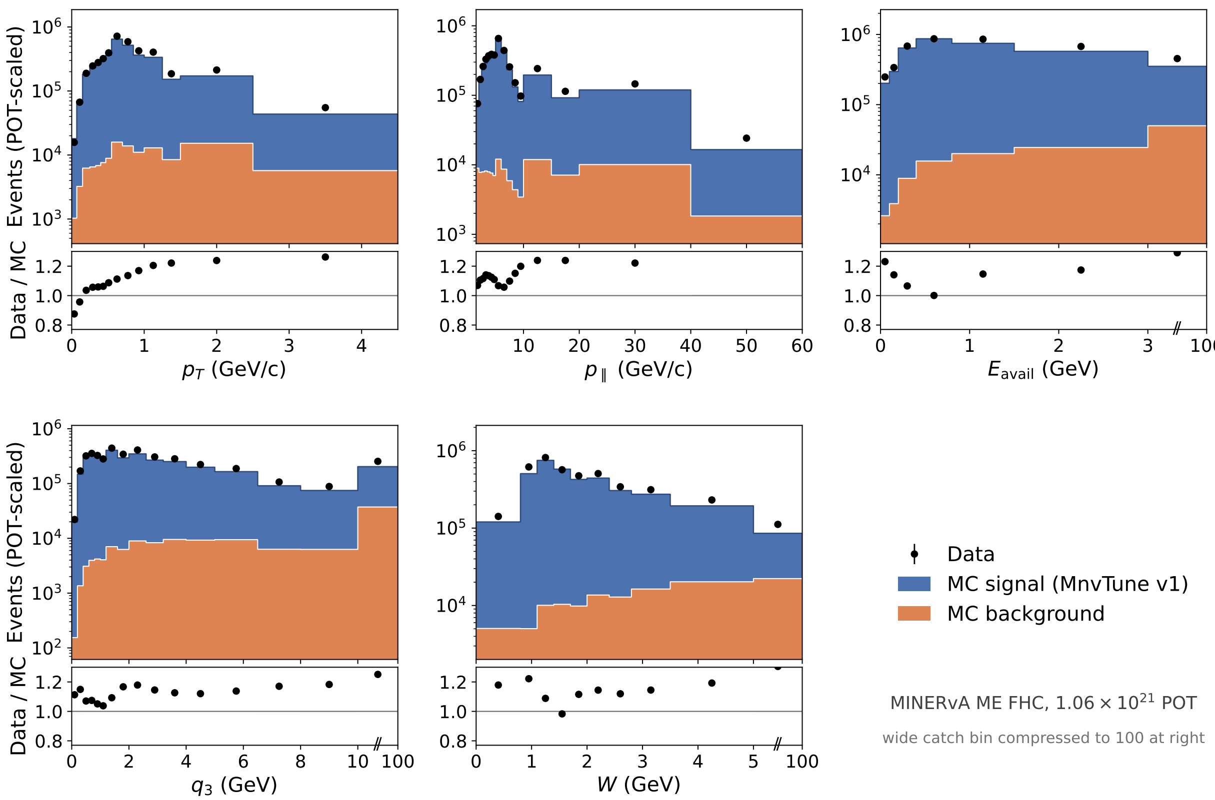

Figure 1: Reco-level control distributions for the analysis selection: data (points) versus POT-scaled simulated signal (blue) stacked on background (orange), with the data/MC ratio below, for p_T, $p_\parallel$, E_{avail}, q_3 and W. On the hadronic axes the wide catch bin is drawn compressed at the right of each panel, running out to 100 GeV. The gray band on the ratio is the MC statistical uncertainty, mostly narrower than the points. The $\sim 12\,\%$ normalization excess of the data over MnvTune v1 is the reco-level counterpart of the truth-level generator deficits of §7.5. **[BPN: extend range for q3 and W? Good to see the distribution go back down at edge of phase space.]** **[JRB: Done!]**

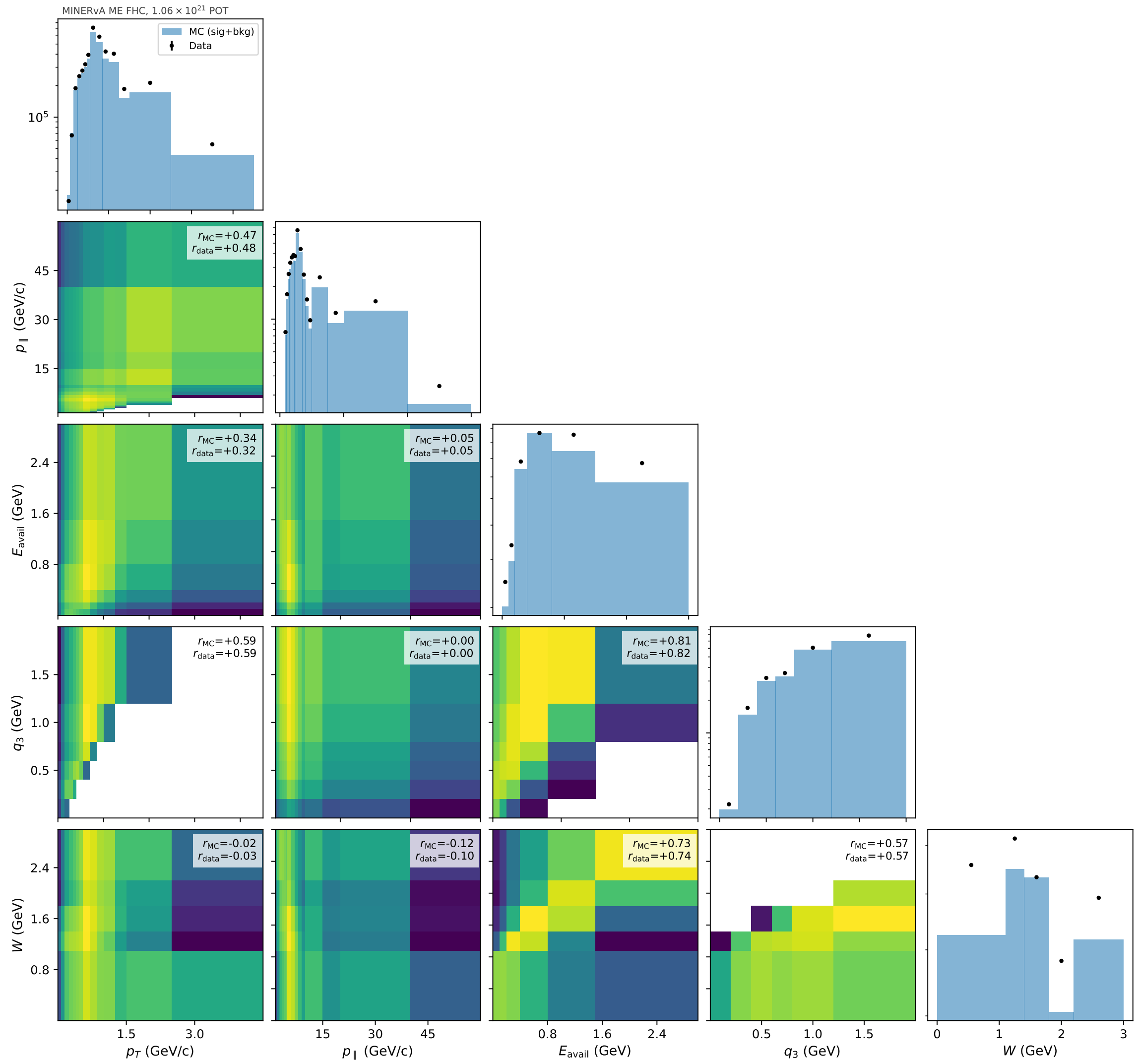

Figure 2: Reco-level corner plot of the five observables: pairwise POT-scaled MC (signal+background) densities (lower triangle, logarithmic color scale; wide catch bins dropped) with the weighted Pearson correlation of the simulation and the corresponding data value annotated per panel, and the 1D marginals (data points over MC) on the diagonal. The strongest correlations are kinematic by construction: q_3 with E_{avail} ($r \approx 0.81$) and W with E_{avail} ($r \approx 0.73$), which share the recoil energy, and q_3 with p_T ($r \approx 0.59$), which shares the muon transverse kick; E_{avail} and W are nearly uncorrelated with the muon variables. The data reproduce every simulated coefficient to within 0.02.

3 The unbinned OmniFold procedure

3.1 Algorithm

OmniFold [1, 2] unfolds by iterated reweighting with machine-learned likelihood ratios, operating on per-event observables rather than on a fixed histogram binning. Each iteration has two steps:

1. **Step 1 (detector space).** A classifier is trained to separate the measured data from the reconstructed simulation (the latter weighted by the current push-weights). Its calibrated

output p yields the likelihood-ratio reweight $w = p/(1-p)$ on every reconstructed simulated event. The weight is *pulled back* to truth level through the event pairing: each simulated event is a (truth, reco) pair, so the reco-space weight attaches directly to the truth record. Truth events with no reconstructed partner (the "misses") receive a step-1 weight from a regressor trained on the reconstructed subsample, w(truth features) — the native OmniFold acceptance treatment, with no separate efficiency correction.

2. **Step 2 (truth space).** A second classifier is trained to separate the original truth-level simulation from the same sample carrying the pulled-back step-1 weights, yielding a smooth truth-space reweight that becomes the next iteration's push-weight.

After n iterations the push-weights define an efficiency-corrected, truth-level reweighted simulation ensemble; histogramming it (in any binning, chosen after the fact) gives the unfolded spectrum. The regularizer is the iteration count together with the model capacity of the learners.

Background subtraction (the one binned step). The measured sample entering step 1 is the full set of selected data events — none is discarded and no negative-weight events are injected. Backgrounds are instead removed by a per-reco-bin *purity* weight, applied event by event: the POT-scaled background prediction (genuine backgrounds plus out-of-phase-space signal, §2.3) is binned on the reconstructed analysis grid, and each data event in reco bin b is weighted by

$$w_{\text{pur}}(b) \;=\; \max\!\big(0,\; N_{\text{data}}(b) - N_{\text{bkg}}(b)\big)\big/N_{\text{data}}(b),$$

the predicted signal purity of that bin (floored at zero in the rare bins where the prediction exceeds the data). The step-1 classifier still sees the full unbinned event kinematics; the analysis binning enters only through this multiplicative correction, an $\sim 3\,\%$ effect overall that is largest where the background concentrates (§2.3).

This correction is, deliberately, the one place the analysis binning enters before the final histogramming. Published unbinned measurements have so far avoided the problem rather than solved it: H1 cut backgrounds to a negligible level, LHCb corrected per track upstream of the unfold, and ATLAS left its $< 0.25\,\%$ background in place and assigned it as an uncertainty, while identifying negatively-weighted simulated events as the natural in-method subtraction [2, 31]; Ref. [18] recommends that route for nontrivial backgrounds but records no analysis that has yet used it. The explicit per-reco-bin weight used here is a conservative middle ground at our sub-percent genuine-background scale. Its fully unbinned realization — injecting the background simulation onto the measured side with negative weight, which targets the identical background-subtracted density $(D-B)/S$ and reproduces the binned correction to $-0.13\,\%$ on the total 2D cross section, and its systematic and statistical covariances to within $2\,\%$ (App. B) — removes the residual binning dependence and is the natural treatment for the full-phase-space extension, where backgrounds are larger.

OmniFold has moved from method paper to production measurement over the last few years. H1 published the first real-data applications — jet substructure in high-Q^2 deep-inelastic scattering, followed by a multi-differential eight-observable jet measurement [24, 25]; LHCb unfolded identified-hadron distributions in Z-tagged jets [32]; and ATLAS performed a simultaneous unbinned measurement of twenty-four Z+jets observables [31]. Ref. [18] collects these and other applications and distills cross-experiment practice; several of the practices audited there appear in our uncertainty budget (the ensemble-mean cross-check of the central value, the bootstrap statistical block, and the unified-throw covariance check of §4.4).

Closest to this analysis is the study of Huang *et al.* [26], the first application of OmniFold to a neutrino cross section: a mock-data unfolding of muon kinematics on the public T2K near-detector simulated dataset, benchmarked against traditional binned approaches. Two of its findings shape expectations for neutrino applications. First, iteration count: Huang *et al.* select 45 iterations for the OmniFold-family methods (20 for their binned benchmark) by a binned χ^2 against the known truth of their mock data — a criterion they note is not feasible on real data. We do not adopt that recommendation, for two reasons. Their mock data injects a large prior–data *shape* discrepancy (an RPA-modified quasielastic model) that many iterations must overcome, whereas our prior already describes the data shapes well (§2.5) and the data pull is dominantly a normalization; and cross-experiment production practice is few iterations (typically $\sim$5 [18]). Our $n = 5$ is instead justified empirically: doubling to $n = 10$ moves the total cross section by 0.026 % (§3.2), and the unfold reproduces an independently published binned measurement of the same data (§5). Second, an ensemble of reweighting trials with different train/test splits controls the ML stochastic component — which we budget explicitly via the seedscan band (§4). The present work differs in unfolding *real* MINERvA data to an absolutely normalized cross section with a complete uncertainty budget, in using GBDT learners suited to a low-dimensional tabular feature set, and in validating the method directly against a published binned measurement of the same data (§5).

3.2 Estimators and configuration

We use gradient-boosted decision trees (GBDTs) rather than the neural networks of the original OmniFold papers. The choice follows the character of the inputs: the scalar unfolds of this note use at most five high-level features per event, a tabular regime in which tree ensembles typically match or outperform neural networks [33] while training deterministically and far faster on CPU. Ref. [34] demonstrates this regime specifically for unbinned unfolding — boosted decision trees are effective with a small number of high-level features, complementing deep models that ingest the full event record — and our own measurements follow the same division of labour: GBDTs for the scalar unfolds, a point-cloud transformer when the input is the full cluster record (§9). §6 validates the choice directly with calibration tests and MLP/transformer cross-checks. The engine (`unbinned_-unfolding/python/omnifold.py`; component terminology in App. D) supports interchangeable backends; the production configuration is **LightGBM** with 100 trees, 8 leaves, learning rate 0.1 (identical settings for the step-1 classifier, step-2 classifier, and miss regressor). Two alternative backends — `exact` (sklearn `GradientBoosting`) and `hist` (sklearn `HistGradientBoosting`, matched 8-leaf configuration) — are retained for method-dependence studies (§5.3). The production iteration count is $n = 5$; doubling to 10 moves the total cross section by 0.026 % (§6). The central value is a single-run unfold with pinned seeds; an ensemble-mean central value (averaging independent reweighting trials) agrees with it to 0.28 %, and ML stochasticity is budgeted explicitly via the seedscan band (§4).

Hyperparameter choice. The learner settings were fixed *a priori* rather than tuned on data: 100 trees, learning rate 0.1, and depth-3 capacity are the sklearn `GradientBoosting` defaults, and the production LightGBM backend is configured to the matched capacity ($\texttt{num\_leaves} = 8 = 2^3$) so that every backend shares the same, deliberately modest, model capacity — capacity being one of OmniFold's regularisers (§3.1). No hyperparameter search was performed on data, avoiding any tuning-on-result loop. The configuration is instead validated after the fact: closure and classifier-calibration tests (§6), insensitivity to doubling the iteration count (quoted above), backend agreement in the method-dependence studies (§5.3), and an explicit seedscan band for residual ML stochasticity (§4).

3.3 MINERvA driver and pipeline conventions

The MINERvA driver is `2d-unfolding/unfold_2d_omnifold_unbinned.py`; the higher-dimensional driver (`nd-unfolding/unfold_nd_omnifold_unbinned.py`) generalizes it to an axis list whose first two entries are always $(p_T, p_{\|})$ (carrying the fiducial gate and the per-p_T flux) and `imports` the same 2D helpers. The unfolded observables are the muon $(p_T, p_{\|})$, extended by E_{avail}, q_3, W in the higher-dimensional runs; the phase space is that of §2.3. The following conventions hold in every production, universe-sweep, and closure run:

- **Reco-space masking.** The measured events entering step-1 training are exactly those passing the reconstructed-level selection; nothing outside the reconstructed phase space reaches the classifier.
- **Fake-free step-1 training.** The step-1 target is the background-subtracted data of the purity-weight construction of §3.1: genuine backgrounds *and* out-of-phase-space signal fakes are removed from the data side, and the simulated reco sample contains no fakes by construction — so neither side of the step-1 training contains events whose truth lies outside the measurement phase space.
- **No double efficiency correction.** When the unfolded sample is histogrammed, each simulated truth event carries the product of its final OmniFold push-weight and its central-value simulation weight (the MnvTune reweighting rides along). No additional efficiency-histogram division is applied: the step-1 miss regression already returns an efficiency-corrected truth spectrum, and dividing again would double-correct — a failure mode that occurred once in development and was caught by the marginal self-validation gate (§6).
- **Closure mirrors production.** In closure tests the pseudo-data are simulated events passing both the reconstructed and truth selections, weighted identically to the production data path, so closure exercises the same code path as the measurement rather than an idealised shortcut.

3.4 Cross-section extraction

Unfolded truth-level counts are converted to the differential cross section via

$$\frac{d^n\sigma}{dx_1 \cdots dx_n} = \frac{U}{c\,\Phi\,N\,\mathrm{POT}\,\Delta x_1 \cdots \Delta x_n} \times 10^4,$$

where U is the unfolded yield; c is the per-bin *completeness*, the fraction of the truth-level sample visible to the unfolding input. Because the event loop appends the truth-only misses explicitly, every truth event enters the unfolding and $c \equiv 1$ bin-by-bin; the factor survives in the formula as a monitored self-check (a configuration that dropped the misses would surface as $c < 1$). Φ is the per-p_T flux integral (broadcast over the other axes), N the fiducial nucleon count (3.2353×10^{30}), and POT the data exposure (1.057×10^{21}). The factor 10^4 converts the flux tabulation per square meter to the reported cm^2 $(1\,\mathrm{m}^2 = 10^4\,\mathrm{cm}^2)$, so the result carries the conventional units of cm^2 per axis unit per nucleon. The implementation is `extract_cross_section_2d` and its N-dimensional generalisation `xsec_nd.py` (self-tested to reproduce the frozen lower-dimensional extractions to $< 10^{-12}$). The flux normalization enters as $\sigma \propto 1/\Phi$, so Φ is itself varied per flux universe when the systematic covariance is built; this variation dominates the Flux band (§4).

3.5 Binning

The reported 2D binning ($14\ p_T \times 16\ p_{\|}$ bins; 205 of 224 reported) follows `minerva_paper_-anc/bin_mapping.txt`; the 19 unreported diagonal cells have $p_T^{\text{hi}}/p_{\|}^{\text{lo}} > \tan 20°$. The extension axes use E_{avail}: $[0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 100]$ GeV; q_3: $[0, 0.2, 0.4, 0.6, 0.8, 1.2, 2.0, 100]$ GeV; W: $[0, 1.1, 1.4, 1.8, 2.2, 3.0, 100]$ GeV — each ending in a wide catch bin so the marginals capture the full CC-inclusive tails (a truncated top edge would break the marginalisation anchors of §7.3). Because the binning is applied only after unfolding, these choices can be revisited without re-running the unfold. The full edge table and reported-bin mapping are given in App. A (§A.3.1).

Neither underflow nor overflow appears in the reported product, by construction rather than by truncation. On the truth side, the muon-axis edges coincide with the measurement phase space itself (§2.3) — beyond them there is no signal definition to measure — and the hadronic axes span $[0, 100]$ GeV, the catch-bin edge sitting far beyond the kinematic reach at $\langle E_\nu \rangle \sim 6$ GeV, so the truth grids are exhaustive. On the reco side, a selected event whose reconstructed kinematics fall outside the grid is treated as unaccepted: it is excluded from the measured sample by the masking convention of §3.3, and its in-phase-space truth content is recovered by the miss treatment exactly as for any other unreconstructed event. Because the unfold is unbinned, there is no response-matrix under/overflow to track — the binning is applied once, at reporting, to events that all lie inside the grids.

Although OmniFold uses no response matrix, the detector smearing the unfold must undo is conveniently summarized by the truth→reco migration on the analysis binning (Fig. 3): the diagonal purity is ~ 0.6 (median) on every axis, i.e. bin-to-bin migrations are substantial and a bin-by-bin correction would be badly biased — the regime requiring unfolding.

The structure differs qualitatively between the muon and hadron sides. The spectrometer muon measurement makes the p_T and $p_\parallel$ maps nearly tridiagonal — migrations rarely exceed one bin, and the purity dips (to ∼0.45–0.5 for p_T and ∼0.35 for $p_\parallel$) occur where the binning is finest, recovering in the wide edge bins. The calorimetric axes are coarser-binned but relatively broader and visibly asymmetric: E_{avail} feeds preferentially *downward* (energy carried by neutrons and other unresponsive particles is not collected), with the diagonal purity falling from 0.61 in the lowest bins to 0.43 in the highest; q_3, built from E_{avail} and the muon, inherits a milder version of the same pattern; and W, which compounds the E_{avail} and muon smearing, is the broadest, with two-bin migrations common in the resonance region while the lowest-W (quasi-elastic) bin stays the purest at 0.71.

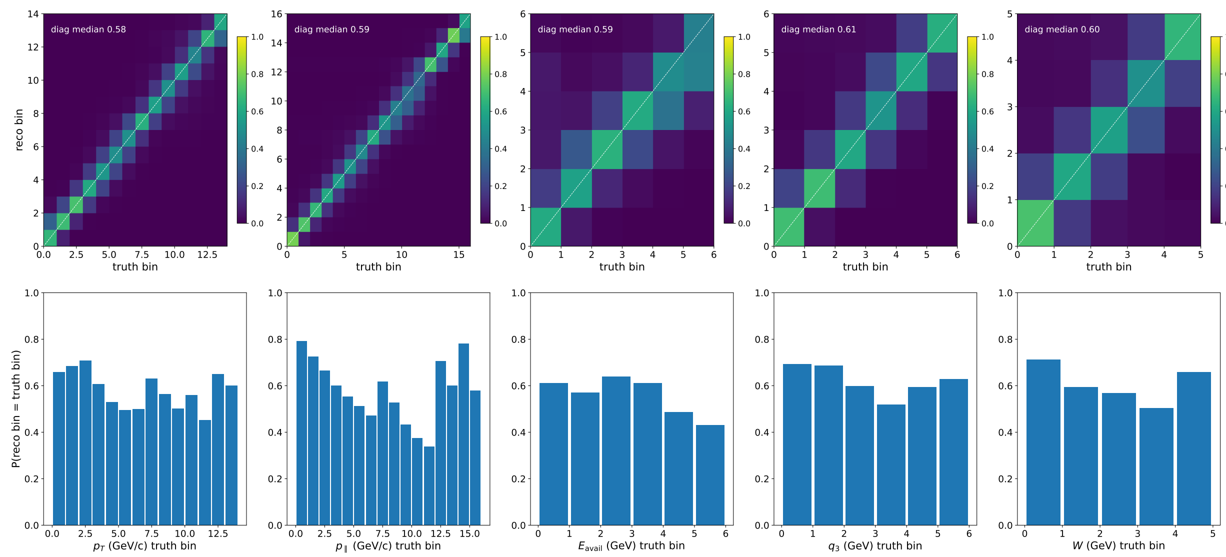

Figure 3: Truth→reco migration of the simulated signal on the analysis binning (top: column-normalized maps; bottom: diagonal purity per truth bin) for the five observables. Median diagonal purity is 0.58–0.61 per axis. Shown for illustration only — the unfold itself is unbinned.

4 Uncertainty quantification

The uncertainty budget combines three independent blocks — systematic (universe), statistical (bootstrap), and ML (seedscan) — summed assuming independence (distinct RNGs / physics sources). This section describes each block and their combination for the 2D measurement; the 3D and 4D budgets (§7.6, §7.8) repeat the identical procedure on the higher-dimensional grids. Full mathematical procedures, with worked examples on the reported numbers, are in App. A. Our *standalone* combined budget matches the published total.

Component	N	$\sqrt{\mathrm{Tr}\, C}$	per-bin median rel. σ
ML noise (lgbm seedscan)	10	5.06×10^{-41}	0.166 %
Statistical (Poisson bootstrap)	300	1.82×10^{-40}	0.549 %
Systematic (universe sweep + 1.4 % norm)	187+1	3.21×10^{-39}	6.830 %
Combined (block sum)	—	3.22×10^{-39}	6.865 %
Paper TotalCov (reference)	—	2.68×10^{-39}	6.86 %

Table 1: Stage-2 UQ envelope on the 205 reported bins. The standalone combined budget (6.865 %) matches the published total (6.86 %). **[BPN: stage 2 undefined as is C.]**

4.1 Systematic universes

The detector and model systematics use the MAT-MINERvA universe machinery: 44 error bands and 187 universes in total, identical to the published analysis's `GetStandardSystematics` set. Each universe is re-unfolded with the central estimator seed. All per-band covariances use the MAT convention: universe-mean centering and biased $1/N$. For a $\pm 1\sigma$ pair this is the outer product of the half-difference; a common displacement of both endpoints from CV is excluded from the variance.

No per-band displacement artifact is stored; only the joint-throw mean shift described below is stored separately. The 100 PPFX universes use the same mean-centered $1/N$ formula.

Dump-all shifted branches are limited to CV event-loop support. For the five genuinely kinematic bands (BeamAngleX/Y, MuonResolution, and Muon_Energy_MINERvA/MINOS), selection-complete active-universe event loops provide the production migration gate; MinosEfficiency and GEANT are weight-only. Top 2D bands: Flux 4.99 %, Muon_Energy_MINOS 2.31 %, MinosEfficiency 1.47 %, Muon_Energy_MINERvA 1.29 %. Figure 4 shows the grouped fractional-uncertainty projections in the published Fig. 6/7 style (the full 205×205 covariance projected exactly, $C_{1D} = PC_{2D}P^{\top}$, before dividing by the 1D central value).

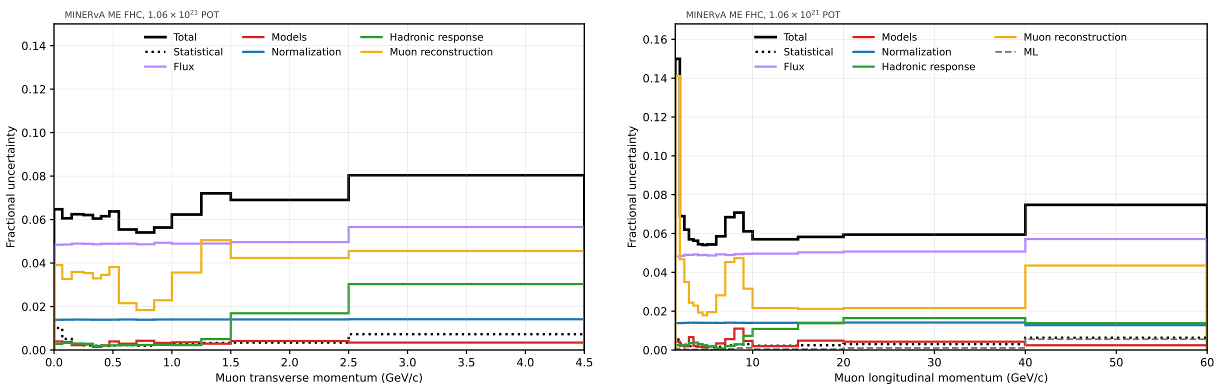

Figure 4: Fractional uncertainty on the 2D cross section projected onto p_T (left) and $p_{\|}$ (right), grouped by source in the style of Figs. 6–7 of Ref. [3]: flux, interaction models, normalization, hadronic response, muon reconstruction, and the statistical band.

Flux $1/\Phi$ normalization. The dominant flux term enters as $\sigma \propto 1/\Phi$; varying Φ per flux universe, rather than dividing every universe by the CV Φ, gives a Flux band of 4.99 % (versus ~ 1 % under the CV convention) and a standalone combined budget of 6.87 %. The 1.4 % target-nucleon normalization is a fully-correlated rank-1 band. The 100-universe flux covariance has a participation-ratio effective rank of ≈ 1 (one coherent normalization mode carries 99.6 % of the variance), and its leading mode correlates at $\rho = 0.900$ with the rank-1 muon-energy-scale band: the rederived Bashyal [35] joint block, whose cross term is rank-2, with two independent estimators giving $\rho = 0.87$–0.90. Adding the block leaves the systematic rank unchanged — its directions already lie within the span of the universe covariance (App. A.4).

4.2 Statistical uncertainty: Poisson bootstrap

The statistical block is a 300-replica per-event Poisson(1) weight bootstrap applied jointly to data and simulation, with two independent sub-RNGs (data and MC are independent variance sources), the MC reco and truth weights riding a *single* per-event draw (preserving the within-event migration correlation), and the GBDT seed pinned per replica so the bootstrap and seedscan variances are separable. Each replica is a full re-unfold; the covariance is the sample covariance over replicas (Fig. 5). Our bootstrap block is genuinely smaller than the published statistical block ($\approx 2.5\times$ by trace) — an unbinned statistical efficiency confirmed by the data-only stream of the data/MC split bootstrap ($C_{\text{data}}/\texttt{StatOnly} = 0.356$), not a missing term (App. A.4).

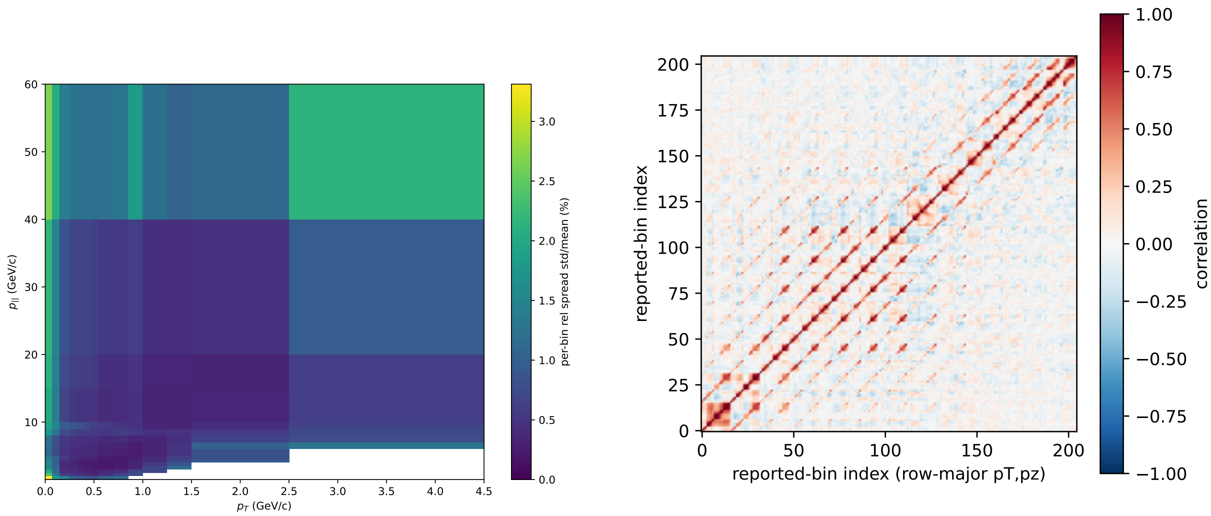

Figure 5: 300-replica Poisson bootstrap: per-bin relative spread of the unfolded 2D cross section (left) and the bootstrap correlation matrix on the 205 reported bins (right).

4.3 ML/optimization uncertainty: seedscan

ML stochasticity is budgeted by re-unfolding with 10 independent estimator seeds at fixed data and fixed universe (Fig. 6); the sample covariance over seeds is the ML block. It is the smallest block (median 0.166 % per bin). Two refinements harden it: (i) a *train/test-split* seedscan, which also re-draws the internal training split, gives a band 1.24× the pure-seed band, moving the combined 3D budget by only +0.04% in $\sqrt{\mathrm{Tr}}$ — the published budgets use the split-inclusive band; (ii) the ensemble-mean central value (averaging the seed ensemble) agrees with the production single-run CV to 0.28 %. For the corrected 4D/5D campaign now in flight, the estimator seed is fixed at 42 and only the train/test split seed varies. Thus its replacement C_{ML} is deliberately a split-response band; pure estimator-seed sensitivity is not an additional component of that higher-dimensional budget. This scope must accompany any quoted replacement result.

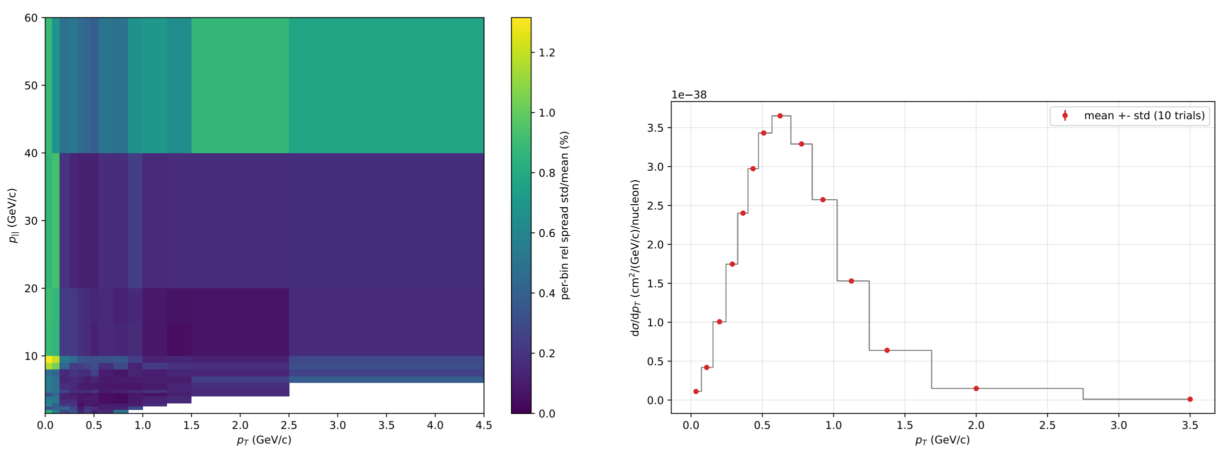

Figure 6: ML seedscan: per-bin relative spread over 10 estimator seeds (left) and the resulting fractional band projected onto p_T (right).

4.4 Combination and the unified-throw test

The nominal covariance decomposition is the block sum $C = C_{\text{syst}} + C_{\text{stat}} + C_{\text{ML}}$, which assumes the blocks are independent *and* that the unfolding responds linearly to simultaneous perturbations (so that per-band covariances add). The second assumption is tested directly by the *unified throw* (§7.8): joint re-unfolds with all reweightable knob bands and the flux multisim perturbed simultaneously per event. The corrected construction uses the actual -1σ and $+1\sigma$ event-weight endpoints, interpolates asymmetrically between them, holds the unfolding-estimator seed fixed for every throw, and forms the joint covariance about the throw mean. The joint displacement of that mean from the CV is stored separately as a per-bin vector and its norm; per-band endpoint displacements are not stored. The matched block comparator uses the MAT convention: universe-mean centering and the biased $1/N$ normalisation, with both endpoints re-unfolded.

The earlier 4D/5D/FPS unified-throw products used a one-sided interpolation, CV-centering, and an estimator-jitter prescription that is no longer the analysis contract. Their adopted per-bin inflation products and all dependent significances are therefore *unquotable pending corrected production.* No scalar jitter subtraction or automatic "adopted inflation" is applied to the corrected stored covariance.

More generally, this reframes the block sum as an approximation — valid only where the unfolding responds linearly — rather than as a default construction. We retain the per-source block decomposition for attribution, but a replacement/adoption decision is made only after the corrected joint throws and matched block endpoints pass their validation gates.

χ^2 caveat. The combined paper+ours $\chi^2/\text{ndf} = 1.481$ double-counts systematics shared with the paper (flux/GENIE/MINOS); the ours-only inverse-covariance χ^2 is evaluated only with a truncated-spectral convention on the well-constrained subspace. We therefore quote per-bin pulls, the regularization scan, and the truncated result together rather than treating an unconstrained pseudo-inverse as a standalone goodness-of-fit (App. A.4).

5 Results

Throughout this section, error bars on the unfolded points are the total uncertainty — the combined systematic (universe), statistical (bootstrap), and ML (seedscan) budget constructed in §4; the published points carry the published total covariance.

5.1 Double-differential cross section

Figure 7 shows the unfolded $d^2\sigma/(dp_T\,dp_\parallel)$ and its 1D projections. The total cross section is $3.073 \times 10^{-38}\,\text{cm}^2/\text{nucleon}$ (published: $3.039 \times 10^{-38}\,\text{cm}^2/\text{nucleon}$).

The shapes are those expected for CC-inclusive scattering at $\langle E_\nu \rangle \sim 6\,\text{GeV}$. The p_T projection vanishes at $p_T = 0$ (phase space), rises to a broad maximum at $p_T \approx 0.55$–$0.70\,\text{GeV/c}$, and falls by more than two orders of magnitude toward the $4.5\,\text{GeV/c}$ edge. The peak position is set by the momentum-transfer scale of few-GeV neutrino scattering: for the forward muons selected here $p_T^2 \simeq Q^2\,E_\mu/E_\nu$, and the cross section concentrates at $Q^2 \lesssim 1\,\text{GeV}^2$ — nucleon form-factor suppression in the quasi-elastic and resonance channels, structure-function kinematics in the deep-inelastic contribution — which with $E_\mu/E_\nu \sim 0.6$ gives the observed $p_T \sim 0.6\,\text{GeV/c}$ scale. The $p_\parallel$ projection rises from the $1.5\,\text{GeV/c}$ threshold to a broad maximum at $p_\parallel \approx 3$–$4.5\,\text{GeV/c}$, the image of the NuMI medium-energy flux peak at $E_\nu \approx 6\,\text{GeV}$, and falls steadily thereafter (two orders of magnitude by $p_\parallel \approx 20\,\text{GeV/c}$). In the 2D map the $\theta_\mu < 20^\circ$ acceptance empties the high-p_T, low-$p_\parallel$ corner (the 19 unreported diagonal cells of §3.5).

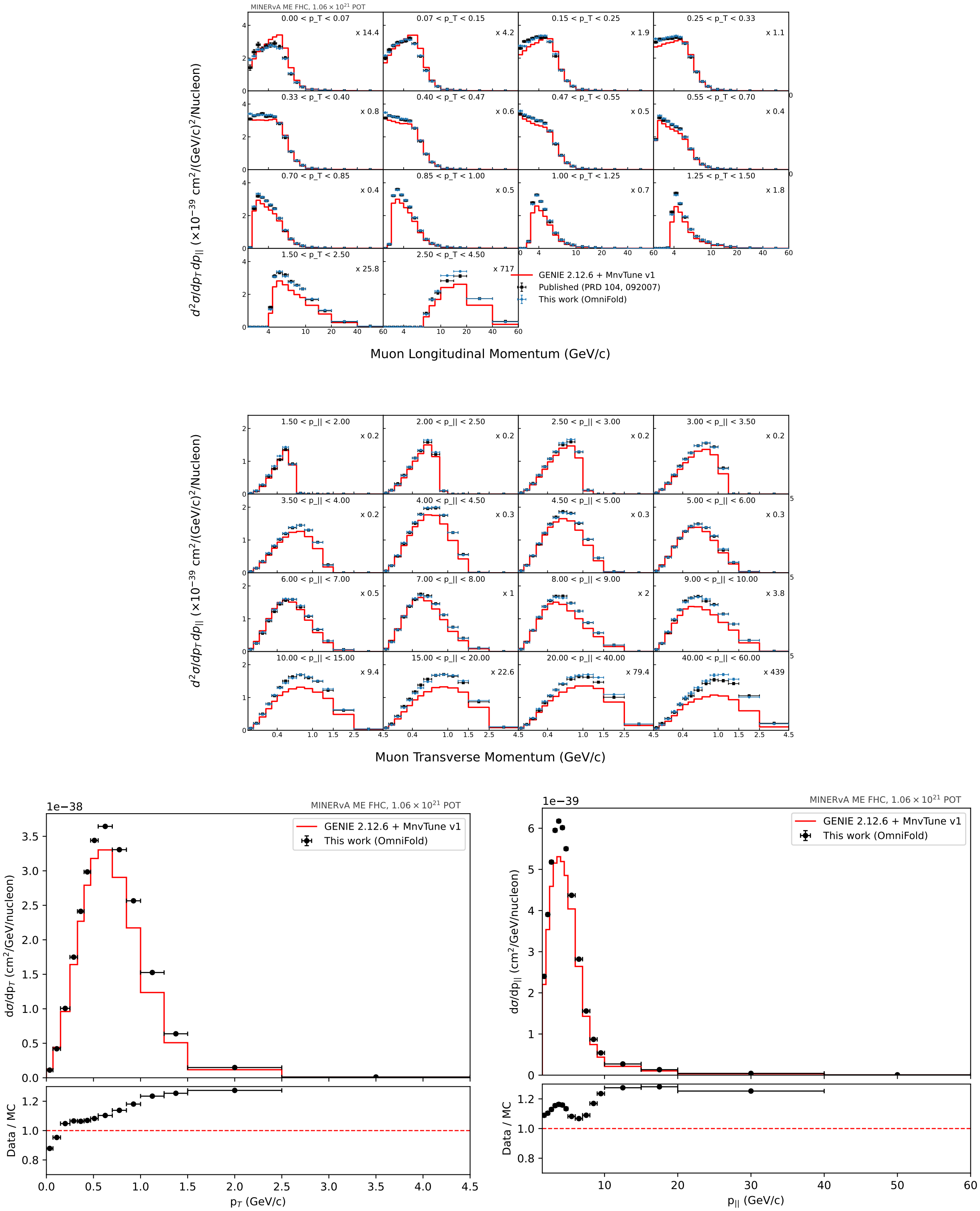

Figure 7: Unfolded 2D cross section (top, Fig. 13 style) and its $p_T/p_{\parallel}$ projections (bottom). The p_T peak at $\sim$0.6 GeV/c reflects the $Q^2 \lesssim 1\,\mathrm{GeV}^2$ momentum-transfer scale and the $p_{\parallel}$ peak the medium-energy flux maximum (see text).

5.2 Comparison to the published measurement

Table 2 quantifies the agreement with Ref. [3] over the 205 reported bins. The total cross section reproduces the published one to 1.1 % (ratio 1.011), with a median per-bin ratio of 1.006; 77.6% of bins agree within 5 % and 94.1% within 10 %. Two χ^2 figures are quoted: against the published covariance alone (3.66 per degree of freedom) and against the combined covariance that also credits this measurement's own uncertainty (1.481); the anatomy of the former is examined in §5.3. The per-bin pull distribution computed with the published covariance (Fig. 8) is centred (mean 0.089) with RMS 0.60, well inside the published uncertainties, and Figs. 9–10 show the same comparison bin-by-bin in p_T and $p_{\|}$ slices.

Quantity (205 reported bins)	Value
$\sigma_{\text{tot}}(\text{ours})/\sigma_{\text{tot}}(\text{paper})$	1.011
Median bin ratio (ours/paper)	1.006
Bins within 5 %/10 %/20 %	77.6/94.1/98.5 %
χ^2/ndf, paper covariance	3.66
χ^2/ndf, combined covariance (paper + ours)	1.481
Pull mean / RMS (paper covariance)	0.089/0.598

Table 2: Agreement with arXiv:2106.16210. Per-bin pulls (Fig. 8) are sub-1σ.

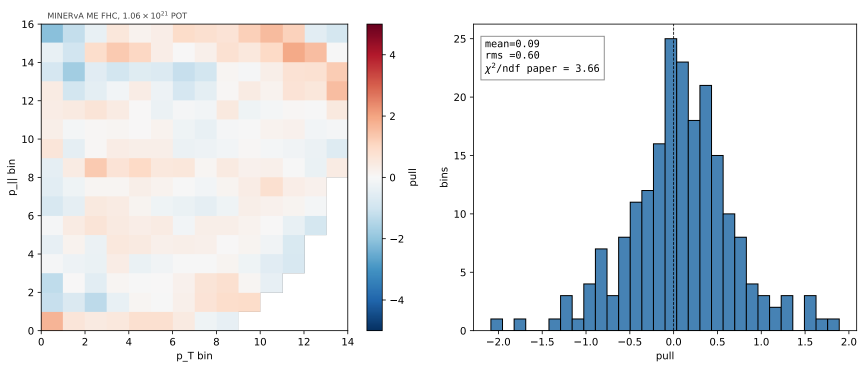

Figure 8: Per-bin pull (ours − paper)/σ on the 14×16 grid and its distribution.

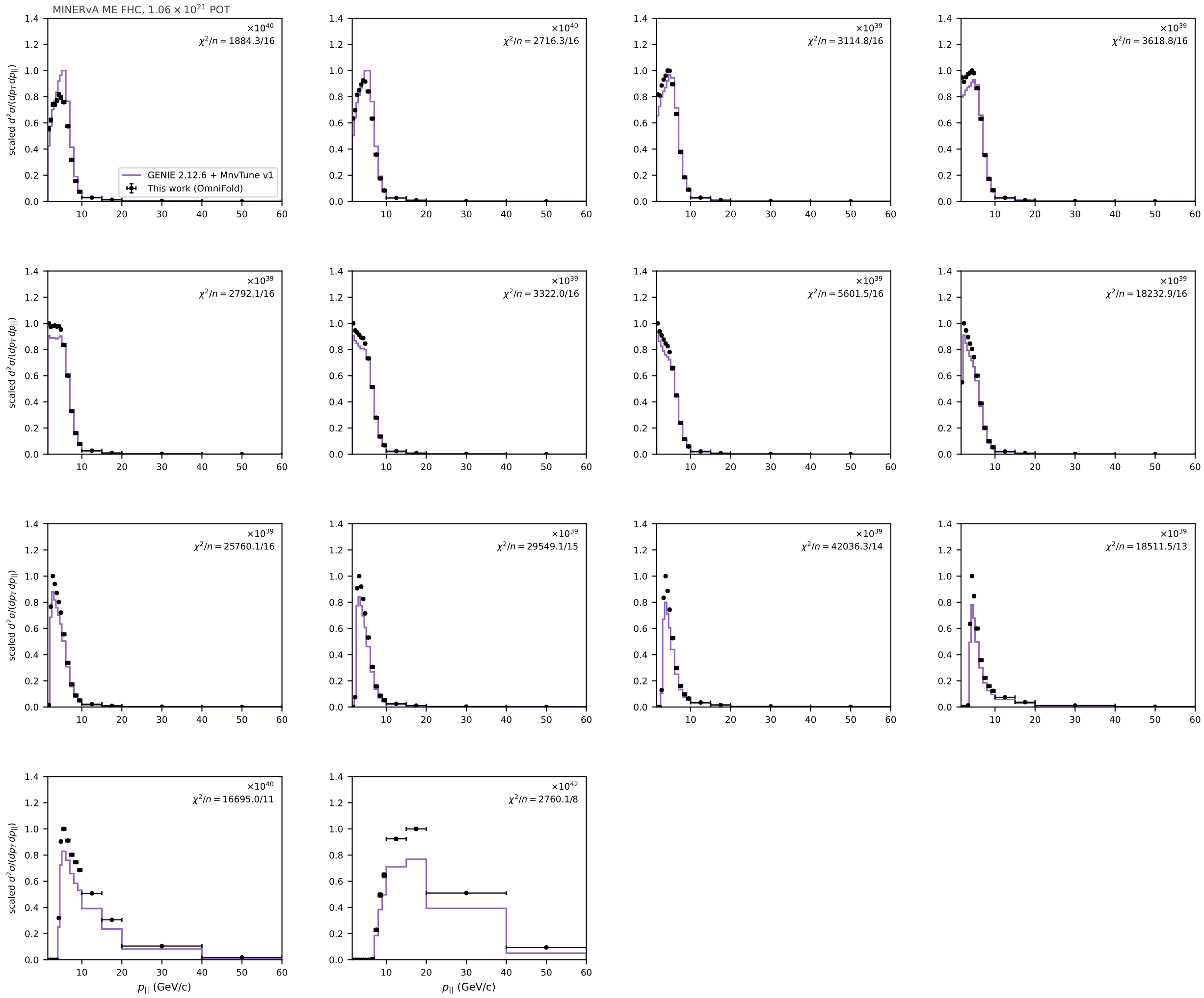

Figure 9: Bin-by-bin comparison to the published measurement: the unfolded cross section (points) overlaid on the published data (histogram) in slices of p_T versus $p_\parallel$.

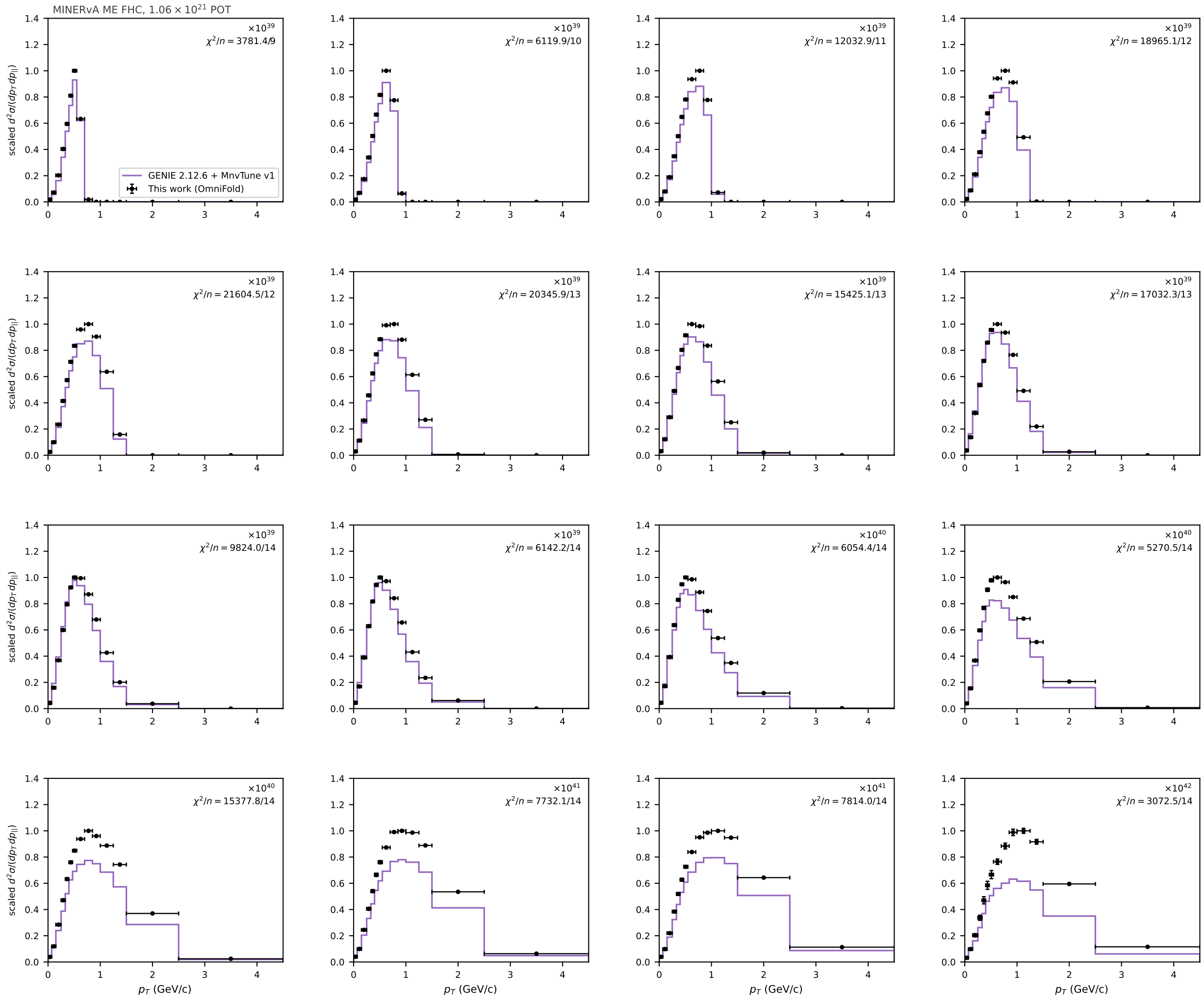

Figure 10: Same comparison as Fig. 9, sliced in $p_{\parallel}$ versus p_T.

5.3 Anatomy of the $\chi^2 = 3.66$ tension

The central values agree (pull RMS 0.598, so a diagonal-only χ^2/ndf would be 0.37), yet the full paper-covariance χ^2/ndf is 3.66: a 10× inflation living entirely in the off-diagonal structure of the published covariance. The eigenmode decomposition (Fig. 11) shows the tension is carried by moderate-λ modes with 5–6σ eigen-pulls (the 10 smallest-λ modes carry only 3 %; keeping the 73 best-measured directions still gives 1.42), so it is *not* a small-λ inversion artifact. It is *shape*, not normalization ($\chi^2_{\text{norm}} = 0.10$ vs $\chi^2_{\text{shape}} = 750$), localised to the low-$p_T$ cross-section peak ridge where the flux and muon-energy bands dominate. A $\sim$ 1-unit methodological band comes from the GBDT estimator (exact-GBT 3.66 vs HistGBT 2.70 ± 0.04 vs LightGBM 2.65; iteration count is negligible, $5 \to 10$ iters move it by +0.01). Full treatment: App. A.5. The eigenmode-localisation maps and the additional model-comparison ratio maps are shown in Figs. 11–12.

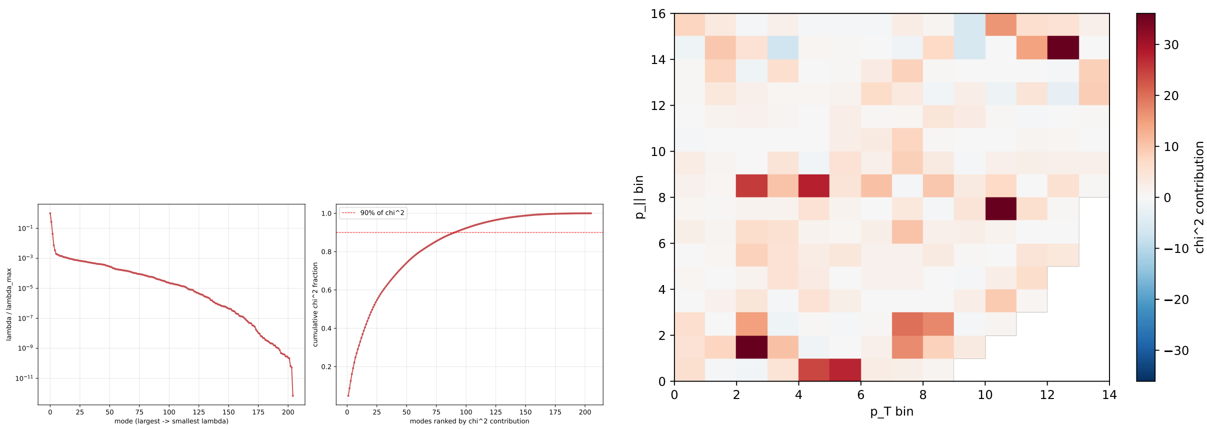

Figure 11: Tension anatomy: eigenvalue spectrum with cumulative χ^2 (left) and the per-bin χ^2-contribution map (right).

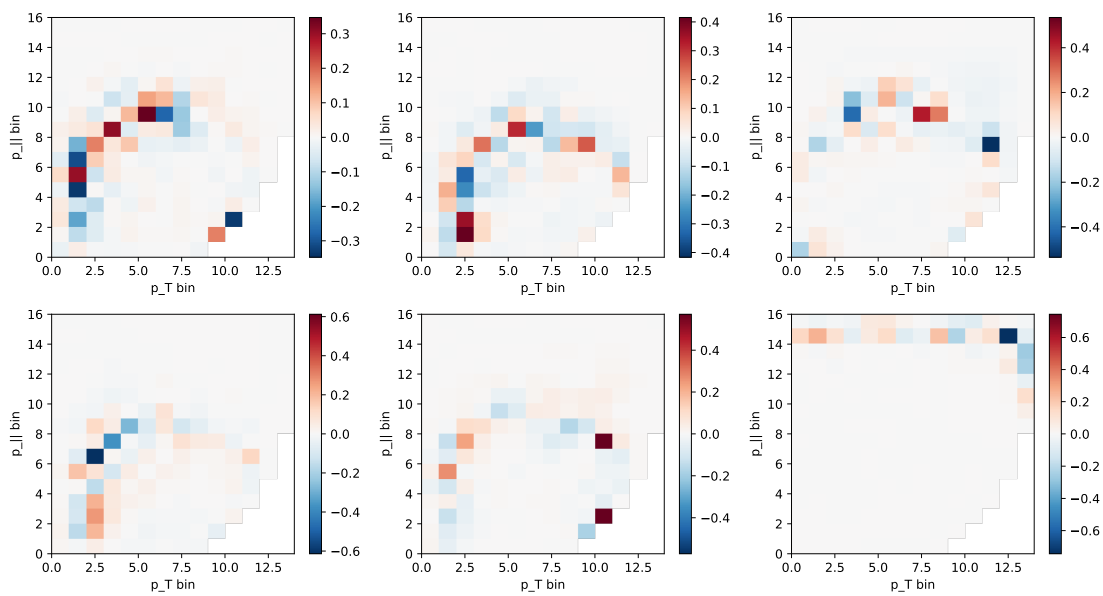

Figure 12: Bin-space maps of the leading tension-carrying eigenmodes of the published covariance: the 5–6σ eigen-pulls live on the low-p_T cross-section-peak ridge where the flux and muon-energy bands dominate.

5.4 Comparison to GENIE MINERvA Tune v1

Both the published data and our unfolded result are compared to the GENIE 2.12.6 [4] MINERvA Tune v1 prediction shipped with the ancillary (Fig. 13). The tune under-predicts the total by $\sim 9\,\%$ ($\sigma_{\text{tune}}/\sigma_{\text{data}} = 0.91$), concentrated in the low-p_T peak, and is strongly disfavoured by the data covariance (χ^2/ndf: data vs tune 33.0, ours vs tune 26.5). Our unfolded result tracks the data far more closely than the model does. The published analysis [3] additionally compares to NuWro [6], GiBUU [7] and NEUT [36]; because OmniFold delivers a per-event reweighted truth ensemble, any such generator prediction can be compared to our result in *any* projection or dimension without re-unfolding — a capability we exploit for the 3D extension (§7).

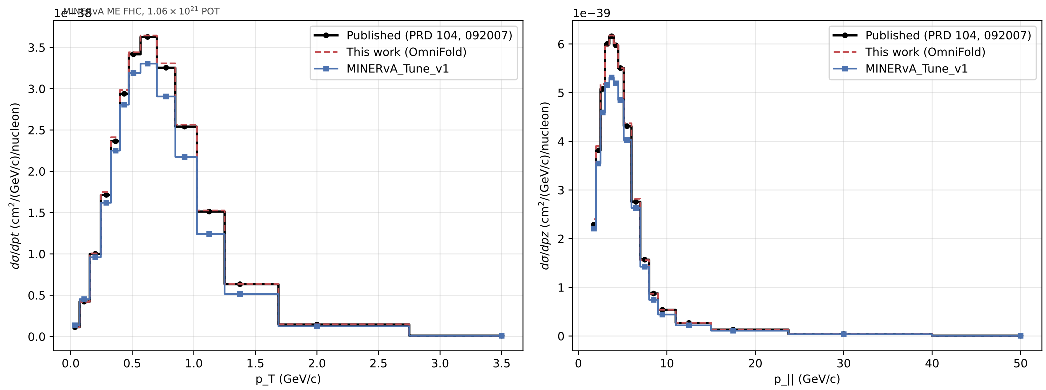

Figure 13: Our result (dashed) on top of the published data (black); MINERvA Tune v1 (orange) dips below in the peak.

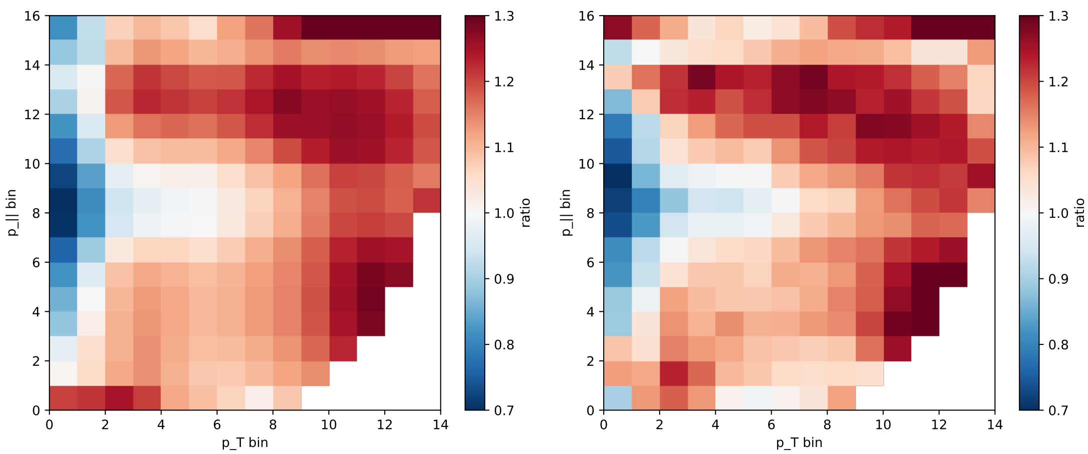

Figure 14: Bin-by-bin ratio maps on the $(p_T, p_\parallel)$ grid: ours/paper (left) and Tune v1/paper (right). The tune's deficit is concentrated in the low-p_T peak; our result is flat near unity.

6 Validation

Closure. Three closure families, all passing (1A lgbm 5-iter; thresholds in parentheses):

- **Truth-reweight** (multiply truth and reco weights by the same factor): gaussian-p_T residual 0.046 %, tilt-$p_\parallel$ 0.013 % (thr 1.5 %) — injected truth recovered.
- **Hidden-variable** (bias on $\Delta p_T = p_T^{\text{reco}} - p_T^{\text{truth}}$, a resolution variable not in the feature set): leakage linear in amplitude, $A = 0.05/0.10/0.30 \to 0.57/1.14/3.43$ % (thr 3.0 %).
- **Alt-model** (train on CV, pseudo-data from an alternative universe): MaCCQE 0.068 %, Flux 0.376 % (thr 2.0 %).

Bottom-line test. Following the unbinned-unfolding guide [18], the unfolded result must recover the data's deviation from the prior with a residual far smaller than the feature itself (the unfolding cannot manufacture discriminating power). Using the closure runs, in the bins carrying the injected feature the unfolded result tracks the injected truth to $\approx 10\times$ better than the feature size: 2D (gaussian-bump truth reweight) residual 1.69 % vs injected 17.2 % (ratio 0.098); 3D (+30 % E_{avail} bump, E_{avail} projection) 1.84 % vs 18.1 % (ratio 0.102). Both pass. (The naive data-vs-prior form—reco vs truth χ^2—instead has truth>reco, as expected for a high-resolution measurement where detector smearing lowers the reco baseline and a stat-only diagonal omits the unfolding covariance; the proper full-covariance goodness-of-fit is the χ^2-vs-paper of §5.3 and the generator comparisons.)

[BPN: Should we use this closure test to set an uncertainty?]

Unbinned goodness-of-fit. As a binning-free check that the unfold actually removes the data $\leftrightarrow$ MC mismatch, we use a classifier two-sample test (C2ST): a classifier is trained to separate the data from the (reweighted) simulation in the full unbinned feature space, and its test-set accuracy is compared to the chance value 0.5. *Before* unfolding the two are separable — accuracy 0.523, AUC 0.535, $z = 33.4$ ($p \approx 5 \times 10^{-244}$); *after* unfolding the classifier is driven to chance — accuracy 0.501, AUC 0.501, $z = 1.4$, $p = 0.17$. Data and unfolded simulation are thus statistically indistinguishable with no detectable residual structure: the unbinned analog of a binned goodness-of-fit, and it passes.

Coverage. A 200-toy closure+bootstrap study gives mean coverage 68.71 % vs Gaussian 68.27 %; mean $|\text{residual}|/\sigma$ 0.794 vs $\sqrt{2/\pi} = 0.798$; and 97.6 % of bins meeting the 65 % target.

Stability. 5-iter total σ within 0.026 % of 10-iter (per-bin shape RMS 1.54 %); HistGBT and exact-GBT agree on total σ to 0.04 %. Efficiency (Fig. 19) reproduces the published Fig. 5.

Classifier choice (GBDT vs. NN). **[BPN: Maybe move to an appendix? The NN in Fig. 15 looks quite bad! Seems a little suspicious. Keep Fig. 17+18 in the main body!]** Published OmniFold analyses use neural-network classifiers [1, 18]; we use gradient-boosted decision trees, whose output must still be a calibrated posterior for $w = p/(1-p)$ to be valid. On the step-1 reco classifier (data vs. MC reco) the reliability curve is near-diagonal (Brier ≈ 0.25 at AUC ≈ 0.537: data and reco MC are nearly indistinguishable, so the reweight is a modest correction). Across the $(p_T, p_\parallel)$ bins the true data/MC ratio still spans ≈ 100 %; the GBDT recovers it to 4.7 % median, at least as well as a dense MLP (20.9 %, untuned), and the two reweights agree with bin-wise correlation 0.92 — the learned reweight is robust to the classifier family. A full scalar keras-MLP OmniFold cross-check on the 3D inputs gives total ratio 1.0078 versus the GBDT reference and median projection differences of 0.7 % to 1.4 %, consistent with the assigned ML band. As a stronger family cross-check, a vendored PET (point-cloud transformer [37]) unfold was run on the actual per-hadron point clouds — the non-muon reconstructed clusters (`cluster_energy`, `cluster_pos`, `cluster_z`, filtered on `cluster_isMuontrack`) read by `CVUniverse::GetRecoClusters`. Area-normalized against the frozen 4D GBDT result, the PET unfold reproduces the scalar shape to a median 2.3 % to 3.9 % per bin (p_T 3.9 %, $p_\parallel$ 2.4 %, E_{avail} 2.6 %, q_3 2.3 %; Fig. 16). In its role here this is a shape/method-robustness check: it shows the learned reweight is stable across both the classifier family (GBDT vs. MLP vs. transformer) and the input representation (engineered scalars vs. raw point cloud). The point-cloud inputs themselves are shown in Fig. 17 (example reco-cluster and truth final-state-hadron clouds for individual events) and Fig. 18 (per-event cardinality of each cloud). Section 9 elevates the PET track to a full-statistics, absolutely-normalized cross section with its own closure gate and systematic budget. Figure 15 shows the underlying classifier-calibration diagnostics (reliability curve and binned ratio recovery) for the GBDT and MLP step-1 classifiers.

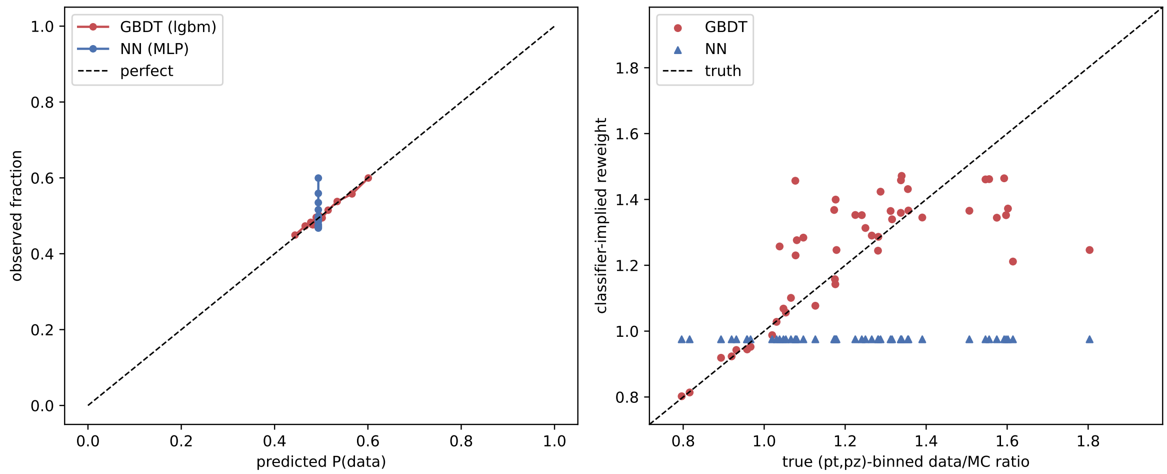

Figure 15: Classifier-calibration diagnostics on the step-1 (data vs. reconstructed MC) task: reliability curves, weight distributions, and binned data/MC ratio recovery for the production GBDT and an untuned dense MLP. The GBDT is near-diagonal in reliability and recovers the binned ratio to 4.7 % median.

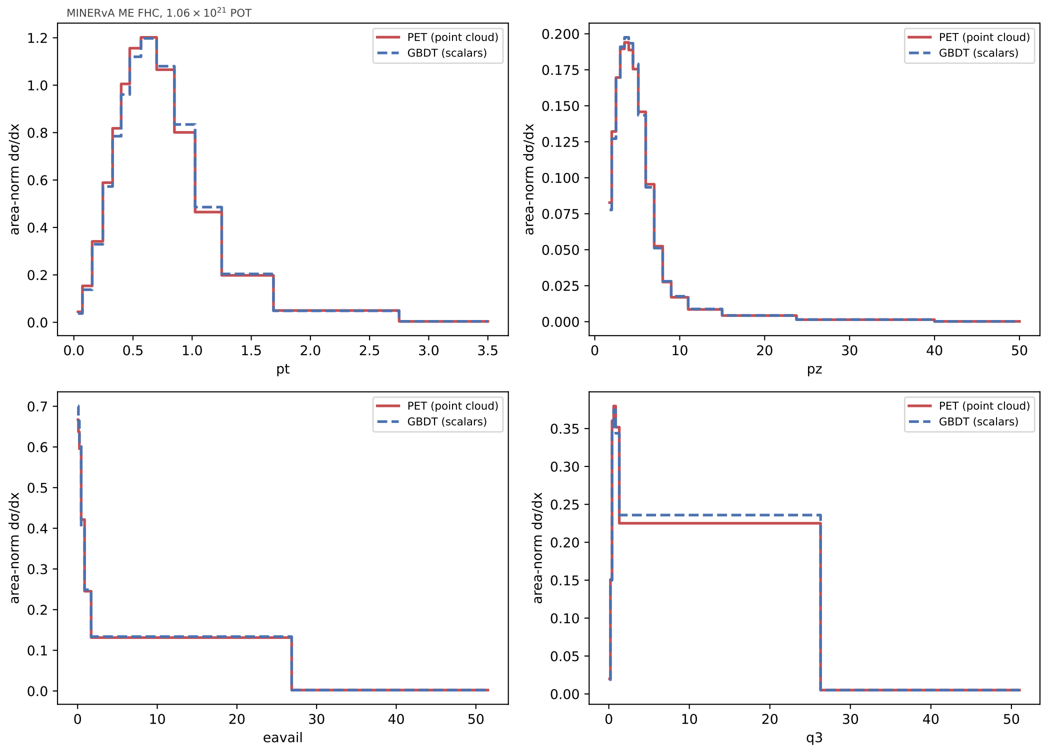

Figure 16: Method-robustness check: area-normalized 4D unfolded shapes from the PET point-cloud transformer (solid) versus the production scalar GBDT (dashed), projected onto each axis. Median per-bin shape differences are 2.3 % to 3.9 %; the only visible deviation is the wide q_3 catch bin, which collects a large, structured high-q_3 tail into one wide bin — a binning artifact of that catch bin rather than a genuine shape disagreement. Shape-only comparison (PET on a 2M-event subsample), not an independent physics result.

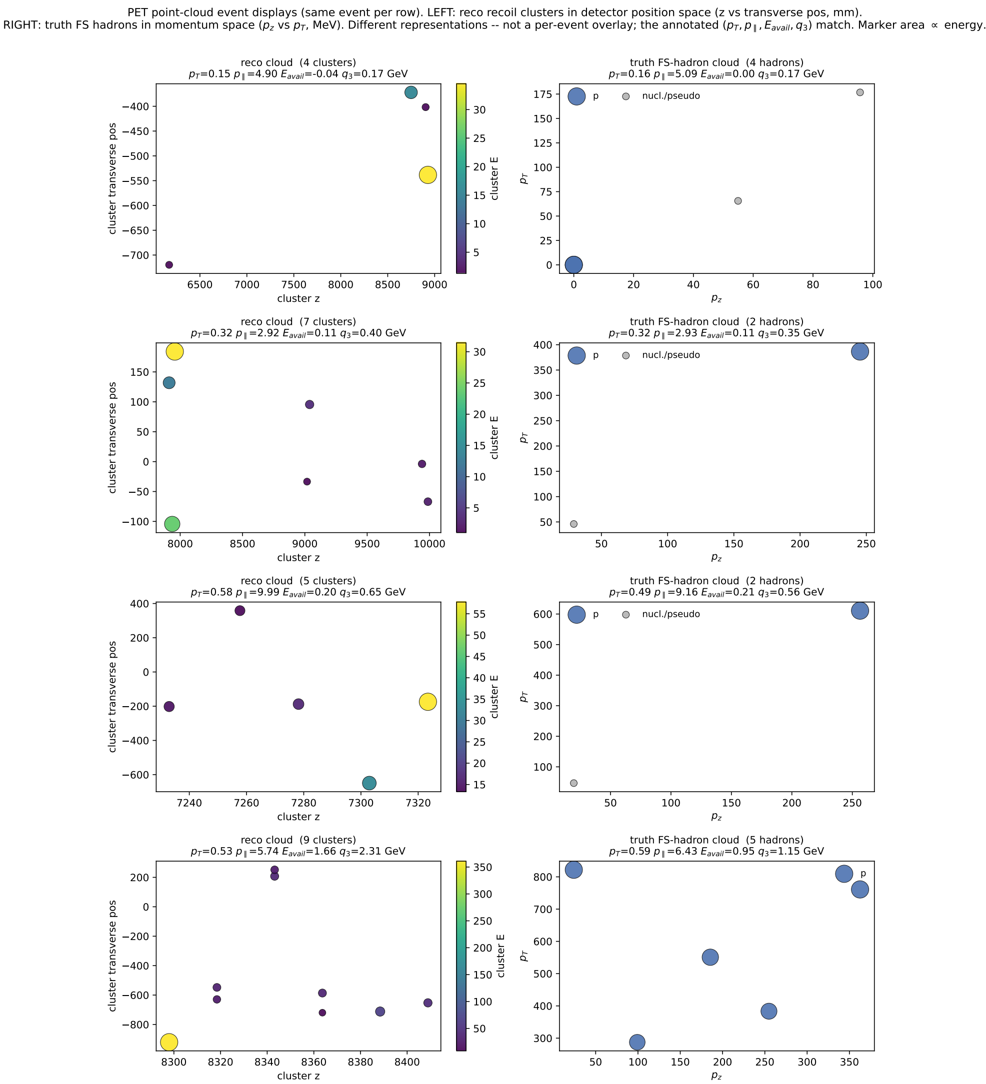

Figure 17: Example PET point-cloud inputs for four individual events (one per row). *Left*: reconstructed recoil clusters in detector position space (cluster z vs. transverse position, mm), color and marker area scaled by cluster energy. *Right*: truth final-state hadrons in momentum space (p_z vs. p_T, MeV), coloured by species. The two panels are complementary representations of the same event, not a per-event overlay; the derived $(p_T, p_{\|}, E_{\text{avail}}, q_3)$ annotated on each panel match between reco and truth. These are the raw clouds the PET unfold consumes in place of the engineered scalars. **[BPN: Great! Can we have the same coords (maybe momentum space) for both levels and overlaid? This will help visualize the distortions as well.]** **[JRB: I'm not sure how we would do this, since the inputs for each level live in different spaces — reco is $(E, \text{position})$ and truth is in momentum space. Maybe an energy vs. angle plot?]**

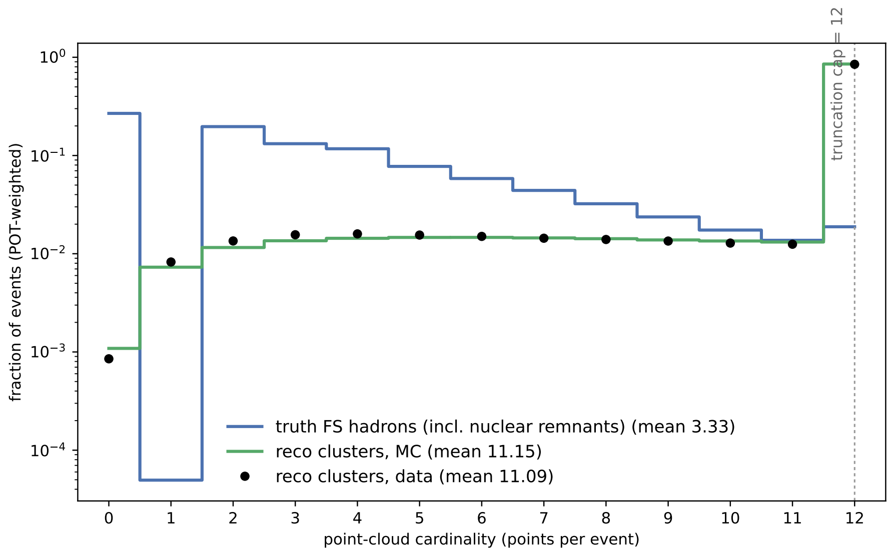

Figure 18: Per-event point-cloud cardinality (number of constituents) for the truth final-state hadrons, the reconstructed clusters in MC, and the reconstructed clusters in data, POT-weighted. The reco-cluster multiplicity greatly exceeds the truth hadron multiplicity — a single hadron deposits energy in many detector clusters — while data and MC reco agree closely (means 11.1 vs. 11.2). Clouds are truncated to the twelve highest-energy constituents per event, so the reco distributions pile up at the cap; the truth clouds are sparse (mean 3.3) and rarely reach it. **[BPN: why strange shape for truth? How did you pick 12?]** **[JRB: 1) The empty k=1 bin is a GENIE bookkeeping artifact: single-real-hadron (mostly quasi-elastic) events almost always carry a residual-nucleus pseudo-particle, which promotes them to two constituents and drains k=1. Dropping those gives the physical final-state-hadron multiplicity (mean 3.2); that version is in `figures/pet_cardinality_real`. This still has a dip at k=1 since soft single quasi-elastic protons fall below threshold into k=0 and FSI means quasi-elastic scattering often results in multiple hadrons. 2) The cap of 12 was chosen because only 2.3 % of truth events exceed it. Its also a compute/memory trade-off, since PET cost grows with constituent count.]**

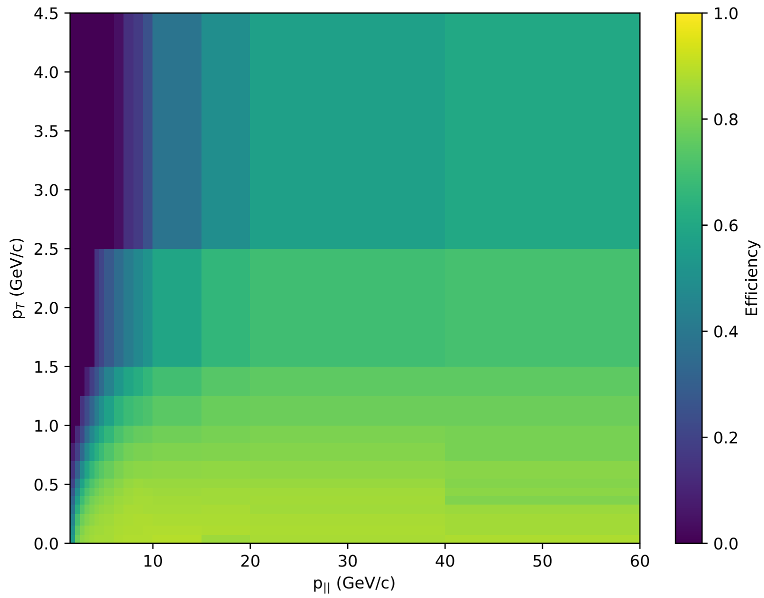

Figure 19: Selection efficiency over the $(p_T, p_\parallel)$ truth grid. Efficiency is diagnostic only — OmniFold's miss treatment performs the acceptance correction natively (§3.1).

7 Higher-dimensional extension: available energy

The enabling property of unbinned OmniFold is that adding an observable is adding an input feature: the step-1/step-2 classifiers simply gain a column, with no migration-matrix or binning penalty. There is no practical D'Agostini IBU analog, which makes this the primary physics motivation for the technique. We exercise it by extending the measurement to $d^3\sigma/(dp_T\, dp_\parallel\, dE_{\text{avail}})$ with available energy as a third axis.

The significance is clear from the measurement record: every published MINERvA differential cross section has unfolded at most two kinematic variables simultaneously (Fig. 20, Table 4), the double-differential measurements all relying on binned IBU. A simultaneous *three*-observable unfold of MINERvA data is, to our knowledge, without precedent, and is a direct consequence of the unbinned formulation.

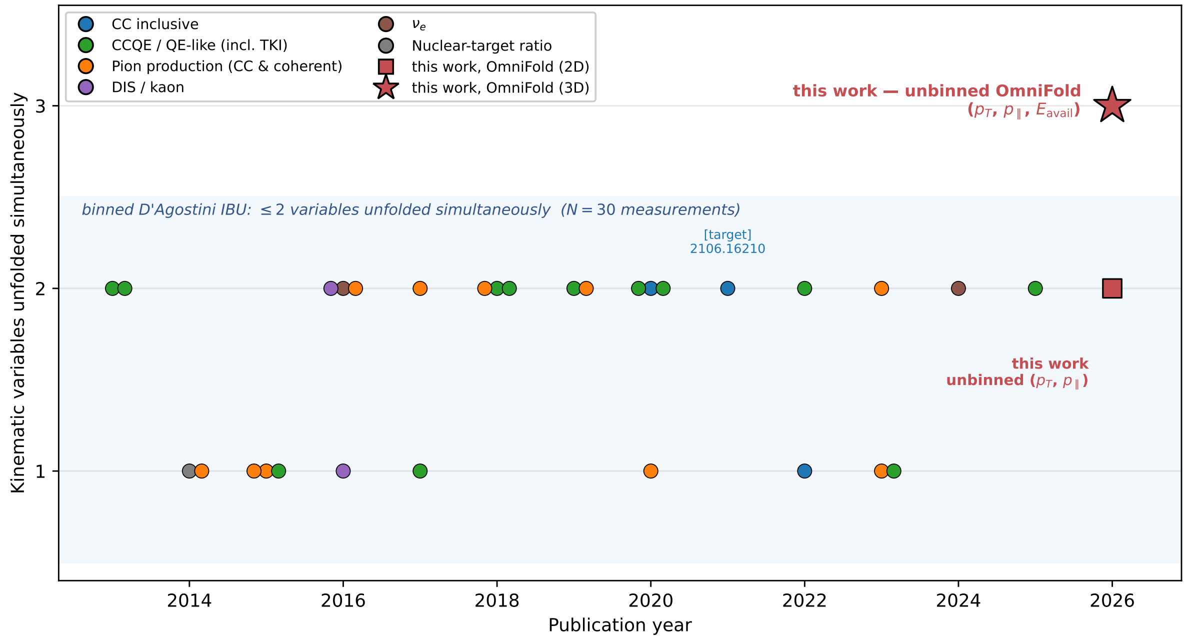

Figure 20: Dimensionality of the full set of MINERvA differential cross-section measurements over time (number of kinematic variables unfolded *simultaneously*; a double-differential $d^2\sigma/dx\,dy$ counts as two), coloured by channel. All $N = 30$ prior measurements use binned D'Agostini IBU and reach at most two variables; this work's unbinned OmniFold adds a third. Source: the MINERvA data-release page; full list in Table 4. Dimensionality is classified from each analysis's headline differential result; flux, reconstruction, generator-tuning and total-only entries are excluded.

7.1 Definitions and pipeline

Available energy follows the MINERvA low-recoil convention (Ref. [38], Eq. 4): truth $E_{\rm avail} = \sum_p \mathrm{KE} + \sum_{\pi^\pm} \mathrm{KE} + \sum_{\pi^0,\gamma} E$ (excluding neutrons and the muon), via `CVUniverse::GetEAvailableTrue()`; reco $E_{\rm avail}$ is the calorimetric tracker+ECAL estimate scaled by 1.17, `CVUniverse::NewEavail()`. The event loop evaluates both accessors over all twelve ME FHC playlists, carrying the $E_{\rm avail}$ branches alongside the $p_T/p_{||}$ branches. The unfolding driver is the 2D pipeline with a three-column feature stack $(p_T, p_{||}, E_{\rm avail})$; the cross-section extraction generalises to a 3D Jacobian $\Delta p_T\,\Delta p_{||}\,\Delta E_{\rm avail}$ with the flux Φ still applied per p_T (broadcast over $p_{||}$ and $E_{\rm avail}$). The $E_{\rm avail}$ binning $[0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 100]$ GeV ends in a wide catch bin so that the $E_{\rm avail}$-marginal captures the full CC-inclusive recoil tail; the global OmniFold completeness is $c = 1.0000$, as in 2D.

7.2 The available-energy spectrum

Figure 21 shows the unfolded $d\sigma/dE_{\rm avail}$ (the 1D projection of the 3D result). It falls monotonically from 2.19×10^{-38} cm^2/GeV/nucleon at the low-recoil peak to 6.8×10^{-41} in the catch bin — the expected CC-inclusive shape, with most of the rate at $E_{\rm avail} \lesssim 0.8$ GeV.

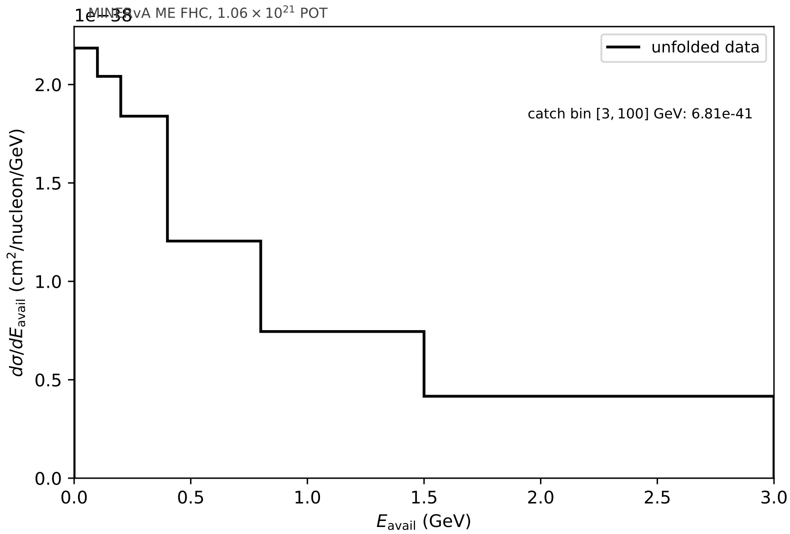

Figure 21: Unfolded available-energy spectrum $d\sigma/dE_{\mathrm{avail}}$, the 1D projection of the 3D measurement. The wide $[3, 100]$ GeV catch bin (value annotated) is excluded from the axis.

7.3 Validation against the 2D measurement

With no published 3D reference, the validation anchor is that marginalising the 3D cross section over E_{avail} must reproduce the 2D result of §5. The marginal recovers the 2D *normalization* closely: total σ agrees to $+0.95\,\%$ and the per-bin marginal/2D ratio has median 1.0016. The *shape* agreement is looser — the marginal versus the published covariance gives $\chi^2/\mathrm{ndf} = 4.98$, above the frozen 2D values (3.66 default, 2.65 LightGBM-CV), driven by a $\sim 4.4\,\%$ per-bin scatter that the third axis injects into the $(p_T, p_{\|})$ marginal (Fig. 22).

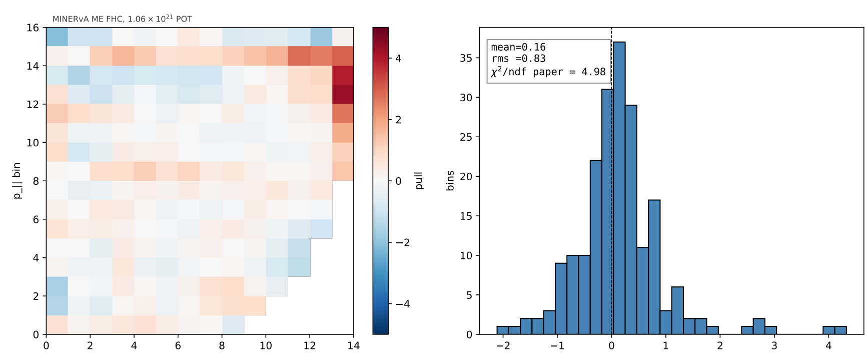

Figure 22: Per-bin pull of the E_{avail}-marginal against the published covariance, on the 14×16 grid and its distribution.

7.4 Closure: the scatter is physical, not a method bias

A reweight closure isolates whether that 4.4 % scatter is method bias or real information. We inject a known $+30\,\%$ Gaussian bump in *truth* E_{avail} (center 0.3 GeV, width 0.15 GeV) into the pseudo-data and require the 3D unfold to recover it. It does: the recovered step-2 mean weight is 1.048 against the injected 1.0485, and the residual of the unfold against the reweighted-truth reference is, on the E_{avail}-marginal, median 0.9999 with standard deviation 0.0006 (0.06 %); the per-3D-bin residual standard deviation is 3.2 % (MC statistics in sparse high-recoil corners). The closure marginal scatter is thus $\sim 70\times$ tighter than the 4.4 % data-versus-2D scatter. The method is therefore unbiased on the new axis, and we read the elevated marginal χ^2 as *genuine data $\leftrightarrow$ MC structure that the available-energy dimension exposes* — information that is invisible to the 2D measurement — rather than a pipeline artifact (the total normalization matches to < 1 %).

7.5 Generator comparison in 3D

Because the published ancillary ships only binned 2D model histograms, a quantitative comparison of the 3D result (or of arbitrary projections) requires generating truth-level events ourselves. No detector simulation is needed — the unfolded result already lives at truth level — so a generator prediction is obtained by histogramming its truth $d^3\sigma$ in the same fiducial phase space and the same truth E_{avail} definition. We compare four predictions: **[BPN: Did you run these simulators?]**

- **GENIE 2.12.10 central value** — generated with `gevgen` on a CH target with the MINERvA ME FHC flux and the DefaultPlusValenciaMEC configuration [4].
- **MINERvA Tune v1** — the analysis tune (base GENIE + RPA, low-recoil 2p2h, and non-resonant-pion reweights), extracted directly from the event-loop MC truth (the `w_truth` weights), which reproduces the shipped ancillary curve to 0.01 % in normalization.
- **NuWro 21.09** [6] — an independent generator (C target), the same flux and phase space.
- **GiBUU 2019** [7] — a transport-theory generator (C target), built and run *natively* on the compute system (no container), same flux and phase space.

NEUT [36] is not included. Each prediction is normalized by its own flux-averaged total CC cross section per nucleon ($\langle\sigma\rangle \approx 3.6$–$4.0 \times 10^{-38}\,\mathrm{cm}^2$).

Figure 23 overlays the four on the unfolded result, with the combined-covariance systematic band of §7.6 on the data. The flux-averaged totals in the measurement phase space order as GiBUU (2.22×10^{-38}) $<$ NuWro (2.34×10^{-38}) $<$ GENIE CV (2.52×10^{-38}) $<$ Tune v1 (2.71×10^{-38}) $<$ data ($3.08 \times 10^{-38}\,\mathrm{cm}^2$/nucleon): all four models under-predict the rate, by 12 % to 28 %. On $(p_T, p_{||})$ the four track the data shape closely; the discriminating axis is E_{avail}, where NuWro, Tune v1 and GiBUU *suppress* the low-recoil quasielastic peak below bare GENIE (RPA and nuclear effects) — yet the data sit *above* all four in the lowest-E_{avail} bins. This low-available-energy excess is invisible to the 2D measurement and is a direct statement of the unbinned method's high-dimensional reach.

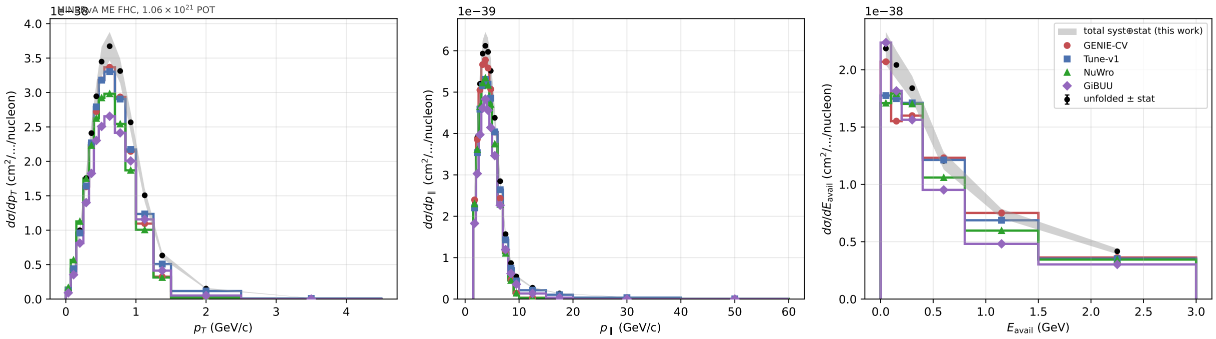

Figure 23: 1D projections of the unfolded 3D cross section (black, with the combined-covariance systematic band) versus four generator predictions produced at truth level: GENIE 2.12 CV, MINERvA Tune v1, NuWro 21.09, and GiBUU 2019. On p_T and $p_\parallel$ the models track the data shape; on E_{avail} they split, and all under-predict the data at low available energy. The wide E_{avail} catch bin is omitted from the axis.

Quantitatively, against the combined 3D covariance (§7.6) the integrated E_{avail} rate (catch bin dropped) places the models at $-7.2\,\%$ (GENIE CV, 1.3σ), $-9.5\,\%$ (Tune v1, 1.8σ), $-15.3\,\%$ (NuWro, 2.9σ) and $-21.9\,\%$ (GiBUU, 4.1σ) below the data. A full-3D goodness-of-fit over all 1431 bins, evaluated with a truncated-spectral χ^2 on the rank-247 constrained subspace of the combined covariance (a raw pseudo-inverse of the rank-deficient matrix is never taken), excludes all four at $p \approx 0$ but ranks them sharply: diagonal-only χ^2/ndf is Tune v1 **4.8**, GiBUU 32.4, GENIE CV 34, NuWro 36; the full correlated metric is Tune v1 131, NuWro 1287, GENIE CV 1512, with GiBUU worst of all because 23.5 % of its residual lies *outside* the constrained subspace (its 3D shape differs in directions the covariance barely constrains) on top of the largest normalization offset. The truncation scan and per-generator eigen-mode residuals are shown in Fig. 24.

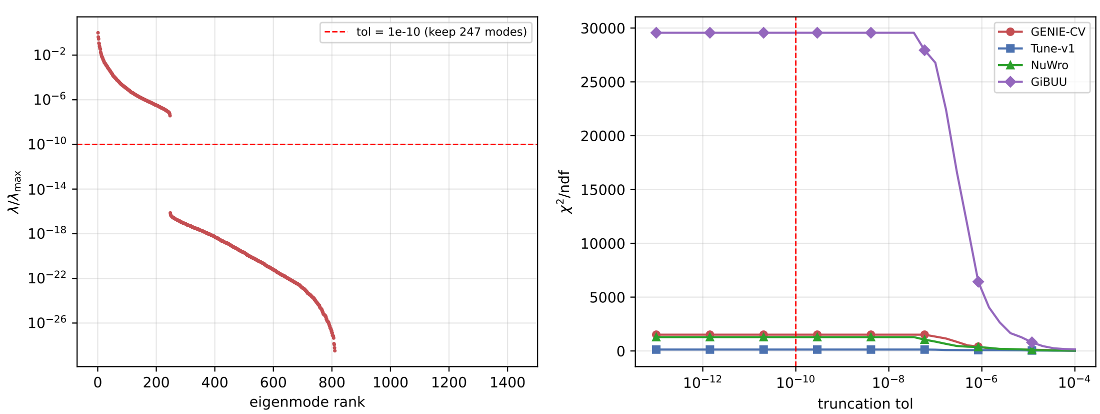

Figure 24: Full-3D truncated-spectral goodness-of-fit: per-generator χ^2/ndf versus eigenvalue-truncation tolerance, with the fraction of the residual captured by the constrained subspace. The ranking (Tune v1 best, GiBUU worst) is stable across the truncation choice.

7.6 The 3D systematic-uncertainty budget

The band on Fig. 23 and the χ^2 above rest on a full 3D analog of the 2D uncertainty budget of §4. The systematic universes are dumped on the three-axis $(p_T, p_{\|}, E_{\mathrm{avail}})$ grid (187 universes), the bootstrap statistical and ML-seedscan covariances propagated, and the three summed into a single combined covariance $C_{\mathrm{syst}} + C_{\mathrm{stat}} + C_{\mathrm{ML}}$ over all 1431 reported bins. The result has $\sqrt{\mathrm{tr}} = 5.72 \times 10^{-39}$, a median per-bin relative uncertainty of 10.4 %, and numerical rank 247 (of 1431); it is flux-dominated, with the same systematic band ordering as the 2D measurement (Fig. 25). This combined covariance is what projects onto each axis for the per-bin bands and onto the full space for the goodness-of-fit, and it is the internally-correct reference for the "ours-only" χ^2 given that our bootstrap statistical block is genuinely smaller than the paper's (an OmniFold efficiency, confirmed by the data/MC split; see App. A.4).

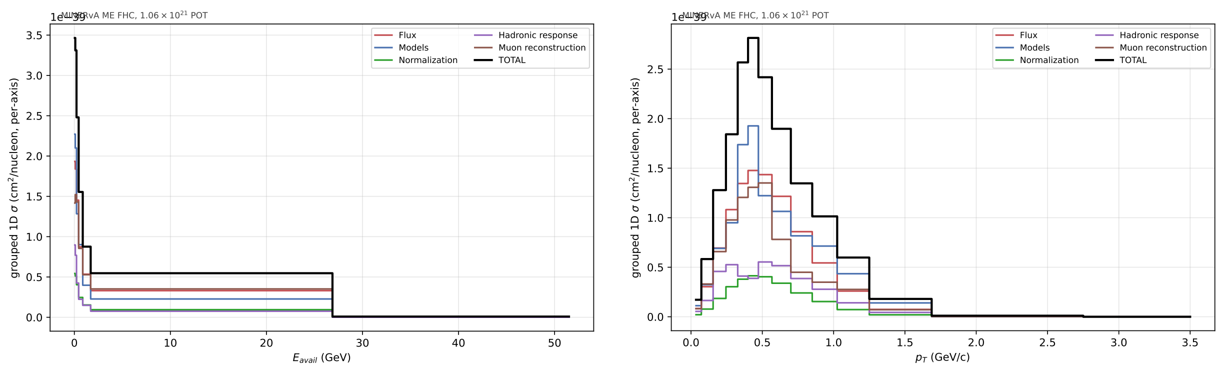

Figure 25: Grouped fractional systematic bands of the 3D combined covariance projected onto E_{avail} (left) and p_T (right): flux-dominated, with the same band ordering as the 2D budget.

7.7 The low-available-energy excess recovers the known low-recoil 2p2h deficit

The low-E_{avail} excess is the most physically interesting feature the third axis exposes, and the 3D framework lets us identify its origin directly by generating the relevant physics ourselves rather than only reweighting. It sits in a well-established MINERvA low-recoil program: the enhancement of the event rate between the quasielastic and Δ peaks, with multi-nucleon final states, was first isolated in the low-three-momentum-transfer subsample [5] and is the effect the empirical MINERvA tune was built to absorb — our measurement re-exposes it on the muon-kinematics $\times\, E_{\mathrm{avail}}$ axes.

Final-state interactions do not explain it. A `grwght1p` reweight pass on the 2M central-value GENIE events varying the pion-FSI dials shows both are sub-percent on $d\sigma/dE_{\mathrm{avail}}$: `FrInel_pi` shifts the spectrum by $\leq 0.74\,\%$ (peak in 0.10–0.20 GeV) and `FrAbs_pi` by $\leq 0.82\,\%$ (peak in the lowest bin), with the integrated rate moving only ±0.02–$0.03\,\%$ at $\pm1\sigma$. A $< 1\,\%$ knob cannot bridge a 7–15 % excess, so the excess points to the initial-state/nuclear model, not rescattering.

The shape and size match 2p2h. A subtlety drives the diagnosis: our base GENIE central value was generated with the default event-generator list, which contains *no* meson-exchange-current (2p2h/MEC) events ($\texttt{mec} = 0$ for all 1.48×10^6 CC events; the modes are QE 11 %, RES 22 %, DIS 66 %, COH 0.9 %). Any 2p2h is therefore *added*, not reweighted. Decomposing the central-value $d\sigma/dE_{\mathrm{avail}}$ by interaction mode (per-bin mode count-fractions $\times$ the bin value) and overlaying the data localises the excess to the quasielastic–Δ dip: the $[0.10, 0.20)$ bin is $+3.9\sigma$ and $[0.20, 0.40)$ is $+2.3\sigma$, while the QE-dominated $[0, 0.10)$ already matches ($+0.8\sigma$) and $[0.40, 1.50)$ match to $\pm0.4\sigma$. 57 % of the integrated deficit sits at $E_{\mathrm{avail}} \leq 0.4$ GeV, exactly where 2p2h lives, and closing it requires a 2p2h component ≈ 43 % of the QE rate (62 % locally) — the standard empirical/Valencia-MEC magnitude, in contrast to the sub-percent FSI dials (Fig. 26).

Enabling Valencia 2p2h confirms it. We regenerated the GENIE central value with the empirical 2p2h turned on (`-event-generator-list Default+CCMEC`, 2M events). Because the Nieves–Simo–Vacas MEC channel, though present in the splines, is absent from the `gspl2root` total-CC graph, the sample is normalized by the non-MEC CC count (anchoring QE+RES+DIS+COH to the known total and letting MEC add on top by $1/(1 - f_{\mathrm{MEC}})$, with $f_{\mathrm{MEC}} = 2.87\,\%$). The added 2p2h lands squarely in the dip and moves GENIE toward the data: the $[0, 0.10)$ bin goes from $+0.8\sigma$ to -0.04σ (onto the data), $[0.10, 0.20)$ from $+3.9\sigma$ to $+2.85\sigma$, and $[0.20, 0.40)$ from $+2.3\sigma$ to $+1.0\sigma$ (Fig. 27). It fills 46 % of the data−CV gap in the dip and 27 % of the integrated deficit (CV $-7.2\,\% \to -5.2\,\%$), and improves the full-3D χ^2/ndf from 1512 to 1145 ($-24\,\%$). It remains well short of Tune v1 (131), because Tune v1 layers MINERvA's empirical low-recoil 2p2h *enhancement* and RPA on top of the stock Valencia MEC — i.e. stock Valencia 2p2h is real but under-strength for MINERvA low recoil, which is precisely the known MINERvA-tune story and is consistent with NuWro and GiBUU (native, un-enhanced 2p2h) sitting low as well. The one feature 2p2h does not touch is a separate $+2.2\sigma$ excess in the high-E_{avail} $[1.50, 3.00)$ bin, a DIS-tail modeling issue resolved on the (E_{avail}, W) plane in §8.

This low-available-energy deficit is the same feature MINERvA's dedicated low-recoil inclusive measurement reports: Ascencio *et al.* [10], which measures $d^2\sigma/(dq_3\, dE_{\mathrm{avail}})$ for $q_3 < 1.2\,\mathrm{GeV}$ at the same $\langle E_\nu \rangle \sim 6\,\mathrm{GeV}$, finds GENIE and NuWro underpredicting the data in the low-E_{avail} region absent an enhanced 2p2h component. Our excess appears in the muon-kinematics extension (integrated over all q_3) rather than at fixed q_3, so the comparison is of the E_{avail} *shape* and direction rather than absolute normalization; the consistency confirms that the OmniFold 3D extension recovers the established MINERvA low-recoil/2p2h feature as a by-product of the muon-kinematics unfolding. A quantitative shape comparison awaits the public [10] data release.

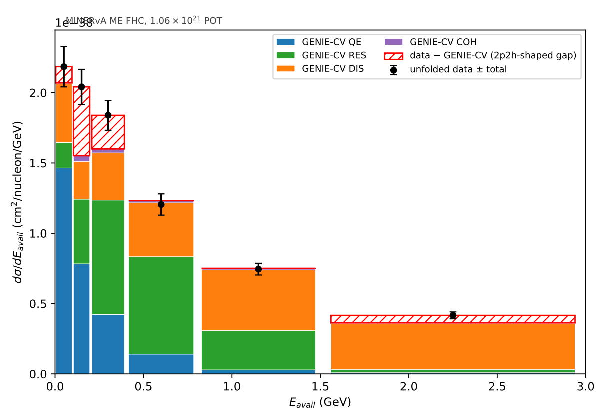

Figure 26: Central-value $d\sigma/dE_{\mathrm{avail}}$ decomposed by interaction mode and overlaid with the unfolded data. The data excess is localised to the quasielastic–Δ dip ($E_{\mathrm{avail}} \leq 0.4\,\mathrm{GeV}$), the shape and size of a missing 2p2h component.

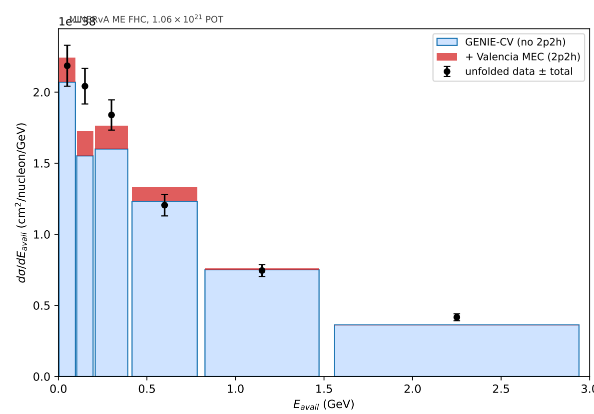

Figure 27: Re-generating GENIE with the empirical Valencia 2p2h enabled (red, stacked on the no-2p2h CV) fills part of the low-E_{avail} dip — about half the data−CV gap, the remainder being the MINERvA-tune enhancement.

7.8 Fourth axis: three-momentum transfer q_3

The same one-feature-per-observable property lets us add a fourth axis at no methodological cost. We choose the three-momentum transfer $q_3 = |\vec{q}|$ because (E_{avail}, q_3) is the plane in which the quasielastic, 2p2h and resonance processes separate, and because it makes the comparison to MINERvA's dedicated low-recoil measurement (Ascencio *et al.* [10], $d^2\sigma/dq_3\, dE_{\text{avail}}$) bin-compatible. Truth q_3 is the generator momentum transfer (`CVUniverse::Getq3True()`); reco q_3 is the calorimetric estimator built from the muon Q^2 and the recoil system (`CVUniverse::RecoQ3()`). The event loop evaluates both accessors over all twelve playlists, exactly as for E_{avail}. Unlike E_{avail}, q_3 is *not* invariant under the lateral systematic shifts, so the shifted-q_3 universes are dumped per universe rather than reused from the central value. The driver feature stack becomes $(p_T, p_\parallel, E_{\text{avail}}, q_3)$; the q_3 binning $[0, 0.2, 0.4, 0.6, 0.8, 1.2, 2.0, 100]$ GeV is deliberately coarse (seven bins) and ends in a catch bin, for $14 \times 16 \times 7 \times 7 = 10\,976$ four-dimensional bins (4830 populated).

Marginal closure. Marginalising the 4D result over q_3 reproduces the 3D E_{avail} spectrum and its low-recoil excess (Fig. 29, right): the E_{avail}-marginal data/GENIE-CV ratio is 1.21 in the lowest bin, dips to 1.01 across the quasielastic–Δ region, and rises again to 1.16–1.23 in the high-E_{avail} DIS tail — the same shape established in 3D (§7.7), now recovered as a by-product of the higher-dimensional unfold (integrated data/GENIE-CV = 1.13). This is the 4D→3D analog of the 3D→2D Jacobian anchor of §7.3.

What q_3 adds. The new content is the resolved (E_{avail}, q_3) plane (Fig. 29, left). Integrated over E_{avail} the q_3-marginal discrepancy is mild (1.03–1.19, center panel), but the joint map is structured: at fixed E_{avail} in the quasielastic–Δ dip the data/GENIE-CV ratio *rises* with q_3, the central value over-predicting the low-q_3, moderate-E_{avail} cells (ratio down to 0.66) while the data excess concentrates at low E_{avail} (all q_3) and toward high q_3. This is precisely the correlation a muon-only or fixed-q_3 measurement cannot expose, and it is the natural input for the bin-compatible Ascencio comparison. We show it here at central-value level (the base GENIE event-generator list contains no 2p2h, as in §7.7); per-cell significances use the combined 4D covariance and are therefore withheld pending its corrected production.

Bin-identical comparison to the published low-recoil measurement. The Ref. [10] supplemental data (44-cell $d^2\sigma/(dE_{\text{avail}}\, dq_3)$ with its full covariance, distributed inside the public arXiv source package) enables a direct quantitative cross-check. Merging both measurements onto their maximal common grid — edges present in *both* binnings, with our $(p_T, p_\parallel)$ marginalisation restricted to $p_\parallel < 20\,\text{GeV/c}$ to mirror the Ascencio muon gate ($\theta_\mu < 20°$, $1.5 < p_\mu < 20\,\text{GeV}$) — gives two super-cells, $E_{\text{avail}} < 0.4\,\text{GeV}$ in $q_3 \in [0.4, 0.6)$ and $[0.6, 1.2)$ GeV: the low-recoil region where the 2p2h physics lives. Our central result sits 9.2 % and 6.3 % above Ascencio. The former pulls, full-covariance χ^2, and consistency claim used the now-quarantined 4D covariance and are withheld until its corrected projection is available (Fig. 28). Caveats: the two share MINERvA flux/detector systematics, treated here as independent; the p_μ-vs-$p_\parallel$ gate differs by $\lesssim 6\,\%$ kinematic smearing at the 20 GeV boundary. A finer 44-cell comparison would require re-unfolding onto the Ascencio edges (trivial for the unbinned method) *and* a systematic sweep on that binning; given the agreement above it is left to future work.

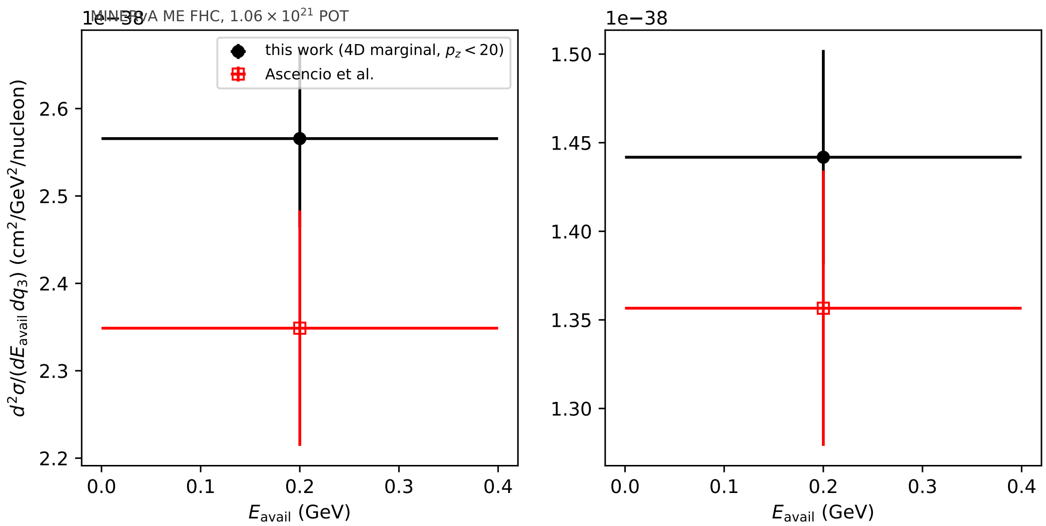

Figure 28: Bin-identical comparison to Ascencio *et al.* [10] on the maximal common (E_{avail}, q_3) grid. The central-value comparison remains current; the panel's uncertainty-derived metrics use the quarantined historical 4D covariance and must not be quoted pending the corrected marginalisation.

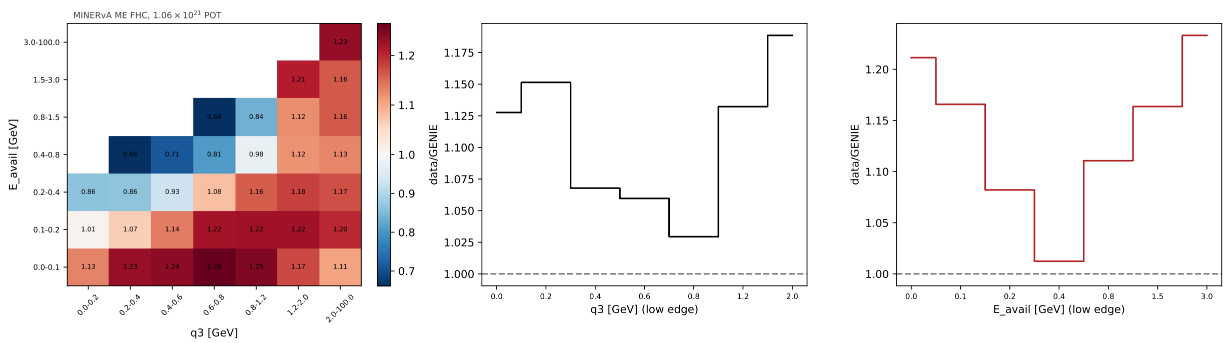

Figure 29: Data/GENIE-CV ratio of the 4D unfold projected onto the (E_{avail}, q_3) plane (left; blank cells are kinematically empty), with the q_3 (center) and E_{avail} (right) marginals. The E_{avail} marginal reproduces the 3D low-recoil excess; the joint plane shows the data $-$ MC discrepancy to carry genuine q_3 structure (the ratio rising with q_3 across the quasielastic–Δ dip). GENIE central value, no 2p2h in the base event-generator list.

Systematic covariance under remediation. The 4D central result and its lower-dimensional anchors remain valid, but the previous 187-universe budget and unified-throw adoption are withdrawn. The replacement uses actual minus/plus endpoints, MAT mean-centered $1/N$ blocks, one fixed estimator seed, mean-centered joint throws with a separately reported mean shift, and no scalar jitter subtraction. Statistical and ML replicas are also being regenerated. Because dump-all lateral branches are CV-support-limited, a targeted selection-complete active-universe campaign is bounding migration effects; full five-band coverage remains the publication gate. No current 4D uncertainty magnitude or unified/block ratio is asserted until the replacement artifact is committed.

8 Fifth axis: the (E_{avail}, W) deep-inelastic-tail study

Enabling Valencia 2p2h resolves the low-recoil excess (§7.7) but leaves a separate high-E_{avail} data excess where 2p2h does not contribute and the sub-percent pion-FSI dials cannot reach. Exact covariance-based significances from the previous construction are withdrawn while corrected 4D/5D uncertainties are regenerated. At the central-value level, GiBUU spreads its deficit across the entire tail (data/generator 1.59 at $E_{\text{avail}} \in [0.8, 1.5)$, 1.36 at $[1.5, 3.0)$) rather than piling it in the deep catch bin like the GENIE variants — an independent, qualitatively distinct confirmation. To test the deep-inelastic hypothesis directly, we added the hadronic invariant mass W as a fifth OmniFold axis.

8.1 W as a fifth axis

Truth W is the experimenter's invariant mass $W = \sqrt{M^2 + 2M(E_\nu - E_\mu) - Q^2}$ with M the struck-nucleon mass (`CVUniverse::GetTrueExperimentersW()`); reco W uses the same calorimetric energy transfer q_0 and muon-kinematic Q^2 as the reco q_3 estimator, with the average nucleon mass (`CVUniverse::RecoW()`). Like q_3 — and unlike E_{avail} — W is not invariant under the lateral systematic shifts, so the event loop dumps shifted-W lateral universes per universe, mirroring the q_3 treatment (§7.8); the dump also carries truth hadronic diagnostics (proton and pion multiplicities, leading-hadron angle). The W binning $[0, 1.1, 1.4, 1.8, 2.2, 3.0, 100]$ GeV brackets the quasielastic peak, the Δ, the transition region, and the DIS tail. The five-axis unfold $(p_T, p_{\|}, E_{\text{avail}}, q_3, W)$ passes the same marginalisation anchors as its predecessors: the W-marginal recovers the frozen 4D cross section to 0.11 % in normalization (5D/4D total = 1.0011), with per-axis median differences of 0.3 % to 1.5 %, and the $d\sigma/dW$ spectrum is finite and non-negative across all six bins.

8.2 Where the excess lives in the (E_{avail}, W) plane

Projecting the unfolded five-axis result onto (E_{avail}, W) and subtracting the GENIE central value localises the excess (Fig. 30): it is *predominantly deep-inelastic*. The two highest-E_{avail} bands carry 57 % of the positive excess; of all positive excess, high-E_{avail} (≥ 0.8 GeV) holds 67 %, and 83 % of *that* sits at $W \geq 1.8$ GeV, with the single largest cell the deep-DIS corner ($E_{\text{avail}} > 3$ GeV, $W > 3$ GeV) at 22 %. A secondary low-W (< 1.1 GeV, QE-like) excess and a small Δ-region ($W \simeq 1.4$–1.8 GeV) deficit are also present.

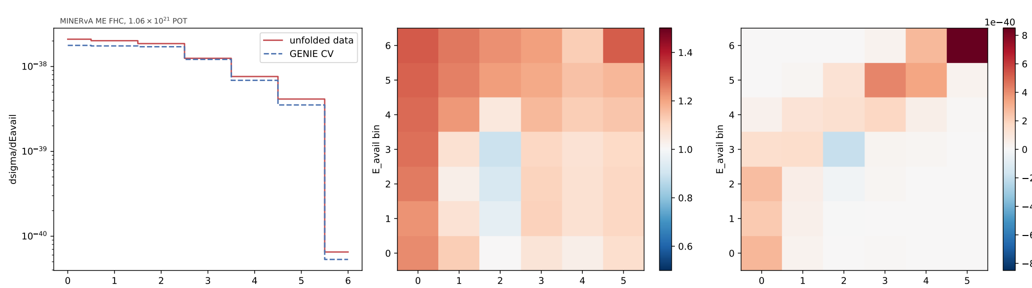

Figure 30: Unfolded-data minus GENIE-CV excess across the (E_{avail}, W) plane. The positive excess concentrates at high E_{avail} *and* high W — the deep-inelastic region — with a secondary low-W (QE-like) component and a small Δ-region deficit.

8.3 Generator band: the corner is generator- and tune-robust

We confirm this against independent spline-normalized (E_{avail}, W) predictions from GENIE CV, GENIE+Valencia 2p2h/MEC, NuWro, and GiBUU (Fig. 31). All four underpredict the high-E_{avail}×high-W corner by 54 % to 58 % (data/generator = 1.54, 1.58, 1.56). Crucially, enabling Valencia 2p2h does *not* close the corner — it slightly worsens it (1.54 → 1.58), since 2p2h populates low W — and the independent NuWro and GiBUU generators miss it by the same or a larger margin (GiBUU is the most deficient overall, integrated $\sigma = 2.22 \times 10^{-38}\,\text{cm}^2$/nucleon against the data's 3.07×10^{-38}); at $W \in [2.2, 3.0)$ GeV all four sit 23 % to 25 % below the data.

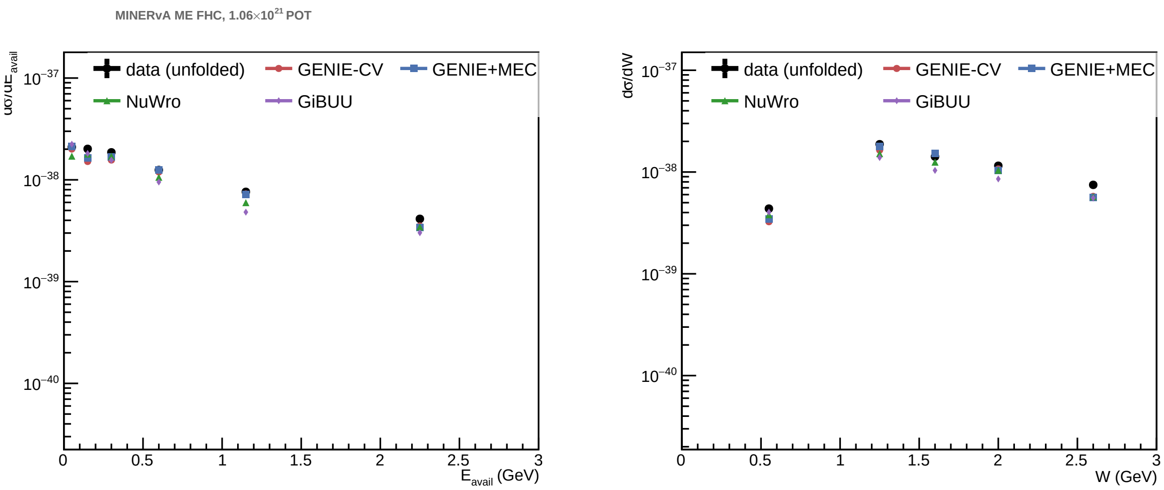

Figure 31: Unfolded (E_{avail}, W) cross section versus the four-generator band (GENIE CV, GENIE+MEC, NuWro, GiBUU). Every generator underpredicts the high-W, high-E_{avail} DIS corner; adding 2p2h does not help there because 2p2h populates low W.

8.4 The (E_{avail}, W) covariance and significances

To turn the band into a number we build a dedicated 42-bin (E_{avail}, W) covariance, $C = C_{\text{syst}} + C_{\text{stat}} + C_{\text{lateral}}$, by a *frozen-reweighter* block sum on the merged five-axis `_universes_full` sample: the central-value OmniFold push-weights are held fixed and only the per-event truth reweight ratios change per systematic universe (the same methodology as the PET budget, §9.2; no per-universe re-inference). The covariance carries 13 knob bands $\pm 1\sigma$, the 100-universe flux multisim, the projected full five-axis bootstrap covariance $C_{EW} = M C_{5D} M^{\top}$, and a W-resolved detector (lateral) block. The former construction summed per-cell standard deviations and squared the result; it is invalid and has been removed. Until the corrected five-axis statistical replicas and corrected systematic products land, the saved (E_{avail}, W) covariance and all significances derived from it are *unquotable*. The detector block uses five shifted-kinematic bands (`Muon_Energy_MINERvA/MINOS`, `MuonResolution`, and `BeamAngleX/Y`) plus weight-only `MinosEfficiency` and GEANT bands. A targeted active-universe campaign is bounding migrations omitted by the CV-support-limited dump-all sample. No replacement uncertainty or significance is asserted until these gates pass.

At the central-value level all four generators underpredict the high-W, high-E_{avail} region, and adding 2p2h does not close it. Whether and by how much each model is statistically incompatible will be reported only after the replacement covariance is committed. The complementary cross-check against MINERvA's published low-recoil measurement is presented in the (E_{avail}, q_3) plane (§7.8):

on the maximal common grid the central values agree, while the covariance-based consistency metric is withheld pending corrected production. **[BPN: left off here]**

9 Point-cloud track: PET on the raw reconstructed clusters

The full promise of OmniFold is unfolding on the event's *raw* representation rather than on engineered scalar summaries. This section documents that track: a point-cloud transformer (PET, the architecture of OmniLearn [37], vendored with the upstream OmniFold implementation) unfolds the same ME FHC data using, on the reconstructed side, the actual calorimetric recoil clusters — per-cluster energy, transverse position and z (`cluster_energy`, `cluster_pos`, `cluster_z`, filtered on `cluster_isMuontrack` via `CVUniverse::GetRecoClusters()`) alongside the muon kinematics. This is a validated *method milestone*: a real, absolutely-normalized, full-statistics cross section with its own closure gate and systematic budget. The network training in this section is subsample-limited (see below), so the production result of record remains the scalar GBDT measurement; the full-phase-space extension (§10.1) subsequently retrains PET at full statistics on the opened truth sample.

9.1 From shape check to absolute cross section

The PET network trains on a GPU on a 2M-event subsample (2 OmniFold iterations); the learned step-2 truth reweight is then applied to the *full* 32.8M-event generator sample (`MultiFold.reweight`), binned on the 4D grid, and pushed through the standard extraction (`extract_cross_section_nd` with the GBDT run's completeness — completeness is reweight-independent — and the same flux/POT/nucleon constants). Three gates validate the absolute path:

1. **Full-statistics reweight sanity**: the pushed weights over the 32.8M generator events are finite with mean 1.028;
2. **Closure (decisive)**: unfolding MC-reco-as-pseudo-data recovers MC truth to total 0.9884 with uniform per-axis median residuals of $\sim 1.1\,\%$ — the absolute machinery is unbiased;
3. **Cross-check vs GBDT**: on data, the PET absolute total is $2.796 \times 10^{-38}\,\mathrm{cm}^2$/nucleon versus the GBDT 3.066×10^{-38}, ratio 0.912, with per-axis median differences of $6.5\,\%$ to $9.9\,\%$ (Fig. 32).

Because the closure is exact to $\sim 1\,\%$, the $\sim 9\,\%$ data-side gap is a genuine *training-configuration* difference (PET trained on the 2M subsample with 2 iterations versus the full-statistics 5-iteration GBDT), not a normalization bug: an under-iterated unfold pushes the result less far from the prior. A higher-epoch retrain moved the total by -1.6% (ratio 0.897), confirming the training-config sensitivity; the residual gap reflects the reduced PET training statistics and iteration count.

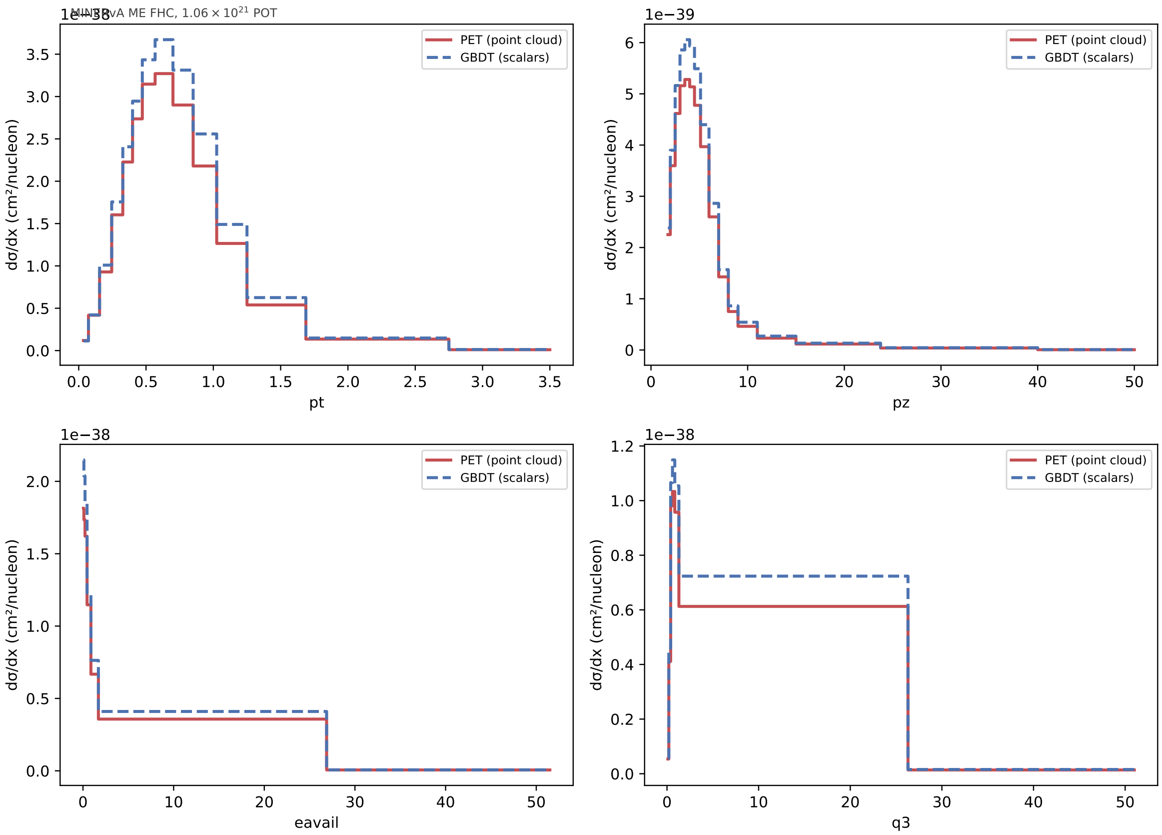

Figure 32: Absolutely-normalized 4D cross section from the PET point-cloud unfold (full-statistics reweight of the generator sample) versus the production scalar GBDT result, projected on each axis. The PET/GBDT total ratio is 0.912; the closure gate (text) shows the absolute machinery itself is unbiased to $\sim 1\,\%$.

9.2 PET systematic budget (frozen reweighter)

Remediation status. The former PET statistical block fluctuated frozen truth weights without fluctuating measured data or retraining PET. It is not a sampling covariance of the estimator. Correct replicas independently Poisson-fluctuate data and MC, retrain PET at a fixed estimator seed, and apply the same coherent MC draw in the final truth binning and completeness. Until a strict complete replica ensemble lands, the old PET C_{stat}, total budgets, and former PET–GBDT precision comparisons are historical diagnostics and *unquotable*.

The current point-cloud input assigns unit weight to every measured event and does not encode the standard reconstructed-background subtraction. Accordingly, the replacement PET bootstrap estimates the statistical covariance of that unsubtracted PET data target. This limitation must accompany any quoted PET uncertainty budget until a background-subtracted point-cloud input is produced.

The replacement PET budget uses actual minus/plus endpoints with the mean-centered MAT convention, an exact 100-universe flux inventory, and a strict ensemble of independently data- and MC-fluctuated replicas. Every replica retrains PET at a fixed estimator seed and carries the same coherent MC draw through final cross-section extraction. Missing endpoints, incomplete replica manifests, non-finite ratios, and partial event coverage are fatal.

Until that ensemble and its GPU-jitter-floor control pass, this note makes no claim that PET is more, less, or equally precise than the scalar GBDT. The old uncertainty overlays and ratio maps are retained only as superseded diagnostic artifacts and are not evidence for a precision conclusion. Detector response is evaluated with the PET-native shifted-universe construction; full per-universe PET retraining remains a distinct, computationally expensive response study.

9.3 Truth-cloud projection: one unfold, many hadronic observables

The truth side of the PET unfold is *already* a point cloud: the generator-level final-state hadron set (`part_gen`, the top-12-by-energy truth hadrons with their PDG codes, built by `CVUniverse::GetTruthFSHadrons`) is carried per event alongside the learned step-2 push weight. Any truth observable that is a function of those 4-vectors can therefore be obtained after the unfold by re-binning the push-weighted truth events — no re-training and no re-unfolding. We demonstrate this *reweight-then-project* capability on three observables computed post-hoc from the stored cloud: available energy $E_{\rm avail}$, the hadronic-system invariant mass $W_{\rm had} = \sqrt{(\sum E)^2 - |\sum \vec{p}|^2}$, and the proton/charged-pion multiplicities. The exercise also quantifies the two limitations of the stored cloud.

Truncation is essentially lossless. Because $E_{\rm avail}$ recomputed from the cloud uses the same formula and inputs as the value stored at dump time, the cloud-vs-stored residual isolates the cost of keeping only the 12 highest-energy hadrons. Over the full 32.8 million truth events (every event now carries a cloud, below) the residual has median 0 MeV and mean -6.4 MeV, with 98.78 % of events within 10 MeV (Fig. 33). The entire deficit lives in the 2.31 % of events that saturate the 12-particle cap (median deficit -35 MeV there, exactly zero below it), and because the ranking is by energy the dropped constituents are the softest mesons — a small, one-sided, bounded effect. The 12-particle cap therefore does not limit $E_{\rm avail}$-scale observables and does not need to be raised.

Coverage: closing the acceptance gap. Originally the cloud was filled only in the reco-signal event loop, so the truth-only "miss" events appended for the unfold (37.9 % of the truth sample) carried no cloud — only 72.6 % of truth events had one, and cloud projections were correspondingly acceptance-limited, landing exactly on the stored-$E_{\rm avail}$ projection restricted to the has-cloud subset but sitting 1.5–2× below the full unfold per bin. This is a truth-side instance of the same dangling-miss-row dump issue tracked for the systematic bank. We fixed it at source by carrying the truth final-state hadrons through the truth-denominator loop and filling `part_gen` for the appended miss rows (the reco cloud `part_reco` correctly stays empty for a miss), then re-dumping, rebuilding the inputs, and re-training. Coverage is now complete: 99.999 % of truth events carry a cloud (only 174 residual empties out of 32.8 million), and the cloud-recomputed $d\sigma/dE_{\rm avail}$ now reproduces the full unfold to $\lesssim 0.5$ % in every $E_{\rm avail}$ bin (Fig. 34, center, where the cloud, stored, and full-unfold curves coincide). The projection is no longer acceptance-limited.

Projection yields genuinely new observables. $W_{\rm had}$ is not a lossy copy of the leptonic "experimenter's" W stored at dump time — the latter is reconstructed from muon kinematics assuming a single struck nucleon at rest, while $W_{\rm had}$ is the true invariant mass of the hadron system. The two differ by a median 0.96 GeV, and the 2D correlation (Fig. 34, left) makes the physics explicit: a sharp locus at $W_{\rm had} \approx 0.94$ GeV (single-baryon final states, whose hadronic mass is just the nucleon rest mass irrespective of leptonic W) and diagonal bands stepped above it by ∼one nucleon mass per extra final-state baryon (multi-nucleon knockout, 2p2h), Fermi-smeared. Projecting the cloud thus delivers a new, physically meaningful hadronic observable from the existing unfold.

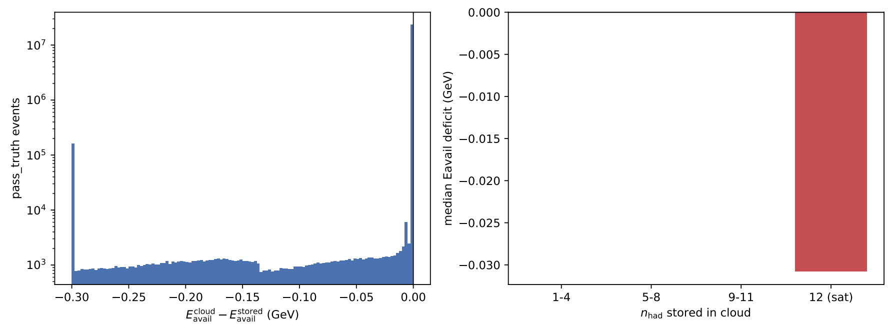

Figure 33: Truncation validation: cloud-recomputed $E_{\rm avail}$ minus the value stored at dump time, over the full 32.8 million truth events. The residual is zero to 98.78 % within 10 MeV; the small negative tail is confined to the 2.31 % of events that saturate the 12-hadron cap. Keeping only the 12 highest-energy hadrons is essentially lossless for $E_{\rm avail}$-scale observables.

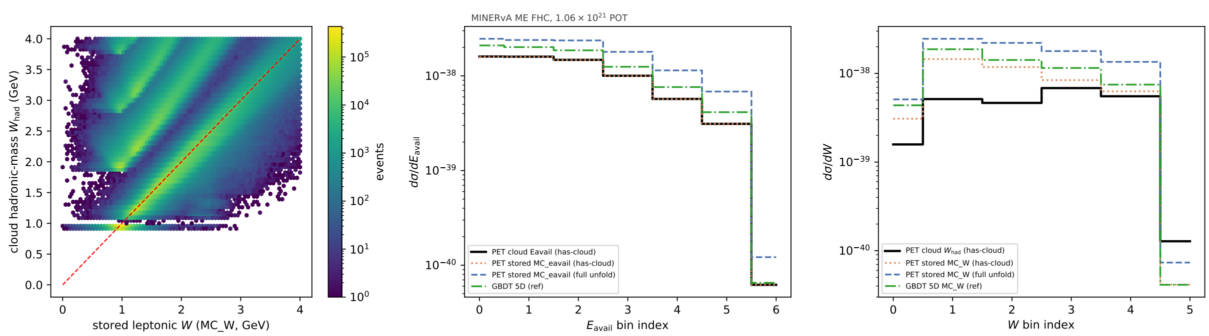

Figure 34: Reweight-then-project demonstration. *Left:* hadronic-system invariant mass $W_{\rm had}$ versus the leptonic stored W — they differ by definition (median 0.96 GeV), with the QE locus at 0.94 GeV and multi-baryon bands above $y = x$. *Center:* projected $d\sigma/dE_{\rm avail}$ from the push-weighted cloud now reproduces the full unfold to $\lesssim 0.5\,\%$ per bin after the miss-row coverage fix — the cloud, stored-$E_{\rm avail}$, and full-unfold curves coincide; the GBDT 5D reference is shown for comparison. *Right:* projected $d\sigma/dW_{\rm had}$, whose shape differs from the leptonic-W reference as expected.

The extrapolation limit. Reweighting can only redistribute weight among truth configurations the generator already produced; it cannot populate phase space the prior never sampled — the same extrapolation limit flagged for the full-phase-space measurement (§10). Within that limit, every observable above is projectable over the full truth spectrum from the current dump. Two extensions lie beyond re-projection from the stored cloud (they would require regenerating the event dump): adding the muon back as a tagged constituent for muon-dependent hadronic observables (it is deliberately excluded from the recoil cloud but retained as a stored scalar); and raising the 12-particle cap for the rare very-high-multiplicity tail (unnecessary at $E_{\rm avail}$ scale per the validation above). Full details and reproduction are in `nd-unfolding/pet/POINTCLOUD_PROJECTION.md`.

9.4 What the PET track demonstrates

The point-cloud track closes the methodological loop begun by the scalar GBDT/MLP cross-checks (§6): the unfolded physics is stable across classifier family (GBDT, MLP, transformer) *and* input representation (engineered scalars, raw reconstructed clusters), now including the absolute normalization. It also establishes the infrastructure — point-cloud DataLoader, GPU training, full-sample reweighting, frozen-reweighter budgets — for unfolding directly on low-level MINERvA data, the natural endpoint of the unbinned program. Finally, because the truth side is itself a point cloud, a single unfold yields not one cross section but a reusable truth object from which any hadronic observable can be projected after the fact (§9.3).

10 First full-phase-space results

This section reports the current central value, a three-prior dependence envelope, and an extension region validated by a hidden-variable closure and 200 coverage toys. The former full systematic covariance and unified-throw adoption are withdrawn while the corrected uncertainty production is run; uncertainty-derived FPS comparisons are not quoted in the interim.

The published phase space ($\theta_\mu < 20^\circ$, $1.5 < p_\parallel < 60\,\mathrm{GeV/c}$, $p_T < 4.5\,\mathrm{GeV/c}$) hides a third of the signal: weighting the true ν_μ CC rate in the tracker fiducial volume, 66.4 % lies inside the cuts, 22.3 % at $p_\parallel < 1.5\,\mathrm{GeV/c}$, 11.3 % at $\theta_\mu > 20^\circ$, and the high-$p_T/p_\parallel$ tails are negligible (Fig. 35). Because OmniFold's miss treatment performs acceptance correction natively, removing the truth-level kinematic cuts is a configurational change (an event-loop switch enlarging the truth denominator from 32.8M to 49M events; the reconstruction-level selection, including the MINOS match, is untouched), and the unfold extrapolates into the unmeasured regions through the step-1 miss regressor. The question is not *whether* it can extrapolate but *at what prior cost*; this section quantifies that.

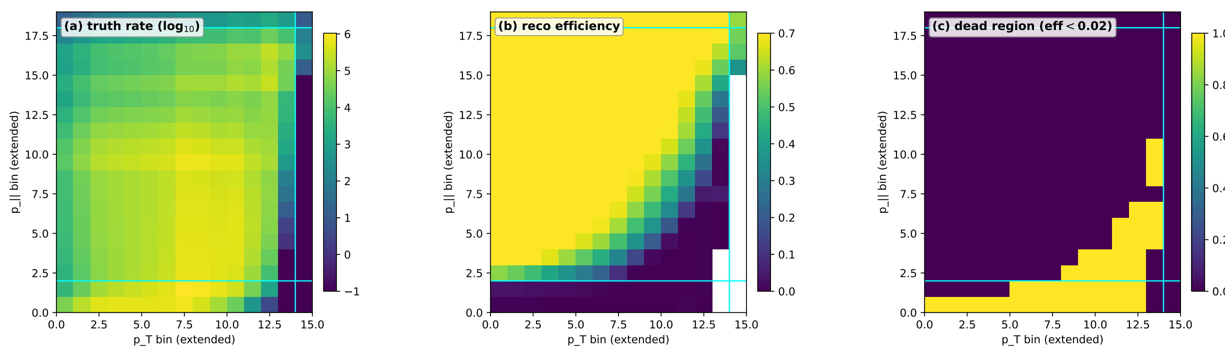

Figure 35: Full-phase-space acceptance study (all twelve playlists): (a) weighted truth rate ($\log_{10}$), (b) reconstruction efficiency, and (c) the dead region with $\epsilon < 2\,\%$, on the extended $(p_T, p_\parallel)$ grid (bin number; the catch bins are too wide for a linear axis). Cyan lines mark the published phase-space boundary. Efficiency is ~ 0.65 and flat through the published region, degrades smoothly across the $\theta > 20^\circ$ wedge, and dies in the $p_\parallel < 1.5\,\mathrm{GeV/c}$ strip and the high-p_T corner; the dead region carries 27.7 % of the total truth rate.

Validation battery. Three results establish the extension (Fig. 36):

1. **Anchor.** Restricted to the published-phase-space sub-block, the full-phase-space unfold reproduces a control unfold (standard omnifile, paper grid, identical settings) to 0.9994 in integral, with a median per-cell ratio of 1.0013 and median deviation 0.57 % over all 185 reported cells — opening the phase space does not damage the measurement where it was already made. The control itself reproduces the frozen production total ($3.073 \times 10^{-38}\,\mathrm{cm}^2/\mathrm{nucleon}$) exactly.

2. **Closure.** A self-closure on the extended grid (simulated reconstruction as pseudo-data) returns recovered/truth = 1.0000 in every cell including the extension — no bookkeeping or normalization bias in the enlarged-denominator configuration. A *hidden-variable* closure goes further: injecting a Gaussian truth bump in true E_{avail} (an axis the $(p_T, p_{\|})$ unfold is blind to) and unfolding, the recovered $(p_T, p_{\|})$ cross section reproduces the bump-reweighted truth to median 0.17 % (90th percentile 0.65 %) in the published cells and median 0.77 % (max 4.05 %) in the extension — both well inside the tier-2 prior band.

3. **Three-prior envelope.** Repeating the unfold with three priors — MnvTune v1, bare GENIE, and a NuWro-shaped prior (built by reweighting the truth $(p_T, p_{\|})$ shape to two million raw NuWro events, which carry no phase-space cut) — spans the total cross section by only ± 1.5 %. Per cell, the half-spread/mean is median 2.9 % inside the published region but median 7.9 % (90th percentile 62 %) in the extension, concentrated in the zero-acceptance cells (Fig. 37).

4. **Coverage.** 200 closure+bootstrap toys give per-bin coverage (fraction of toys with $|r| \leq 1$) of mean 68.9 % against the 68.27 % target ($\langle |r| \rangle = 0.79$, target 0.80); the published (68.5 %) and extension (70.0 %, marginally conservative) regions are both correctly calibrated — the bootstrap band means what it says in the extended grid.

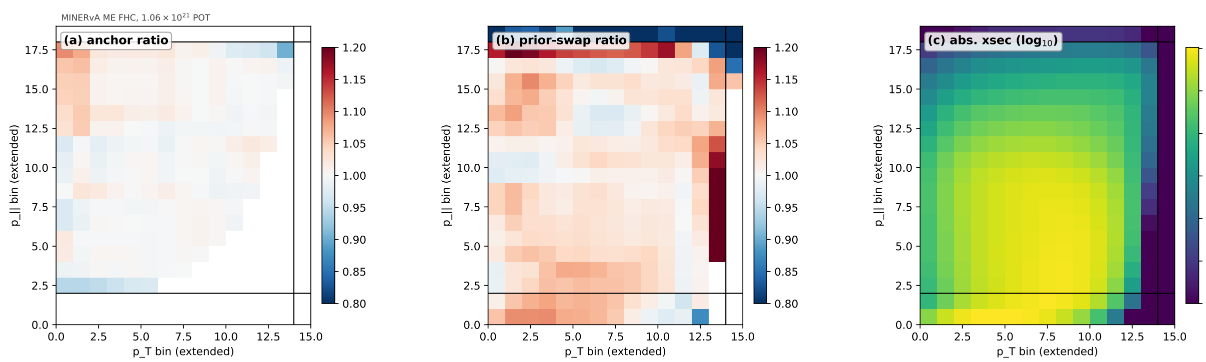

Figure 36: Full-phase-space validation battery (all playlists): (a) anchor ratio in the published sub-block (median 1.0013), (b) the MnvTune/bare-GENIE prior-swap ratio, and (c) the full-phase-space cross section ($\log_{10}$). Panels (a) and (b) are *ratios* (diverging scale centred at unity); (c) is an *absolute* magnitude (sequential scale) — hence the two color schemes. Black lines mark the published boundary; axes are bin number on the extended grid.

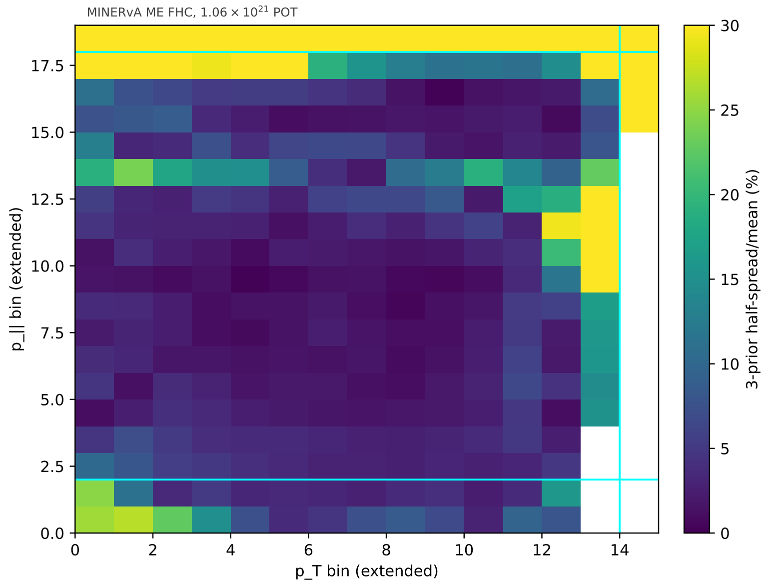

Figure 37: Three-prior dependence envelope (half-spread/mean per cell, in percent). Prior dependence is at the few-percent level wherever the detector has acceptance and reaches tens of percent only in the dead cells — the quantitative basis for the two-tier reporting.

Result and reporting convention. The full-phase-space CC-inclusive cross section on the tracker fiducial volume is

$$\sigma_{\text{FPS}} = 4.502 \times 10^{-38}\,\text{cm}^2/\text{nucleon} \quad \text{(central value)},$$

46 % above the restricted-phase-space $3.073 \times 10^{-38}\,\text{cm}^2$/nucleon. We adopt **two-tier reporting**: *tier 1* (the measurement) covers cells with $\epsilon \gtrsim 2\%$ — the published region plus the measurable extension ($\sim 6\,\%$ of the rate, chiefly the $\theta > 20^\circ$ wedge and the $p_\parallel \in [0.75, 1.5]$ GeV/c edge); *tier 2* (prior-extrapolated) covers the dead cells ($\sim 28\,\%$ of the rate), published but flagged, carrying the three-prior envelope as an explicit additional band.

The FPS central value, restricted-region anchor, and three-prior envelope remain current. Its former universe/bootstrap/ML budget and unified-throw adoption used the superseded one-sided, CV-centered, jitter-subtracted construction and are unquotable. Corrected FPS uncertainty production must use the contract in §4.4 and selection-complete lateral support appropriate to the enlarged phase space. No replacement uncertainty magnitude is asserted here.

10.1 The point-cloud realization: full statistics on the 5D grid

The result above establishes the full-phase-space extension on the two-dimensional $(p_T, p_\parallel)$ scalar grid. The point-cloud track (§9) repeats the exercise on the full 5D headline grid $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$, with two differences from the GBDT case. First, because OmniFold's step-2 network is a learned function of the truth phase space it was trained on, opening the kinematic cuts requires a genuine PET retrain on the enlarged truth sample rather than a reweight of the restricted-phase-space network. Second, because retraining separately for each prior hypothesis is prohibitively expensive for this network, the model-dependence envelope reuses the frozen-reweighter transfer that underlies the PET systematic budget (§9.2): a cheap GBDT re-unfold under three priors — MnvTune v1, bare GENIE, and the NuWro-shaped prior — supplies per-bin fractional prior-dependence ratios, applied to the frozen PET headline weights rather than driving three additional PET trainings.

Extending the grid and the truth phase space. Two lifts are applied in sequence, each validated by re-training PET on the enlarged sample: (1) the $\theta_\mu < 20°$ truth gate is lifted consistently across the point-cloud dump, the GBDT dump, and the PET retrain, enlarging the truth sample from 32.9 million to 38.6 million events; (2) the reporting grid is then extended with catch bins for the low-$p_\parallel$ strip and the high-p_T corner (p_T: 15 bins, $p_\parallel$: 19 bins, against the published 14×16), giving a final $15 \times 19 \times 7 \times 7 \times 6 = 83\,790$-bin grid on $49\,152\,885$ truth events. Restricting the extended-grid PET cross section to the 9752 bins fully interior to the published $\theta_\mu < 20°$ rectangle reproduces the restricted-phase-space PET headline to integral ratio 0.99, median per-bin deviation 1.6 % (90th percentile 4.2 %) — consistent with ordinary retrain-to-retrain noise between two independently-trained networks (quantified directly below), not a distortion from the extended grid or the lifted gate.

Model-dependence envelope and two-tier reporting. Of the 11 875 bins the PET headline populates on the extended grid, 9307 clear a reconstruction-efficiency completeness of ≥ 0.5 (tier 1, measured) and 2568 fall below it (tier 2, extrapolated), concentrated in the low-$p_\parallel$ catch rows and the high-p_T column. The three-prior model-dependence envelope (half-spread/mean per bin) has median 18.7 % (90th percentile 61.6 %) in tier 1 and median 16.3 % (90th percentile 45.4 %) in tier 2 — flat across the completeness range at large, rather than blooming in the extrapolated cells the way the coarser 2D pilot's efficiency-binned prior-swap did (that comparison used two priors and a different disagreement metric on the 14×16 grid; the finer 5D binning combined with the three-prior GBDT transfer evidently constrains the catch bins comparably well to the interior). The combined model-dependence covariance has $\sqrt{\operatorname{tr} C_{\text{modeldep}}} = 8.56 \times 10^{-37}$; because C_{modeldep} is rank 2 by construction (built from two prior-difference vectors), it is stored and combined as those two vectors rather than as a dense matrix.

Retraining-response floor. The flat 16–19% envelope raises an obvious question: how much of that is a genuine three-prior disagreement, and how much is ordinary PET training stochasticity? To isolate the latter, six additional PET replicas were trained varying only the random seed and the training-sample size (four replicas at 10M events, two at 5M), evaluated through the identical extraction chain, and their per-bin spread compared against the model-dependence envelope in each tier. At 10M events the tier-2 retraining-response floor is median 1.14 % (90th percentile 2.17 %), a factor of ~ 14 below the 16.3 % tier-2 envelope, and it shrinks further from the 5M-event value (1.37 %), the expected direction for genuine convergence rather than an already-saturated noise floor. Retraining noise is therefore a negligible contributor next to the three-prior systematic, justifying the frozen-reweighter transfer used for both the model-dependence envelope above and the PET systematic budget (§9.2): training PET once and propagating prior/systematic variation through fractional reweighting, rather than retraining per universe, does not discard resolving power that a full per-universe retrain would have recovered.

11 Summary and outlook

Summary. Unbinned OmniFold with gradient-boosted trees reproduces the published MINERvA ME FHC CC-inclusive $d^2\sigma/(dp_T\, dp_\parallel)$ (σ_{tot} ratio 1.011; 94.1 % of bins within 10 %; pull RMS 0.60) with a standalone uncertainty budget that matches the published total (6.87 % vs 6.86 % median per bin) and passes a full validation suite: three closure families, a bottom-line test, frequentist coverage, binned and unbinned (C2ST) goodness-of-fit, and classifier-family cross-checks. On that foundation we performed the first simultaneous three-, four- and five-observable unfolds of MINERvA data ($+E_{\text{avail}}$, $+q_3$, $+W$), each validated by recovering its lower-dimensional predecessor under marginalisation. The physics output:

- the low-E_{avail} excess over all four truth-level generators recovers the established MINERvA low-recoil/2p2h deficit through an independent method, with the expected FSI insensitivity and Valencia 2p2h response (§7.7);
- the (E_{avail}, q_3) plane resolves the deficit's momentum-transfer structure and shifts the systematic budget from flux- to model-dominated (§7.8);
- the unified-throw machinery now tests the block sum with asymmetric endpoints, a fixed estimator seed, and mean-centered joint throws; corrected 4D production is pending (§4.4);
- the (E_{avail}, W) study localises the remaining high-E_{avail} excess to the deep-inelastic region, where all four generators underpredict the central result; corrected significances are pending (§8);
- the PET point-cloud track yields a full-statistics, absolutely-normalized cross section from the raw reconstructed clusters and passes central-value closure; its uncertainty and PET–GBDT precision comparison are being rebuilt (§9).

Outlook. The full-phase-space measurement (§10) yields a validated central value $\sigma_{\text{FPS}} = 4.502 \times 10^{-38}\,\text{cm}^2/\text{nucleon}$ (46 % above the restricted phase space), carrying its anchor, closure and three-prior envelope. Its universe/bootstrap/ML budget and unified-throw construction are quarantined pending corrected production (§4.4); its point-cloud realization on the full 5D grid (§10.1) carries the two-tier prior envelope and a retraining-response floor an order of magnitude below it. Natural refinements are: NEUT as a fifth generator, per-lateral PET re-inference for a fully self-contained point-cloud budget, and the finer 44-cell Ascencio comparison (the maximal-common-grid comparison is presented in §7.8). Two paper-independent cross-checks would further strengthen the analysis — a flux $\leftrightarrow$ muon posterior from MAT-MINERvA for a fully self-derived ρ, and published precedent for the truncated-spectral "ours-only" goodness-of-fit convention (App. A.4).

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A Statistical methods

This appendix is the complete uncertainty-quantification methodology of the measurement — the seedscan, bootstrap and universe-covariance procedures, the MAT-conformant covariance construction, the $1/\Phi$ flux normalization, the comparison conventions (χ^2, pulls, shape-only), the ill-conditioning/rank analysis, the tension anatomy, the coverage toys, and the closure tests. Each procedure gives the recipe, the math, the implementation pointer, and a worked example using a reported number. We document only the analysis-specific numerical conventions, not a general statistics primer; the χ^2 convention referenced throughout is Eq. (10) in §A.3.2.

A.1 Numerical conventions: inverting a covariance (Cholesky, ridge jitter, pseudo-inverse)

Equation (10) requires inverting $\mathbf{C}$. Three practical issues come up in this analysis.

Positive-definiteness check (Cholesky). A matrix $\mathbf{C}$ is positive-definite (PD) iff there is a lower-triangular $\mathbf{L}$ with $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$ (the Cholesky factorisation). If Cholesky succeeds, $\mathbf{C}$ is invertible and Eq. (10) is well-posed.

Ridge jitter. Numerical noise can leave a PD matrix barely non-PD. The standard fix is to add an isotropic ridge $\varepsilon\,\mathbf{I}$. The `uq/analyze_uq.py` script does exactly this:

$$\varepsilon \;=\; 10^{-12}\,\frac{\mathrm{Tr}\mathbf{C}}{n}, \tag{1}$$

where the trace sets the natural scale of the eigenvalues. Twelve orders of magnitude below the mean diagonal is far below any real signal in $\mathbf{C}$ and safely above floating-point round-off.

Rank deficiency at $N \le n$. If you estimate $\mathbf{C}$ from N replicas in a space of n bins with $N \le n$, the resulting sample covariance has rank at most $N-1$: there are not enough independent "directions" to fill the full $n \times n$ matrix. This was the regime of the Stage-1 bootstrap ($N = 50$ replicas, $n = 205$ bins): Cholesky failed without jitter, and the required jitter, 1.6×10^{-94}, was a numerical artifact of the rank deficiency rather than a real near-singularity. The Stage-2 bootstrap ($N = 300 > 205$) is full-rank. The *systematic* covariance suffers a more stubborn version of this problem that survives at any universe count — see §A.4.

Pseudo-inverse. For the χ^2 comparison against the paper covariance, `compare_to_paper_full-cov.py` replaces $\mathbf{C}^{-1}$ with the Moore–Penrose pseudo-inverse $\mathbf{C}^{+}$, computed via SVD. Writing $\mathbf{C} = \mathbf{U}\Sigma\mathbf{V}^\top$ with $\Sigma = \mathrm{diag}(\sigma_1, \dots, \sigma_n)$,

$$\mathbf{C}^{+} \;=\; \mathbf{V}\,\Sigma^{+}\,\mathbf{U}^\top, \qquad \Sigma^{+}_{ii} \;=\; \begin{cases} 1/\sigma_i & \sigma_i > \tau, \\ 0 & \text{otherwise,} \end{cases} \tag{2}$$

with a small threshold τ (NumPy's default is $\tau = n \cdot \sigma_{\max} \cdot \varepsilon_{\text{machine}}$). This silently drops singular modes that are numerically indistinguishable from zero, replacing them with 0 in the inverse rather than ∞. The result is a stable quadratic form even when $\mathbf{C}$ has zero or near-zero eigenvalues.

A.2 Variance-source procedures

Each subsection below follows the same template: *What it estimates · Procedure · Math · Code pointer · Worked example.*

A.2.1 ML stochasticity: the seedscan

What it estimates. The variance in the unfolded cross section due to random choices inside the GBDT (gradient-boosted decision tree) optimiser. OmniFold is deterministic given the input weights, the network architecture, and the random seed; flipping only the seed produces slightly different fits to the per-event reweighting functions, and that spread is the "ML noise floor".

Procedure. Run $n = 10$ full 5-iteration MEFHC unfolds with everything fixed except the GBDT `random_state`. The driver `sbatch_unfold_2d_MEFHC_5iter_seedscan.sh` pins three seeds per trial via `--seed N`:

$$(\text{step-1 classifier}, \text{step-2 classifier}, \text{miss regressor}) \;=\; (N,\, N+1,\, N+2), \qquad N \in \{1, \ldots, 10\}. \quad (3)$$

The canonical seedscan uses the `lgbm` (LightGBM) backend, so it is backend-matched to the production unfold.

Math. For each reported bin i, build the n-vector $X_i^{(1)}, \ldots, X_i^{(n)}$ across trials and compute the unbiased sample standard deviation

$$\sigma_i^{\mathrm{ML}} \;=\; \sqrt{\frac{1}{n-1}\sum_{k=1}^{n}\left(X_i^{(k)} - \bar{X}_i\right)^2}, \qquad \bar{X}_i = \frac{1}{n}\sum_k X_i^{(k)}. \quad (4)$$

The headline diagnostic is the per-bin *relative* spread $\sigma_i^{\mathrm{ML}}/\bar{X}_i$, summarized by its median over the 205 reported bins.

Code pointer. `seedscan/analyze_seedscan.py` (rollup); `sbatch_unfold_2d_MEFHC_5iter_seedscan.sh` (driver); covariance output `uq/seedscan_lgbm_ml/uq_covariance_ml.root`.

Worked example (UQ envelope). With $n = 10$ `lgbm` trials on the 205-bin set:

- Total-σ relative spread: $\boxed{0.0055\%}$.
- Per-bin median relative spread: $\boxed{0.166\%}$.
- Component trace $\sqrt{\mathrm{Tr}\mathbf{C}^{\mathrm{ML}}} = 5.061 \times 10^{-41}$.

The paper's per-bin total-uncertainty median is 6.86%; the ML noise is therefore $\approx 2.4\%$ of the paper's quoted uncertainty and far subdominant to data-stat and systematics. Going from $n = 4$ to $n = 10$ trials moved the headline numbers within rounding, so the seedscan envelope is converged.

A.2.2 Statistical uncertainty: the Poisson bootstrap

What it estimates. The combined data-statistical and MC-statistical variance, propagated *coherently* through the full OmniFold pipeline. Coherent means: if some data event happens to be "up-weighted" in a replica, the same event flows through both the step-1 classifier training and the final reweighted-MC cross-section construction; if an MC event is up-weighted, the same up-weight is applied to its reco-tree and truth-tree copies, preserving the response matrix.

Procedure. For each replica $k = 1, \ldots, N$:

1. Draw per-event Poisson(1) factors $b_i^{\mathrm{data},(k)}$ for every data event and $b_i^{\mathrm{MC},(k)}$ for every MC event, using two independent sub-RNGs (a data RNG seeded from k, and an MC RNG seeded from $k + 10^7$). The two sources are uncorrelated by construction.
2. Multiply data weights by b^{data} and MC weights (both reco and truth copies of each MC event) by the *same* b^{MC}. This keeps the migration matrix self-consistent.
3. Run OmniFold (5 iterations) with the GBDT seed *pinned*. Pinning the seed isolates the bootstrap signal from the ML noise of §A.2.1.
4. Save `hXSec2D` (and 1D projections).

Stage-1 used $N = 50$ replicas (rank-deficient on 205 bins); Stage-2 scaled to $N = 300 > 205$ for a full-rank statistical covariance.

Caveat (seed-varying replicas). The $N = 300$ covariance was produced with replicas that varied *both* `--bootstrap-seed N` and `--seed N`, so it carries a small ML-stochastic contribution on top of the pure Poisson statistics. The leak is numerically small relative to the systematic block (the ML floor is 0.166% vs the stat 0.564%); a purely Poisson block would fix `--seed` while varying only `--bootstrap-seed`.

Math. Flatten each replica's 14×16 histogram into a 205-vector $\mathbf{X}^{(k)}$ over reported bins (row-major over $(p_T, p_\parallel)$). The bootstrap covariance is the unbiased sample covariance:

$$\left(\mathbf{C}^{\mathrm{boot}}\right)_{ij} = \frac{1}{N-1} \sum_{k=1}^{N} \left(X_i^{(k)} - \bar{X}_i\right)\left(X_j^{(k)} - \bar{X}_j\right), \qquad \bar{X}_i = \frac{1}{N} \sum_k X_i^{(k)}. \tag{5}$$

This is exactly `np.cov(Xrep, rowvar=False, ddof=1)` in `uq/analyze_uq.py`. The rank-deficiency check is the Cholesky-plus-jitter pattern from §A.1.

Why Poisson(1) is the right primitive. A non-parametric bootstrap with replacement assigns event i a multinomial count n_i with $\mathbb{E}[n_i] = 1$, $\mathrm{Var}[n_i] = (N-1)/N \to 1$, and $\mathbb{E}[\sum n_i] = N$. In the large-N limit the n_i become independent and each is Poisson(1) (Appendix A.9). For a weighted analysis you multiply $w_i \to w_i n_i$; since n_i may equal zero, some events are dropped from a given replica, exactly mimicking the "with replacement" resample.

Code pointer. Bootstrap weight draws and seed pinning: `unfold_2d_omnifold_unbinned.py` (`--bootstrap-seed`, `--seed`).
Covariance rollup: `uq/analyze_uq.py`.
Output: `uq/bootstrap_MEFHC_300/uq_covariance_boot300.root`.

Worked example ($N = 50$ vs $N = 300$, MEFHC lgbm).

Metric (205 bins)	$N = 50$	$N = 300$
Total σ relative std	0.068%	$\boxed{0.061\%}$
Per-bin median relative spread	0.532%	$\boxed{0.564\%}$
Per-bin p_{84}	1.235%	1.215%

The per-bin median is stable from $N = 50$ to $N = 300$ (0.53% → 0.56%); the $N = 300$ block is the one used in the headline because it is full rank ($N > 205$), whereas $N = 50$ was rank-deficient and its Cholesky needed jitter 1.6×10^{-94}. The bootstrap median is $\approx 3.4\times$ the seedscan median (0.166%), as expected for a sample of $\mathcal{O}(10^6)$ data events. Across datasets the per-bin median scales as $\sqrt{N_{\text{events}}}$: playlist $1A$ ($\approx N/12$ of MEFHC) gives 1.46%, consistent with $\sqrt{12} \times 0.56\% \approx 1.9\%$ at the same backend.

A.2.3 Systematic uncertainty: universes

What it estimates. Model uncertainty, by re-running the full unfold under each of several physically-motivated reweightings of the MC. Where the bootstrap shakes the *events*, the universes shake the *model*: a different flux → different per-event weights → different unfolded cross section.

Procedure. The C++ event loop (`runEventLoopOmniFold.cpp`) dumps, for each MC event, alternative weight branches $w^{\text{truth}}_{B,u}$ and $w^{\text{reco}}_{B,u}$ for every (band B, universe index u) pair. The Stage-2 sweep has **187 universes across 44 bands**. Five bands are genuinely kinematic: BeamAngleX/Y, MuonResolution, and Muon_Energy_MINERvA/MINOS. They require selection-complete `MNV101_-ACTIVE_UNIVERSE` event loops when migration across the event selection matters. MinosEfficiency and GEANT are weight-only. The dominant ensemble is the 100-universe PPFX flux band; almost all other bands are $\pm 1\sigma$ pairs.

Band (examples)	Variation	N_u
`Flux`	PPFX hadron-production sampling	100
`Muon_Energy_MINOS`	Muon energy scale, $\pm 1\sigma$	2
`Muon_Energy_MINERvA`	Muon energy scale (lateral), $\pm 1\sigma$	2
`MinosEfficiency`	MINOS muon-match efficiency, $\pm 1\sigma$	2
`MaRES`, `MvRES`, `MaCCQE`	GENIE knobs, $\pm 1\sigma$	2 ea.
`GEANT_{Neutron,Pion,Proton}`	FSI response (lateral), $\pm 1\sigma$	2 ea.

For each (B, u), `unfold_2d_omnifold_unbinned.py --universe B:u` reweights all MC events (truth and reco) by the matching branch and runs OmniFold with a fixed seed. A separate matched "CV" job runs without `--universe`, with the *same* omnifile, backend, seed, iteration count, and flux, so that ML noise cancels between CV and universe.

Math (MAT-conformant convention). For universe (B, u), define the per-bin value $\mathbf{X}^{(B,u)}$, a 205-vector over reported bins. MINERvA's MAT framework (`PlotUtils/MnvVertErrorBand::CalcCovMx`, see §A.2.4) builds the per-band covariance as the *mean-centered, biased* sample covariance, referenced to the **universe mean** (not the CV) and normalized by $1/N_u$:

$$\boxed{\mathbf{C}^B \;=\; \frac{1}{N_u}\sum_{u=1}^{N_u}\big(\mathbf{X}^{(B,u)} - \langle\mathbf{X}\rangle_B\big)\big(\mathbf{X}^{(B,u)} - \langle\mathbf{X}\rangle_B\big)^{\top}, \qquad \langle\mathbf{X}\rangle_B = \frac{1}{N_u}\sum_u \mathbf{X}^{(B,u)}.} \tag{6}$$

This single formula is applied *uniformly* to all $N_u \geq 2$ — there is **no special-case $\pm 1\sigma$ pair formula**. For a $\pm 1\sigma$ pair ($N_u = 2$) it collapses to the rank-1 $\frac{1}{4}(\mathbf{X}_+ - \mathbf{X}_-)(\mathbf{X}_+ - \mathbf{X}_-)^{\top}$; mean-centering kills the symmetric component, so a one-sided or asymmetric pair correctly contributes near-zero variance rather than the rank-2 contribution a naive CV-centered formula would give (Appendix A.10).

Target-nucleon normalization (rank-1 add-on). MINERvA quotes a 1.4% overall target-nucleon normalization uncertainty (Aliaga NIM, arXiv:1305.5199) that lives outside the `MnvVertErrorBand` machinery. We add it as a fully-correlated rank-1 outer product on top of the band sum:

$$\mathbf{C}^{\text{norm}} \;=\; (\sigma_N\,\mathbf{X}^{\text{CV}})(\sigma_N\,\mathbf{X}^{\text{CV}})^{\top}, \qquad \sigma_N = 0.014, \tag{7}$$

via `analyze_universes.py --add-norm 0.014`.

Total systematic covariance. Bands are assumed independent (different physics sources, different RNGs), so

$$\mathbf{C}^{\text{syst}} \;=\; \sum_B \mathbf{C}^B \;+\; \mathbf{C}^{\text{norm}}. \tag{8}$$

Independence is a modeling assumption with one known correlated exception — the Bashyal flux $\leftrightarrow$ Muon_Energy_MINOS joint block (§A.4), which we have since rederived ($\rho = 0.900$, `uq/flux_muonE_cross.root`); it is real but rank-preserving, so for the 2D budget the block sum above is retained as the headline and the cross term is carried as a diagnostic.

There is, however, a second and larger correction that Eq. (8) omits: the band sum is linear in the per-band covariances, whereas the OmniFold reweighter is *nonlinear* in the input perturbations, so jointly perturbing all bands and re-unfolding produces a cross term beyond any pairwise correlation. We measure this directly with a *unified throw* — N joint re-unfolds in which every reweightable band and the flux multisim are perturbed simultaneously per event ($w_{\text{CV}}\prod_B \rho_B^{g_B}$, $g_B \sim \mathcal{N}(0,1)$) and the cross section re-extracted. The corrected implementation uses asymmetric $\pm 1\sigma$ endpoint interpolation, one fixed estimator seed, and covariance about the throw mean; the mean shift from CV is stored separately. It does not subtract a scalar run-to-run jitter floor (`nd-unfolding/unified_throw_cov.py`). The old 4D/5D/FPS numerical examples and adoptions are quarantined pending corrected production. A single-throw superposition probe is insufficient to test the nonlinear response; a strict joint ensemble and matched endpoint comparator are required. Any later adoption must document the low-rank treatment, mean shift, and selection-support bound before transferring information onto a sweep block, $\mathbf{C}^{\text{syst}} \to (\mathbf{C}^{\text{syst}} - \mathbf{C}^{\text{vert}}) + \mathbf{G}\,\mathbf{C}^{\text{vert}}\mathbf{G}$ ($\mathbf{G} = \operatorname{diag} g$), which is positive-semidefinite by construction (a direct rank-N swap is not) and never falls below the block baseline.

Code pointer. Event-loop dump: `runEventLoopOmniFold.cpp` (env `MNV101_DUMP_UNIVERSES`).
Per-universe unfold: `unfold_2d_omnifold_unbinned.py --universe BAND:IDX`.
Per-band rollup: `uq/analyze_universes.py` (`--add-norm 0.014`).
Output: `uq/universe_stage2_MEFHC_full_matcorr/uq_universe_covariance_full_matcorr.root`.

Worked example (full matcorr sweep). Top contributing bands, per-bin median relative σ_i/X_i^{CV} over 205 reported bins:

Band	Per-bin median rel. σ
Muon_Energy_MINOS	2.305%
Muon_Energy_MINERvA	1.285%
MinosEfficiency	1.472%
Flux (PPFX, $N_u = 100$)	1.008%
MaRES	0.548%
MvRES	0.377%
MaCCQE	0.357%
Systematic total (44 bands + norm)	$\boxed{4.783\%}$ (median); $\sqrt{\mathrm{Tr}\mathbf{C}} = 2.463 \times 10^{-39}$

Grouped by category the medians are: Muon reconstruction 3.366%, Models 2.034%, Flux 1.008%, Hadronic response 0.758%, Statistical 0.563%, Normalization 0.081%. Muon energy scale (MINOS+MINERvA) and the MINOS-efficiency band dominate; these are exactly the bands the Ruterbories paper flags as flux-correlated (§A.4).

A.2.4 How MINERvA builds its published covariance

MINERvA's covariance construction is not spelled out in the paper; it is defined in the MAT C++ framework. The takeaway: **MAT samples per-band exactly as we do**, using the mean-centered $1/N$ sample covariance, and sums bands element-wise.

What the Ruterbories 2021 paper says (Section V). Three uncertainty categories — flux, detector response, neutrino-interaction model — evaluated by "re-extracting the cross section using modified simulations", i.e. the universe method, but with *no covariance formula given.* Specifics quoted: flux below 4% across most of phase space thanks to the ν-e-scattering constraint (Valencia, arXiv:1906.00111); 1.4% target-nucleon normalization; in-situ test-beam hadron response with per-particle GEANT inelastic variations; muon energy scale 1% constrained by the low-recoil fit of Bashyal (JINST 16 P08068), with the explicit statement that *"the fit causes the flux and muon energy scale uncertainties to be correlated and that correlation is propagated to the final result"* — the only non-block-diagonal piece the paper admits. The only covariance that appears in equations is V in the model-test χ^2 (Eq. 1, "the measurement covariance matrix"), with 205 degrees of freedom (Table I), and a log-normal χ^2 variant motivated by the highly-correlated flux normalization (Peelle's Pertinent Puzzle).

What the paper does *not* say. No per-band covariance construction; no $\pm 1\sigma$ pair formula; no band-independence/block-sum statement; nothing about rank, condition number, or regularization of V; no bootstrap or coverage validation (the paper uses D'Agostini iterative unfolding). All of this is delegated to MAT or simply unstated.

Per-band covariance: `MnvVertErrorBand::CalcCovMx`. Reading the MAT source establishes three facts, encoded in Eq. (6):

- the reference is the **universe mean** $\langle x \rangle_u$, not the central value X_{CV};
- the normalization is $1/N$ (biased), not Bessel's $1/(N-1)$ — a 1% offset for PPFX ($N = 100$);
- there is **no special case for** $N = 2$ — the mean-centered N-universe form is applied uniformly. The "MINERvA-101 $\pm 1\sigma$ pair" formula $\frac{1}{2}(\mathbf{\Delta}_+\mathbf{\Delta}_+^\top + \mathbf{\Delta}_-\mathbf{\Delta}_-^\top)$ is a folk re-derivation that agrees only for CV-symmetric shifts.

We verified our rollup against MAT byte-for-byte: populating a `PlotUtils::MnvH1D` with the same 44 bands / 188 universes and comparing `MnvH1D::GetTotalErrorMatrix` against our `--add-norm 0` rollup gives a max element-wise relative difference of 5.5×10^{-17} (machine ε), with $\sqrt{\mathrm{Tr}}$ matching to seven digits (`uq/verify_matcorr_vs_mnvh1d.py`).

Total covariance: `MnvH1D::GetTotalErrorMatrix`. A straight element-wise sum across bands plus the statistical matrix. Band-independence is assumed; cross-band correlations exist only if encoded inside a single combined band (e.g. the Bashyal joint fit, handled outside this loop). MAT does *no* ridge / shrinkage / SVD truncation on the published total cov and no special handling of rank deficiency — the published V is full-rank because it includes the *statistical* block, not because it sums richer systematics (its systematic-only rank is $120 < 140$; see §A.4).

Upstream references that set each band's $\pm 1\sigma$ amplitude.

Input	Reference
Flux / PPFX (100 universes)	Aliaga, PRD 94 092005 (2016), arXiv:1607.00704
ν-e flux constraint ($< 4\%$)	Valencia, PRD 100 092001 (2019), arXiv:1906.00111
Muon-E scale $\leftrightarrow$ flux fit	Bashyal, JINST 16 (2021) P08068
Test-beam / GEANT hadron response	Aliaga, NIM A 789 28 (2015), arXiv:1501.06431
GENIE reweight knobs	Andreopoulos, NIM A 614 87 (2010); GENIE manual arXiv:1510.05494
1.4% normalization (detector mass)	Aliaga, NIM A 743 130 (2014), arXiv:1305.5199

A.2.5 Block-sum combination

What it estimates. The total uncertainty covariance, combining the three statistically independent components above.

Independence assumption. The bootstrap samples *events* (with two distinct sub-RNGs for data and MC). The universes vary *model weights*, with the GBDT seed pinned. The seedscan varies the *GBDT seed*, with all weights fixed. The three Monte Carlos are independent by construction:

$$\mathbf{C}^{\mathrm{tot}} \;=\; \mathbf{C}^{\mathrm{ML}} + \mathbf{C}^{\mathrm{stat}} + \mathbf{C}^{\mathrm{syst}}. \tag{9}$$

The combined-cov hook in `compare_to_paper_fullcov.py` (`--omnifold-cov ROOT:HIST`, repeatable) adds OmniFold- derived covariances on top of the paper's `TotalCovariance` through exactly this rule. $\mathbf{C}^{\mathrm{ML}}$ is more than an order of magnitude below the other two; the headline includes it but it is numerically negligible.

Worked example (UQ envelope). Component traces and per-bin median relative spreads:

Component	$\sqrt{\mathrm{Tr}\mathbf{C}}$	Per-bin median rel. σ
ML noise (lgbm seedscan, $n = 10$)	5.061×10^{-41}	0.166%
Statistical (Poisson bootstrap, $N = 300$)	1.828×10^{-40}	0.564%
Systematic (matcorr sweep + norm)	2.463×10^{-39}	4.783%
Combined (block sum)	2.470×10^{-39}	$\boxed{4.822\%}$
Paper `TotalCov` (reference)	2.676×10^{-39}	6.86%

The combined OmniFold envelope is $\approx 70\%$ of the paper's per-bin median uncertainty, dominated by the systematic block. (The combined $\sqrt{\mathrm{Tr}}$ is the quadrature sum of the three independent component traces; the per-bin median 4.822% is taken directly from the matcorr rollup summary.)

A.3 Comparing to the paper

A.3.1 The 205-bin selection

The paper reports $\mathrm{d}^2\sigma/(\mathrm{d}p_T\,\mathrm{d}p_\parallel)$ on a 14-bin p_T axis $\times$ 16-bin $p_\parallel$ axis = 224 cells (edges in `minerva_paper_anc/bin_mapping.txt`). Of those, 205 have $x^{\mathrm{paper}} > 0$ in `data_result_ptpl_-2D_minerva_inclusive_6GeV.txt`; the remaining 19 are paper-unreported cells that the paper's analysis suppresses via an efficiency cut. We drop those 19 from both sides of every χ^2 and pull — that is the unambiguous "apples-to-apples" choice.

A secondary diagnostic mask, used only as a sensitivity check, is the strict-interior set: cells where $p_T^{\mathrm{high}}/p_\parallel^{\mathrm{low}} \le \tan 20°$. Some cells in the 205-bin set straddle the $\theta_\mu < 20°$ kinematic cut and have wildly suppressed paper efficiencies that we do not replicate; pulls on those cells run to $\mathcal{O}(100)\sigma$ and dominate the χ^2. The 185-bin interior cut removes them, and we quote it only as a sensitivity check that the headline tension is not an artifact of those few boundary cells. **The canonical metric is the 205-bin set** used everywhere below.

A.3.2 Covariance-aware χ^2/ndf

Math. With $\boldsymbol{\Delta} \equiv \mathbf{x}^{\mathrm{ours}} - \mathbf{x}^{\mathrm{paper}}$ on the 205-bin set and $\mathbf{C}$ the chosen covariance,

$$\chi^2 \;=\; \boldsymbol{\Delta}^\top\,\mathbf{C}^+\,\boldsymbol{\Delta}, \qquad \mathrm{ndf} \;=\; n_{\mathrm{reported}} = 205, \qquad \chi^2/\mathrm{ndf} \;=\; \chi^2/205. \tag{10}$$

The implementation (`chi2_with_cov` in `compare_to_paper_fullcov.py`) builds the 205×205 block, inverts with `np.linalg.pinv` (§A.1), and forms the quadratic form.

Code pointer. `compare_to_paper_fullcov.py` (headline); `--omnifold-cov ROOT:HIST` (repeatable) adds each OmniFold covariance via Eq. (9); `--subtract-stat` removes the paper's stat block to avoid double-counting the bootstrap (§A.4).

Worked example: comparison to paper (headline numbers).

Metric (205 bins)	Paper	OmniFold (combined)
Width-weighted total σ, $\sqrt{\mathbf{w}^\top\mathbf{C}\,\mathbf{w}}/(\mathbf{w}^\top\mathbf{x})$	4.61%	2.26%
Per-bin median relative σ	6.86%	4.822%
χ^2/ndf	3.661 (paper-only)	$\boxed{1.699}$ (paper+ours)
Log-normal χ^2/ndf	—	1.688

The 3.661 → 1.699 drop (−54%) is the central UQ result: once the OmniFold-side error budget is included, the agreement with the paper improves to $\chi^2/\mathrm{ndf} \approx 1.7$. (A perfectly correct model with correct uncertainties gives exactly 1; 1.7 tolerates small residual shape mismatch in high-$p_\parallel$ tails, see §A.3.4.) **This combined χ^2 double-counts systematics shared with the paper** (flux, GENIE, RPA, 2p2h, MINOS); see §A.4 for why it remains the quoted headline and why we report per-bin pulls rather than a single unqualified ours-only χ^2. The log-normal variant (matching the paper's Table I treatment of the correlated flux normalization) gives 1.688, essentially identical.

Edge-bin sensitivity ($p_\parallel$-min cut). `compare_to_paper_fullcov.py` ships a side-table that recomputes χ^2/ndf on progressively shrinking subsets, using the paper covariance only (no OmniFold add):

$p_\parallel \geq$ (GeV/c)	N_{bins}	χ^2	χ^2/ndf	median ratio
1.5	205	750.49	3.661	1.0064
2.0	196	685.36	3.497	1.0059
2.5	186	545.87	2.935	1.0059
3.0	175	474.84	2.713	1.0059
3.5	163	452.95	2.779	1.0073
4.0	151	428.72	2.839	1.0068

The residual is concentrated in the low-$p_\parallel$, near-edge cells; the high-$p_\parallel$ tails carry sub-2% shape disagreement. This is a robustness check, not a separate method.

A.3.3 Pull distribution

Math. For each of the 205 reported bins,

$$\text{pull}_i \;=\; \frac{x_i^{\text{ours}} - x_i^{\text{paper}}}{\sqrt{C_{ii}^{\text{tot}}}}, \qquad \mathbb{E}[\text{pull}_i] = 0, \quad \text{Var}[\text{pull}_i] = 1 \text{ under the null.} \tag{11}$$

Summary diagnostics: pull mean, RMS, the 14×16 pull heatmap, and a pull histogram (in `compare_to_paper_fullcov.py`).

Worked example (headline numbers). With the combined covariance (paper TotalCov + matcorr universe + bootstrap-300 + ML):

- Pull mean: $\boxed{0.069}$ (centred on zero; no global offset).
- Pull RMS: $\boxed{0.466}$.

Pull RMS < 1 means the combined error band is *conservative* relative to the actual disagreement — the band is wider than it needs to be to absorb the observed scatter. No bins have $|\text{pull}| > 3$ in the reported set. Compare with paper-cov-only: mean 0.089, RMS 0.598 — similar mean (no bias), slightly wider spread before we include our own uncertainties.

A.3.4 Shape-only χ^2 with Jacobian propagation

What it estimates. Whether the *shape* of the $(p_T, p_\parallel)$ cross section matches the paper, independent of absolute normalization. Useful because the absolute normalization depends on the flux integral, target nucleon count, and POT (all of which can drift by $\mathcal{O}(\%)$), whereas the shape is a pure unfolding diagnostic.

Definition. With bin widths $w_j = (\Delta p_T)_j\,(\Delta p_\parallel)_j$ and total $T(\mathbf{x}) = \sum_{j\in\text{mask}} w_j x_j$, the unit-area shape is

$$s_i(\mathbf{x}) \;=\; \frac{x_i}{T(\mathbf{x})}. \tag{12}$$

By construction $\sum_i w_i s_i = 1$.

Jacobian. We need the covariance of $\mathbf{s}$ given the covariance of $\mathbf{x}$. Differentiating,

$$\frac{\partial s_i}{\partial x_j} \;=\; \frac{\delta_{ij}}{T} - \frac{x_i\,w_j}{T^2} \;=\; \frac{1}{T}\big(\delta_{ij} - s_i\,w_j\big). \tag{13}$$

In matrix form (in `normalize_xsec_shape.py`),

$$\mathbf{J} \;=\; \frac{1}{T}\big(\mathbf{I} - \mathbf{s}\,\mathbf{w}^{\top}\big), \qquad \mathbf{C}^{\text{shape}} \;=\; \mathbf{J}\,\mathbf{C}^{\text{paper}}\,\mathbf{J}^{\top}. \tag{14}$$

Rank reduction. $\mathbf{J}$ is rank $n-1$: the unit-area constraint $\sum_i w_i s_i = 1$ kills one degree of freedom. Algebraically, $\mathbf{w}^{\top}\mathbf{J} = \mathbf{w}^{\top}(\mathbf{I} - \mathbf{s}\mathbf{w}^{\top})/T = (\mathbf{w}^{\top} - \mathbf{w}^{\top})/T = \mathbf{0}^{\top}$, so the "all-bins-up" direction is the kernel of $\mathbf{J}$ and lives in the null space of $\mathbf{C}^{\text{shape}}$. Accordingly:

- $\mathbf{C}^{\text{shape}}$ has one zero eigenvalue and rank $n-1$.
- $\chi^2_{\text{shape}} = (\mathbf{s}^{\text{ours}} - \mathbf{s}^{\text{paper}})^{\top}(\mathbf{C}^{\text{shape}})^{+}(\mathbf{s}^{\text{ours}} - \mathbf{s}^{\text{paper}})$ (pseudo-inverse drops the constrained direction).
- $\text{ndf} = n_{\text{bins}} - 1 = 204$ on the 205-bin set.

Code pointer. `normalize_xsec_shape.py`: `shape_jacobian`, `shape_chi2`.

Worked example (headline numbers). Shape χ^2/ndf on 205 bins with the paper's `TotalCovariance` (Jacobian-propagated): $\boxed{3.596}$. This is essentially the absolute-normalization χ^2/ndf of 3.661 *minus* the contribution of overall scale. Total-σ ratio is $\sigma^{\text{ours}}_{\text{tot}}/\sigma^{\text{paper}}_{\text{tot}} = 1.0111$ — 1% scale difference. Shape disagreement at the $\sim 1\%$ level (median bin ratio 1.0064, 94% of bins within 10% of the paper) accounts for the remaining shape residual.

A.4 Ill-conditioning and rank deficiency

The obstruction. For publication we would like an *ours-only* χ^2/ndf — the data-to-model agreement evaluated with our own covariance $\mathbf{C}^{\text{ours}} = \mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}} + \mathbf{C}^{\text{ML}}$ rather than the paper's V. That requires inverting $\mathbf{C}^{\text{ours}}$, and it is rank-deficient:

$$\text{rank}(\mathbf{C}^{\text{syst}}_{\text{sample}}) = 140/205, \qquad \text{cond}(\mathbf{C}^{\text{syst}}) = 8.7\times 10^{8}. \tag{15}$$

The deficiency is set by **band count, not universe count**: under the MAT formula Eq. (6) each $\pm 1\sigma$ pair is rank 1 and the 100-universe PPFX band is rank ≤ 99 (and *effectively* rank-1: a single normalization mode holds 99.6 % of its variance, §4.1), so the 44-band sum cannot fill the 205-dimensional bin space. *This is not a missing-bands problem.* The paper's own systematic-only covariance (`TotalCov` − `StatOnlyCov`) is rank 120 — *lower* than our 140 — so we propagate at least as many independent systematic directions as the published analysis (we use the same `GetStandardSystematics`). The paper's V is full-rank (rank = 204, cond = 1.5×10^{12}) purely because it **includes the statistical block** (`StatOnlyCov`, rank 205, near-diagonal), not because it sums richer systematics. Adding our 300-replica bootstrap does the same for us: $\text{rank}(\mathbf{C}^{\text{syst}}+\mathbf{C}^{\text{boot}}) = 201$, cond $= 3.8\times 10^{14}$.

The real obstruction is statistical, not rank. Even at rank 201 a defensible single ours-only χ^2 is out of reach — but for a *statistical-precision* reason, not a missing-band one. Our OmniFold bootstrap stat block is 2.5× smaller by trace than the paper's stat block ($\sqrt{\text{Tr}\,\mathbf{C}^{\text{boot}}} = 1.8\times 10^{-40}$ vs $\sqrt{\text{Tr}}$ `StatOnlyCov` $= 4.6\times 10^{-40}$), so it regularises the small systematic modes 2.5× less. The naive pseudo-inverse then gives $\chi^2/\mathrm{ndf} = 252$ for $\mathbf{C}^{\mathrm{syst}} + \mathbf{C}^{\mathrm{boot}}$, dominated by poorly-determined directions, against the paper's 3.66 with its larger stat floor. The lever toward an ours-only χ^2 is therefore the *statistical* block — not the hunt for more systematic bands. The gap is *not* a missing MC-statistical term: our bootstrap already draws independent Poisson(1) multipliers on *both* data and MC per replica (§A.2.2; Practical Guide 2507.09582 §5.1), so data-stat and MC-stat contribute jointly. With that excluded, the smaller stat covariance most likely reflects genuinely higher unbinned-OmniFold statistical efficiency relative to the published binned D'Agostini unfold (whose iterative regularization amplifies statistical fluctuations), or a difference in what the paper's `StatOnlyCov` encompasses — the split-bootstrap test below resolves this.

Resolution (data/MC split bootstrap). A 200+200-replica *split* bootstrap Poisson-fluctuates the data weights only (`boot_data`) or the MC weights only (`boot_mc`) with independent RNGs, so in expectation $\mathbf{C}^{\mathrm{boot}} = \mathbf{C}^{\mathrm{boot}}_{\mathrm{data}} + \mathbf{C}^{\mathrm{boot}}_{\mathrm{mc}}$. Three findings settle the question. (i) *Closure holds:* $\sqrt{\mathrm{Tr}(\mathbf{C}_{\mathrm{data}} + \mathbf{C}_{\mathrm{mc}})}/\sqrt{\mathrm{Tr}\mathbf{C}^{\mathrm{boot}}} = 1.07$, with the variance split 77% data / 23% MC. (ii) The *data stream alone* gives $\sqrt{\mathrm{Tr}\mathbf{C}_{\mathrm{data}}}/\sqrt{\mathrm{Tr}\,\texttt{StatOnly}} = 0.356$ ($\mathbf{C}^{\mathrm{boot}}/\texttt{StatOnly} = 0.392 = 1/2.55$): the 2.5× deficit is present in the data-only stat error, so it is *not* an MC-statistics omission — it reflects genuinely higher unbinned-OmniFold statistical efficiency than the binned D'Agostini unfold. (iii) There is *also* a structure difference: our stat block is correlated (mean $|\text{off-diag}\,\rho| = 0.108$, leading-eigenvector overlap with `StatOnly` 0.46) whereas the paper's is near-diagonal (0.009); scaling our block to the paper trace still leaves $\chi^2/\mathrm{ndf} = 76.7$, so the gap is shape as well as magnitude. **Consequence:** the correct ours-only χ^2 uses our *own* $\mathbf{C}^{\mathrm{ours}}$, not the paper's stat floor. The truncation-dependence below remains (the inversion is still regularizer-set), so we lead with per-bin pulls and the scan; the ours-only goodness-of-fit is reported on $\mathbf{C}^{\mathrm{ours}}$ via a truncated-spectral χ^2 (retain $\lambda > 10^{-10}\lambda_{\max}$), the same convention carried into the 3D generator comparison (§A.4 is the 2D anchor; the 3D comparison applies it on all 1431 reported bins).

Regularization scans (all cutoff-dependent). On $\mathbf{C}^{\mathrm{ours}}$ over the 205 reported bins, no regularizer gives a stable single number:

Method	Setting	Result
Eigenvalue truncation	$\lambda/\lambda_{\max} > \{10^{-3}, 10^{-4}, 10^{-5}\}$	rank $\{19, 49, 82\}$; $\chi^2/205 = \{0.13, 1.12, 3.08\}$
Ridge $\mathbf{C} + \varepsilon\lambda_{\max}\mathbf{I}$	$\varepsilon = \{10^{-3}, 10^{-4}, 10^{-5}\}$	$\chi^2/205 = \{0.35, 1.38, 4.30\}$
Diagonal-target shrinkage	$\lambda = 0.05$	rank $178 \to 204$, cond 1.3×10^{11}, $\chi^2/205 = 3.29$

Each method spans roughly two decades of $\chi^2/205$ across a two-decade scan of its knob, with no defensible single cutoff. The useful conclusion is diagnostic, not a number: the naive inverse is dominated by tiny, poorly-determined modes, and in the retained high-variance modes the central-value difference is not large. We therefore report direct per-bin pulls (§A.3.3) and this scan rather than a single unqualified ours-only χ^2.

Bootstrap/paper-stat double-count. The combined headline (§A.3.2) adds $\mathbf{C}^{\mathrm{boot}}$ on top of the paper's `TotalCov`, which already contains the paper's statistical block — a double count. `--subtract-stat` replaces the paper baseline by `TotalCov` − `StatOnlyCov` before adding our covariances, driving χ^2/ndf $1.699 \to 23.96$. That *over*-corrects: our $\sqrt{\mathrm{Tr}\mathbf{C}^{\mathrm{boot}}} = 1.83 \times 10^{-40}$ is far smaller than the paper's published stat block ($\sim$ the 2.7×10^{-39} scale of `TotalCov`), so subtracting the latter without a comparable replacement removes too much. The headline therefore keeps the double-counted 1.699

as the well-defined "what does OmniFold UQ add on top of the published V" statement, with the de-double-counted number on file as a secondary diagnostic.

Rank and conditioning: summary. The 140/205 systematic rank is richer than the paper's systematic-only 120 (not a missing-bands problem); the Bashyal flux $\leftrightarrow$ muon-energy joint block is rederived ($\rho = 0.900$) and carried as a diagnostic — adding it leaves the systematic rank unchanged, its directions already lying within the span of the universe covariance (§4.1); the PPFX band is effectively rank-1 (§4.1); and the 2.5× smaller bootstrap is explained by the data/MC split as genuine unbinned-OmniFold efficiency, not a missing MC term. Two paper-independent refinements are possible: a fitted $(\Phi_{\rm norm}, \Phi_{\rm shape}, \mathrm{MuonE}_{\rm scale})$ posterior from MAT-MINERvA for a self-derived ρ, and published precedent for the truncated-spectral "ours-only" χ^2 convention.

Adopted conventions. The publication-grade headline uses the combined covariance for $\chi^2/\mathrm{ndf} = 1.699$ with the double-count caveat documented; the per-band rollup is MAT-conformant and byte-for-byte verified (§A.2.4); the 1.4% normalization is included as a rank-1 band; and the ours-only cov is reported via the truncated-spectral χ^2 on $\mathbf{C}^{\rm ours}$ (rank 201) with the regularization scan disclosed.

A.5 Anatomy of the paper-covariance tension

The paradox. The central values agree well: $\sigma_{\rm tot}^{\rm ours}/\sigma_{\rm tot}^{\rm paper} = 1.011$, median bin ratio 1.006, and the *per-bin pull RMS* against the paper covariance is only 0.598 (§A.3.3). A diagonal-only χ^2 — ignoring all off-diagonal correlations — is therefore

$$\chi^2_{\rm diag}/\mathrm{ndf} = \frac{1}{n}\sum_i \frac{\Delta_i^2}{C_{ii}} = \mathrm{RMS(pull)}^2 + \mathrm{mean(pull)}^2 \approx 0.366, \tag{16}$$

yet the full-covariance $\chi^2/\mathrm{ndf} = 3.661$. The tension is a 10× inflation that lives *entirely* in the off-diagonal / small-eigenvalue structure of the published `TotalCov` (`diagnose_tension.py`). The rest of this section dissects it.

Eigenmode decomposition. Diagonalise the reported-block paper covariance, $\mathbf{C} = \sum_k \lambda_k \mathbf{u}_k \mathbf{u}_k^\top$ ($\lambda_1 \geq \cdots \geq \lambda_n$), and project the residual $\boldsymbol{\Delta}$ onto each eigenvector, $c_k = \mathbf{u}_k^\top \boldsymbol{\Delta}$. Then

$$\chi^2 = \boldsymbol{\Delta}^\top \mathbf{C}^+ \boldsymbol{\Delta} = \sum_k \frac{c_k^2}{\lambda_k}, \qquad \text{“eigen-pull”}_k \equiv \frac{c_k}{\sqrt{\lambda_k}}, \tag{17}$$

verified to reproduce the `pinv` $\chi^2 = 750.49$ to 3×10^{-9}. The decomposition shows the tension is **neither a few pathological null directions nor the absolute smallest eigenvalues**:

- The top contributing modes sit at $\lambda_k/\lambda_{\rm max} \sim 10^{-4}$–$10^{-5}$ — the lower-middle of the spectrum by magnitude, but still $\sim$ 7–8 decades above the cond-floor ($\lambda/\lambda_{\rm max} \sim 10^{-12}$) — with genuine eigen-pulls of 5–6σ.
- The ten *smallest*-λ modes carry only 3% of the χ^2; about 90 modes are needed to reach 90%. The χ^2 is broadly distributed, not localised to the ill-conditioned tail.

Regularization robustness. A `pinv`-rcond scan confirms the tension is not a numerical artifact: retaining only the 73 best-determined directions ($\lambda_k/\lambda_{\rm max} > 10^{-4}$) still gives $\chi^2/\mathrm{ndf} = 1.42$; 139 directions ($> 10^{-6}$) give 2.79. A rank-truncation scan keeping the r largest-λ modes rises smoothly 0.69 (r=50) → 2.35 (100) → 3.30 (180) → 3.66 (205) — no late jump, i.e. the tension does not ride on the last few least-determined modes.

It is shape, not normalization. Splitting $\mathbf{\Delta}$ into a component along the measured spectrum $\mathbf{x}^{\text{paper}}$ (an overall scale) and the orthogonal remainder, the scale piece contributes $\chi^2_{\text{norm}} = 0.10$ and the shape piece $\chi^2_{\text{shape}} = 750.1$. The +1.1% normalization offset is statistically irrelevant to the tension; it is a correlated *shape* difference. (Consistent with the Jacobian shape-only $\chi^2/\text{ndf} = 3.60$, §A.3.4.)

Where it lives. Mapping the dominant eigenvectors and the per-bin contribution $\Delta_i(\mathbf{C}^+\mathbf{\Delta})_i$ back onto the 14×16 grid (`tension_mode_maps.png`, `tension_chi2_map.png`) localises the tension to the **low-p_T cross-section peak ridge** ($p_T \lesssim 0.4$ GeV/c, p_T-bins 2/7/10 carry 16/11/12%) and the low-$p_\parallel$ near-edge cells ($p_\parallel$-bins 0/1/8). The carrier modes are sign-alternating *shape* oscillations across the peak, not vertical p_T-columns (which a flux $1/\Phi(p_T)$ normalization error would produce) and not the $\theta_\mu < 20°$ acceptance boundary. This is exactly where the flux and Muon_Energy bands — the two largest, and the only named cross-band correlation (Bashyal, §A.4) — dominate the covariance: tight bin-to-bin correlation there leaves little freedom to absorb a coherent $\sim$ 1–2% shape difference, so a small OmniFold-vs-IBU shape difference becomes a large χ^2.

A methodological component (estimator and iteration). Part of the tension is regularization-dependent — the same data, same normalization ($\Sigma\sigma$ within 0.1%), unfolded with a different GBDT backend moves the paper-cov χ^2 by $\sim$ 1 unit while central values barely change (iteration count, by contrast, is negligible):

OmniFold configuration	paper-cov χ^2/ndf	pull RMS
exact-GBT, 5 iter (frozen production)	3.66	0.598
HistGBT, 5 iter (10-seed mean)	2.70 ± 0.04	0.56
LightGBM, 5 iter	2.65	0.563
LightGBM, 10 iter	2.66	0.565

Two things separate cleanly. (i) **Iteration count is negligible**: LightGBM 5 $\to$ 10 iters moves the χ^2 by +0.01 (2.645 $\to$ 2.657), so OmniFold is well-converged at 5 iters and the tension is *not* an under-iteration artifact. (ii) **The GBDT estimator carries a $\sim$ 1-unit band**: the frozen exact-GBT production (3.66) is the outlier, while both histogram-binned backends agree at 2.65–2.70 (HistGBT is a 10-seed mean, range 2.63–2.74; the ±0.04 ML-seed jitter is $\sim 25\times$ smaller than the 0.96 exact$\to$hist gap, so that gap is genuinely the estimator, not seed noise). The exact-gradient trees fit finer shape structure (less implicit smoothing) and so deviate more from the paper's IBU-regularized result. This is direct evidence that ~ 1 χ^2-unit of the tension is OmniFold's implicit regularization rather than a physical data–MC disagreement (`tension_iter/`).

Conclusion. The tension is (i) a genuine, broadly-distributed, *correlated shape* difference — not a normalization offset and not a small-λ inversion artifact — concentrated in the low-p_T peak where the flux/Muon_Energy bands dominate, plus (ii) a $\sim$ 1-χ^2-unit methodological band from the GBDT estimator/iteration regularization. Both point to the same structure: the Bashyal flux$\leftrightarrow$Muon_Energy joint block (rederived at $\rho = 0.900$, §A.4) is exactly the covariance structure governing the region that carries the tension.

A.6 Calibration: coverage toys

What it estimates. Whether the bootstrap covariance band actually covers the truth at the rate it claims. A bootstrap band that is wrong — too tight or too loose — would give the right shape but the wrong χ^2. Coverage toys provide an independent calibration check.

Procedure (`sbatch_coverage_toys_MEFHC_200.sh`). Generate $N_{\text{toy}} = 200$ closure unfolds, each one combining two existing modes of `unfold_2d_omnifold_unbinned.py`:

- `--closure`: pseudo-data are MC reco events (weighted), so the reference truth marginal $\mathbf{X}^{\text{truth}}$ is known exactly and is written to `hTruthXSec2D` in every output file.
- `--bootstrap-seed` k (seeds chosen to avoid collision with the production replicas): per-event Poisson(1) draws, same as the production bootstrap.

Each toy is independent.

Math. Stack the toys into an $N_{\text{toy}} \times 14 \times 16$ array $\mathbf{U}^{(t)}$. For each reported bin (where $\bar{X}_i^{\text{truth}} > 0$, i indexing the 205 reported cells):

$$\sigma_i \;=\; \operatorname{std}\big(\mathbf{U}_i^{(t)};\, \text{ddof} = 1\big) \quad \text{(spread across toys)}, \tag{18}$$

$$r_i^{(t)} \;=\; \frac{U_i^{(t)} - \bar{X}_i^{\text{truth}}}{\sigma_i} \quad \text{(per-toy, per-bin standardised residual)}, \tag{19}$$

$$c_i \;=\; \frac{1}{N_{\text{toy}}} \sum_{t=1}^{N_{\text{toy}}} \mathbf{1}\big\{|r_i^{(t)}| \le 1\big\} \quad \text{(per-bin coverage)}. \tag{20}$$

The expected value of c_i under a well-calibrated Gaussian band is the central-σ probability of a unit normal:

$$c^{\text{expected}} \;=\; \Phi(1) - \Phi(-1) \;=\; \operatorname{erf}(1/\sqrt{2}) \;\approx\; 0.6827. \tag{21}$$

A second diagnostic is the mean absolute pull, $\langle |r| \rangle$, which for a unit normal is $\sqrt{2/\pi} \approx 0.798$.

Code pointer. `uq/coverage_toys.py` (rollup); driver `sbatch_coverage_toys_MEFHC_-200.sh`; outputs `uq/coverage/2d_xsec_MEFHC_5iter_lgbm_coverage_toy*.root` and `uq/coverage/coverage_summary.txt`.

Worked example (coverage). With $N_{\text{toy}} = 200$ on the 205 reported bins:

- Mean coverage: $\boxed{68.71\%}$ (target 68.27%; $\Delta = +0.44\%$); median 68.50%.
- $\langle |r| \rangle = 0.794$ (target 0.798; within 0.5%).
- Signed residual mean: $+0.006 \pm 0.082\,\sigma$ (no bias).
- Fraction of bins above the 65% target: $\boxed{97.6\%}$ — only 5 of 205 bins fall below.

The agreement of $\langle c \rangle$ and $\langle |r| \rangle$ with their expected values at the half-percent level is the cleanest demonstration that the Poisson-bootstrap band is properly calibrated. The 200-toy ensemble reduces the per-bin binomial uncertainty to $\sqrt{0.68 \cdot 0.32/200} \approx 3.3\%$, which is why 97.6% of bins clear the 65% target.

A.7 Closure tests: detecting bias

The variance methods of §A.2 answer the question "how much does the unfold wobble." Closure tests answer a different question: "is the unfold biased — does it return a value that is consistently offset from the right answer when we inject a known truth?" Bias does not show up in the variance procedures; you have to look for it.

The three closure tests inject a known deformation into the MC, unfold, and check that the answer reproduces the deformation. Each test has its own "acceptance threshold," set comfortably above the ML-noise floor of 0.166%.

A.7.1 Truth-reweight closure

Procedure. For a known shape $f(p_T, p_\parallel)$, multiply both the MC reco weights *and* the MC truth weights by f at the event level. Unfold the reweighted pseudo-data against the unchanged (CV) response. The reference is the reweighted truth marginal — the unfolded answer should match it.

Two shapes tested.

- `gauss_pt`: $f = 1 + A \exp\big[-((p_T - p_T^0)/\sigma)^2\big]$ with $A = 0.2$, $p_T^0 = 0.4$ GeV/c, $\sigma = 0.1$ GeV/c.
- `tilt_pz`: $f = 1 + \alpha(p_\parallel - p_\parallel^{\text{ref}})$ with $\alpha = 0.1$, $p_\parallel^{\text{ref}} = 5$ GeV/c.

Acceptance. Median per-bin |residual| $\leq 1.5\%$.

Code pointer. `uq/closure/closure_truth_reweight.py`.

Worked example (1A lgbm 5-iter).

- `gauss_pt`: median |residual| = $\boxed{0.046\%}$ (PASS).
- `tilt_pz`: median |residual| = $\boxed{0.013\%}$ (PASS).

Both are $\sim 100\times$ tighter than the threshold; OmniFold recovers the injected truth shape essentially perfectly.

A.7.2 Hidden-variable closure

Procedure. This is the subtle one. The OmniFold feature set sees $(p_T, p_\parallel)$ on reco and truth, but *not* the resolution variable $\Delta p_T \equiv p_T^{\text{reco}} - p_T^{\text{truth}}$. So we inject a bias on Δp_T only — a kinematic axis the network cannot see — and ask: how much of that bias leaks into the measured $(p_T, p_\parallel)$ marginal?

The pseudo-data weight bump is

$$w^{\text{reco}} \;\rightarrow\; w^{\text{reco}} \cdot \Big(1 + A \exp\big[-((\Delta p_T - c)/\sigma_{\Delta p_T})^2\big]\Big), \tag{22}$$

with $c = +0.1$ GeV/c, $\sigma_{\Delta p_T} = 0.05$ GeV/c. Truth weights are *unchanged*, so the reference is the unaltered CV truth marginal. The amplitude is scanned: $A \in \{0.05, 0.10, 0.30\}$.

Leakage coefficient. Define the signed residual $\bar{r}(A)$ as the median of the per-bin signed residual at amplitude A. The leakage coefficient is the slope of a linear fit:

$$\bar{r}(A) \;\approx\; \kappa_{\text{leak}}\, A \quad \Rightarrow \quad \kappa_{\text{leak}} \;=\; \frac{\partial \bar{r}}{\partial A}. \tag{23}$$

κ_{leak} tells you what fraction of an unseen Δp_T-shaped bias contaminates the visible answer.

Acceptance. Median per-bin |residual| $\leq 3.0\%$ (looser than truth- reweight because Δp_T and p_T are weakly correlated, so the unfolding can never fully kill the leak).

Code pointer. `uq/closure/closure_hidden_var.py`.

Worked example (1A lgbm 5-iter, 205 reported bins).

- Per-amplitude median residuals: 0.57%/1.14%/3.43% for $A = 0.05/0.10/0.30$ — linear in A, as expected for small perturbations.
- Linear fit slope: $\kappa_{\text{leak}} = \boxed{0.113}$.

About 11% of a hidden Δp_T bias leaks into the measured $(p_T, p_\parallel)$ marginal — under the 3% threshold for the largest realistic deformation ($A = 0.30$). Larger leakage would indicate that Δp_T ought to be added to the feature set; the present number is small enough to be safely ignored.

A.7.3 Alternative-model closure

Procedure. Train the OmniFold response on the CV weights. Build the pseudo-data from an alternative-universe reweighting (e.g. `MaCCQE:0` $+1\sigma$, or `Flux:50` PPFX index). The reference is the in-acceptance-extrapolated alt truth:

$$\tilde{X}_i^{\text{ref}} \; = \; X_i^{\text{CV,truth}} \cdot \frac{X_i^{\text{alt,in-acc}}}{X_i^{\text{CV,in-acc}}}. \tag{24}$$

This scales the CV truth marginal by the ratio of in-acceptance counts between alt and CV, isolating the unfolding's response to a model-shape change from acceptance/efficiency effects. MaCCQE probes an interaction-model shape change; Flux probes flux reweighting — two distinct kinds of model dependence.

Acceptance. Median per-bin |residual| $\leq 2.0\%$.

Code pointer. `uq/closure/closure_alt_model.py`.

Worked example (1A lgbm 5-iter).

- `MaCCQE:0` ($+1\sigma$): median |residual| = $\boxed{0.068\%}$ (PASS); in-acceptance bias -1.68% folded out by the extrapolation.
- `Flux:50` (PPFX idx 50): median |residual| = $\boxed{0.376\%}$ (PASS); in-acceptance bias -7.32%.

Both pass with $\gg 5\times$ headroom. The unfolding is robust to model variations on the order of $\pm 1\sigma$ of GENIE and full PPFX flux universes.

A.8 How the pieces fit together

The combined result, in one sentence: *across* 205 *paper-reported bins, the combined OmniFold + paper error budget gives* $\chi^2/\text{ndf} = 1.699$ *and pull mean/RMS* 0.069/0.466*, with a pull-RMS-*< 1 *band that has been independently checked at the* 0.5% *level by* 200 *coverage toys, on top of a* 0.166% *ML-noise floor that is more than an order of magnitude below the systematic envelope.*

Audit trail of the headline

Number	Section	Procedure
$\chi^2/\text{ndf} = 1.699$	§A.3.2	Eq. (10) with $\mathbf{C}^{\text{paper}} + \mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{stat}} + \mathbf{C}^{\text{ML}}$
$\mathbf{C}^{\text{syst}}$	§A.2.3	187 universes, MAT-conformant Eq. (6) + norm
$\mathbf{C}^{\text{stat}}$	§A.2.2	Poisson bootstrap, 300 replicas, Eq. (5)
$\mathbf{C}^{\text{ML}}$	§A.2.1	lgbm seedscan, $n = 10$ trials
$\mathbf{C}^{\text{paper}}$	§A.2.4	Paper's published `TotalCovariance`
Pull RMS $= 0.466$	§A.3.3	Per-bin standardised residual on combined $\mathbf{C}$
Combined median rel. $= 4.822\%$	§A.2.5	$\sqrt{\text{diag}\mathbf{C}^{\text{tot}}}/\mathbf{X}^{\text{CV}}$
Mean coverage $= 68.71\%$	§A.6	Coverage toys, $N_{\text{toy}} = 200$

Standing assumptions

1. **Gaussian approximation.** Pulls and χ^2 treat each bin as Gaussian. Justified because per-bin entries are weighted sums over $\mathcal{O}(10^4)$ MC events; central limit theorem applies.
2. **Independence of error sources.** Eq. (9) requires no cross-correlations between ML, statistical, and systematic. The bootstrap pins the GBDT seed (so ML noise cancels at first order); universes pin the seed too; the seedscan pins the bootstrap. Cross-terms are higher-order. (The available $N = 300$ bootstrap block carries a small residual ML leak, §A.2.2.)
3. **Bootstrap calibration.** §A.6's mean coverage $= 68.71\%$ certifies $\mathbf{C}^{\text{stat}}$ to half a percent.
4. **Band independence in $\mathbf{C}^{\text{syst}}$.** Different physics sources are taken as uncorrelated; the one known exception, the Bashyal flux $\leftrightarrow$ muon-E joint block, is rederived and carried as a diagnostic — adding it leaves the systematic rank unchanged (§A.4).
5. **Combined-χ^2 double-count.** The headline shares systematics with the paper's V and is the "what OmniFold UQ adds on top of V" statement, not an independent ours-only test (§A.4).
6. **Closure passes are valid for the perturbations tested.** Nothing guarantees they pass for unphysical deformations far outside the training distribution.

A.9 Multinomial → independent Poisson bootstrap limit

A non-parametric bootstrap with replacement draws N events from the original N-event sample. Let n_i be the number of times event i is drawn. Then $(n_1, \ldots, n_N) \sim \text{Multinomial}(N, \mathbf{p})$ with $p_i = 1/N$ uniform, $\sum_i n_i = N$.

The marginal $n_i \sim \text{Binomial}(N, 1/N)$, whose mean is 1 and variance $(N-1)/N \to 1$. For large N, the binomial converges to Poisson(1) (Poisson limit theorem: many trials, small success probability, fixed expected count).

The full multinomial has the constraint $\sum n_i = N$, which introduces $\mathcal{O}(1/N)$ correlations between the n_i. In the large-N limit those correlations vanish and the n_i become *independent* Poisson(1) variables. This is the Poissonisation of the multinomial: the conditioned N-event multinomial is a Poisson process *conditioned* on hitting N counts, and the unconditional version is independent Poissons.

In practice, the independent-Poisson form is what we implement (one `np.random.default_rng().poisson(1.0)` draw per event), and the $\sum n_i = N$ constraint is dropped. For $N \gtrsim 10^6$ (our regime), the relative error from the dropped constraint is $\mathcal{O}(1/\sqrt{N})$ on the total event count — far smaller than the per-bin Poisson variance.

A.10 The $\pm 1\sigma$ pair covariance: mean-centered (MAT-conformant)

For a single universe band with two universes $u = 0, 1$ at $\pm 1\sigma$, write the two value vectors as $\mathbf{X}_0$ and $\mathbf{X}_1$, with universe mean $\langle \mathbf{X} \rangle = \frac{1}{2}(\mathbf{X}_0 + \mathbf{X}_1)$. The MAT-conformant estimator Eq. (6) at $N_u = 2$ is

$$\mathbf{C}^B_{\text{mean}} \;=\; \tfrac{1}{2} \sum_{u=0,1} (\mathbf{X}_u - \langle \mathbf{X} \rangle)(\mathbf{X}_u - \langle \mathbf{X} \rangle)^\top \;=\; \tfrac{1}{4}(\mathbf{X}_0 - \mathbf{X}_1)(\mathbf{X}_0 - \mathbf{X}_1)^\top, \tag{25}$$

a rank-1 outer product of the *half-difference*. It is mean-centered, so any common offset of the pair away from X_{CV} contributes nothing.

A naive alternative centers on the CV rather than the pair mean, with $\boldsymbol{\Delta}_u = \mathbf{X}_u - \mathbf{X}_{\text{CV}}$ and decomposition $\boldsymbol{\Delta}_0 = \mathbf{m} + \mathbf{a}$, $\boldsymbol{\Delta}_1 = \mathbf{m} - \mathbf{a}$:

$$\mathbf{C}^B_{\text{CV-centered}} \;=\; \tfrac{1}{2}\big(\boldsymbol{\Delta}_0 \boldsymbol{\Delta}_0^\top + \boldsymbol{\Delta}_1 \boldsymbol{\Delta}_1^\top\big) \;=\; \mathbf{m}\mathbf{m}^\top + \mathbf{a}\mathbf{a}^\top. \tag{26}$$

The two agree *only* for a CV-symmetric pair ($\mathbf{m} = 0$), where both reduce to $\mathbf{a}\mathbf{a}^\top$. For an asymmetric or one-sided pair the CV-centered form adds a spurious rank-1 $\mathbf{m}\mathbf{m}^\top$ term — it treats the common offset as a second variance direction, which mean-centering correctly collapses to near zero. We mean-center throughout, matching MAT (§A.2.4); the bands where the distinction matters are the asymmetric ones (BeamAngle, MuonResolution, and the one-sided NC band).

B Unbinned background subtraction by negative-weight injection

The background subtraction of §3.1 is the one step in which the analysis binning enters before the final histogramming: each data event in reconstructed bin b is down-weighted by the purity $w_{\text{pur}}(b) = \max(0, N_{\text{data}}(b) - N_{\text{bkg}}(b))/N_{\text{data}}(b)$. This appendix documents the fully unbinned realization of the same correction — injecting the background simulation onto the measured side of the fit with negative weight — shows that it targets an identical estimand, and quantifies its agreement with the binned correction on the 2D measurement. It is the continuous version of the purity weight recommended for nontrivial backgrounds by Ref. [18], and the natural treatment for the full-phase-space extension, where the background is larger.

B.1 Estimator

Step 1 of OmniFold trains a classifier that separates the simulated reconstruction (class 0, weighted density S) from the measured sample (class 1, weighted density ρ_1), then reweights the simulation by the learned density ratio $r(x) = p(x)/(1 - p(x))$, whose population optimum is $r^\star(x) = \rho_1(x)/S(x)$. The background treatment is entirely a choice of ρ_1:

- the binned purity weight sets $\rho_1(x) = D(x)\, w_{\text{pur}}(x) = \max(0, D(x) - B(x))$, realized by a per-reco-bin scalar applied to each data event, with D the data density and B the POT-scaled background density;
- negative-weight injection appends the background-simulation reconstruction events to the measured side with weight $-w_{\text{bkg}} \times$ (POT scale), leaving the data events at $+1$, so that $\rho_1(x) = D(x) - B(x)$ *continuously*, with no analysis binning.

Both give the same background-subtracted estimand $r^\star(x) = (D(x) - B(x))/S(x)$; they differ only in whether the subtraction is performed per reco bin or event by event in the full reconstructed phase space. The injected background is restricted to the same reconstructed fiducial window as the data, so the two constructions act on identical support.

The floor. Where the background locally exceeds the data, $D - B < 0$, the binned method floors the correction at zero through the $\max(0, \cdot)$. In the unbinned case the classifier output $p \in (0, 1)$ forces $r = p/(1 - p) \geq 0$, so a negative reweight cannot arise; the floor is therefore structural rather than imposed. The fit is not well-posed in such a region, however — the class-1 weighted mass is locally negative — so the principled treatment there is the "Stay Positive" refinement of Ref. [39], which resamples the signed-weight sample into an equivalent positive-weight one before training. Whether the plain injection and the refinement coincide in practice depends on the classifier. With the histogrammed gradient-boosted density estimator used for the production result they agree to the median of 1.000 reported below, at the sub-percent genuine-background scale of this measurement. With an *exact* (tree-boosted) classifier the raw signed weights instead destabilize training — a full low-p_T row of the cross section inflates by up to a factor $\sim 5 \times 10^4$ where the local background is large — and the Stay-Positive resampling removes the pathology, recovering agreement with the binned purity to a median of 0.999 (141/148 bins within 5 %). The refinement is therefore what makes the unbinned subtraction robust across estimators, and it matters most for the larger, more structured backgrounds of the full-phase-space extension.

B.2 Closed-form toy

The estimator is verified on a one-dimensional problem with a known answer: signal simulation drawn from $\mathcal{N}(0, 1)$, a data signal from $\mathcal{N}(0.3, 1.1)$ with equal normalization, and a background $\mathcal{N}(2, 0.7)$ contributing 16.7 % of the pseudo-data, so that the target reweight $(D - B)/S = \phi(x; 0.3, 1.1)/\phi(x; 0, 1)$ is a smooth positive function. A gradient-boosted classifier trained with the signed-weight construction recovers this ratio to a median 6.8 % relative error — matching the binned purity construction's 6.7 % on the identical problem, and reproducing it bin-for-bin — while omitting the subtraction leaves a 10.9 % bias from the residual background. The result is stable across classifier seeds (± 0.3 %), and the negative sample weights introduce no numerical instability. In a stress configuration with the background moved into the sparse signal tail and deliberately over-subtracted, the learned reweight floors cleanly at zero over the region where $B > D$, as anticipated above.

B.3 Comparison on the 2D measurement

Table 3 compares the two background treatments on the ME FHC data, unfolded with identical settings and a common classifier seed so that the difference isolates the background treatment. The two agree to -0.13 % on the total cross section — an order of magnitude inside the ~ 3 % purity correction itself — with a median per-bin ratio of 1.000 and a 1.4 % RMS scatter across the plane (Fig. 38). The only bin differing by more than a few percent is a single low-p_T, high-background edge bin (-12.6 %), exactly where the background concentrates and the subtraction is largest. The unbinned subtraction thus reproduces the adopted binned correction wherever the correction is well determined, and departs from it only in the regime where the binned $\max(0, \cdot)$ floor and the structural floor of §B.1 are themselves the operative choice.

B.4 Propagation through systematics and statistics

A central-value match is necessary but not sufficient: to replace the binned correction the unbinned subtraction must also reproduce the *uncertainty* budget. The full 2D ME FHC covariance was therefore rebuilt end to end under negative-weight injection — the systematic ensemble of 187 universes, with the genuine background varied consistently within each universe rather than frozen at nominal, and a statistical bootstrap — and compared to the binned-purity construction with identical settings. The two agree at the level the $(D - B)/S$ identity predicts: the systematic

Table 3: Binned purity versus unbinned negative-weight background subtraction on the 2D ME FHC measurement (common estimator and seed). The per-bin figures are the negative-weight/purity ratio over the populated cross-section bins.

	binned purity	negative-weight
total σ (cm^2/nucleon)	$3.0727e-38$	$3.0687e-38$
total ratio (neg/pur)	0.9987 (-0.13 %)	
median per-bin ratio	1.000	
per-bin RMS deviation	1.4 %	
largest single-bin deviation	-12.6 % (low p_T, high background)	

covariance matches in overall magnitude to a $\sqrt{\text{tr}}$ ratio of 0.986 (negative-weight/purity), and the statistical covariance, compared on matched bootstrap seeds, to 0.982. The residual few-per-mille differences are bootstrap and classifier-seed sampling noise, not a method bias. Negative-weight injection thus reproduces not only the central cross section but its full covariance, per bin and in aggregate.

The equivalence at the 2D scale — in the central value, the closed-form estimator recovery, and now the full systematic and statistical covariance — together with the Stay-Positive refinement that keeps the estimator robust on the larger backgrounds of the full-phase-space extension, establishes negative-weight injection as a fully validated unbinned replacement for the binned purity correction. The binned purity is retained as the default for the reproduction measurement of this note, with negative-weight injection carried as the validated unbinned cross-check and as the baseline subtraction for the full-phase-space extension (§10), where its binning-independence and continuous floor are essential.

C The MINERvA measurement-dimensionality survey

The full classification behind Fig. 20: every MINERvA differential cross-section measurement, the number of kinematic variables its headline result unfolds simultaneously, and the reference.

Table 4: Full MINERvA differential cross-section survey: the number of kinematic variables unfolded simultaneously (Dim.), classified from each analysis's headline differential result. Channels: CC-inclusive (incl.), CCQE/QE-like incl. transverse kinematic imbalance (QE), pion production CC and coherent (π), deep-inelastic / kaon (DIS), electron-neutrino (ν_e), nuclear-target ratio (ratio).

Year	Channel; observables	Dim.	Ref.
2013	QE; ν_μ CCQE $p_T, p_\parallel$	2	[40]
2013	QE; $\bar{\nu}_\mu$ CCQE $p_T, p_\parallel$	2	[41]
2014	ratio; CC on C/Fe/Pb to CH	1	[42]
2014	π; coherent $\pi^\pm$ (total & diff.)	1	[43]
2015	π; CC $1\pi^+$ (T_π, θ_π)	1	[44]
2015	QE; μ+p final states	1	[45]
2015	π; $\bar{\nu}_\mu$ single π^0	1	[46]
2016	ν_e; CCQE-like on CH	2	[47]
2016	π; $\nu_\mu/\bar{\nu}_\mu$ pion prod.	2	[48]

continued...

Year	Channel; observables	Dim.	Ref.
2016	DIS; partonic nuclear effects	2	[49]
2016	DIS; CC K^+ production	1	[50]
2017	π; CC single π^0	2	[51]
2017	QE; nuclear-dependence (multi-target)	1	[52]
2018	QE; $\bar{\nu}_\mu$ QE-like $p_T, p_\parallel$	2	[53]
2018	QE; μ-p mesonless TKI	2	[54]
2018	π; coherent $\pi^\pm$ on C	2	[55]
2019	QE; quasielastic-like ν_μ	2	[56]
2019	π; $\bar{\nu}_\mu$ single π^-	2	[57]
2020	incl.; CC-incl $\langle E_\nu \rangle \sim 3.5$ GeV $p_T, p_\parallel$	2	[58]
2020	QE; nuclear binding & TKI	2	[59]
2020	QE; high-statistics QE-like	2	[60]
2020	π; CC π^0 nuclear effects	1	[61]
2021	incl.; CC-incl $\langle E_\nu \rangle \sim 6$ GeV $p_T, p_\parallel$ **(target)**	2	[3]
2022	QE; QE-like muon & proton kinematics	2	[62]
2022	incl.; CC-incl low momentum transfer	1	[10]
2023	π; CC $1\pi^+$ (multi-target)	2	[63]
2023	π; coherent π^+ (multi-target)	1	[64]
2023	QE; $\bar{\nu}_\mu$ multi-neutron low-E_{avail}	1	[65]
2024	ν_e; low-Q^2 (E_{avail}, p_T)	2	[38]
2025	QE; A-dependence TKI (multi-target)	2	[66]
2026	**this work**; unbinned $p_T, p_\parallel$	2	—
2026	**this work**; unbinned $p_T, p_\parallel, E_{\text{avail}}$	**3**	—
2026	**this work**; unbinned $p_T, p_\parallel, E_{\text{avail}}, q_3$	**4**	—
2026	**this work**; unbinned $p_T, p_\parallel, E_{\text{avail}}, q_3, W$	**5**	—

D Codebase layout and terminology

The analysis is a chain of four components; the italicised names are used throughout the note.

- *Event loop* (C++): `runEventLoopOmniFold`, built on the MINERvA Analysis Toolkit (MAT) and the MINERvA-101 framework. It reads the production AnaTuples, applies the selection and the systematic-universe machinery, and writes per-event ROOT trees.
- *Omnifiles*: the event-loop outputs — per-event records for data, truth/reco-paired simulated signal, simulated background, and the truth-level efficiency denominator, with POT bookkeeping; the `_universes_full` variants add per-universe weight and shifted-kinematics branches.
- *Driver* (Python): the per-analysis orchestration — `2d-unfolding/unfold_2d_omnifold_unbinned.py` and its N-dimensional generalisation `nd-unfolding/unfold_nd_omnifold_unbinned.py`. The driver assembles training arrays from the omnifiles, applies the purity-weight background subtraction (§3.1), invokes the engine, and extracts the cross section.
- *Engine* (Python): `unbinned_unfolding/python/omnifold.py` — the OmniFold iteration itself, agnostic to the learner backend (LightGBM in production; sklearn `exact`/`hist` alternates; MLP and point-cloud-transformer cross-checks).

The point-cloud measurements of §9 replace the driver's scalar features with the cluster-level inputs of `nd-unfolding/pet/`, reusing the same omnifile and cross-section machinery.

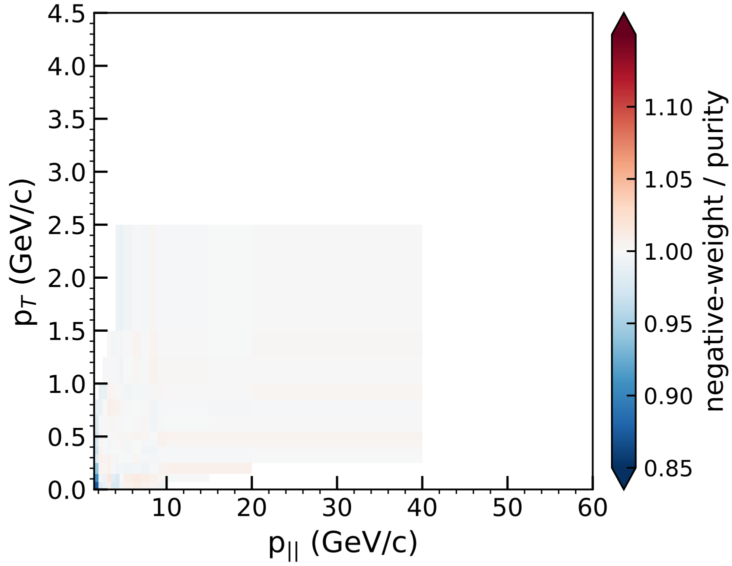

Figure 38: Per-bin ratio of the negative-weight to the binned-purity cross section on the 2D ME FHC measurement (common estimator and seed), over the populated bins. The two agree to a median of 1.000 across the plane (near-white); the only departure beyond a few percent is the single low-p_T, high-background edge bin, where the background is largest and the $\max(0, \cdot)$/structural floor of §B.1 is the operative choice.

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